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RESEARCH AND DESIGN FOR LIFTING REENTRY

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CONTENTS

*This paper expresses the opinions of the author and does not necessarily represent the official view of the Ministry of Defence.

ABSTRACT

Compared with convex or flat-bottomed craft, a lifting reentry vehicle having concavity on the lower surfaces allows a stronger shock wave to be produced and the more highly compressed air can flow beneath the vehicle with much smaller losses in lifting pressure; this is associated with the fact that concavity reduces undersurface crossflows and spillage around the leading edges. The stronger shock and the preservation of lifting pressures result from and illustrate the principle of flow containment. Flow containment should reduce rates of heat transfer at stagnation points, since, at a given wing loading, flight speed and lift/drag ratio, deceleration will occur at higher altitudes and, hence, at given flightspeed, ambient air density will be reduced. It should also reduce heat transfer under the lower surfaces since, at given flight speed, local flow velocities will have been reduced by the stronger shock. Experiments initiated in 1970 have confirmed that at reentry Mach numbers ($6 < M_\infty < 22$) and reentry angles of attack ($25^o < \alpha < 70^o$, say) a body of basically delta planform will produce significant increases in maximum lift coefficient, and in C_L per unit L/D if at least a part of the undersurface heatshield has anhedral. The present paper summarises the experimental data produced so far and demonstrates that existing analyses explain the trends and can indicate improvements in heatshield geometry. It also considers the design of realistic configurations having partly or wholly concave undersurfaces. Two families of vehicle are considered. The first is related to the current Space Shuttle orbiter, in which the specified payload size and shape has resulted in the use of a large fuselage mated to a low wing with slight dihedral and delta planform; it is shown that, when compared with the orbiter at reentry conditions, a high-wing orbiter with moderate anhedral and a partly concave undersurface can provide about 20 % higher C_L at the same angle of attack, Mach number and lift/drag ratio. The second family is of aerodynamically more advanced vehicles, which can provide higher values of hypersonic L/D and, thus, permit greater cross-range, while also offering possibly improved stability at reentry and in post-reentry conditions.

LIST OF SYMBOLS

<u>Symbols</u>

a	(-1) multiplied by shock wave curvature in plane of symmetry (see Section 2.1.)
a_∞	freestream speed of sound
b	see Fig. 47b
c	chord length
c_l	rolling moment coefficient
c_n	yawing moment coefficient
Δc_n	change in yawing moment coefficient
C_D	drag coefficient
C_H	heat transfer coefficient $\equiv \dot{q}/\rho_\infty U_\infty (H_o - H_{Aw})$
C_L	lift coefficient
C_M	pitching moment coefficient
$-C_{M_\theta}$	stiffness derivative $(-\partial C_M/\partial\theta)$
$-C_{M_{\dot\theta}}$	damping derivative $(-\partial C_M/\partial\dot\theta)$
C_p	pressure coefficient
d	model length in Fig. 41
D	drag force

D_p	pressure drag force
eas	equivalent airspeed
f_o	wind-off pitching frequency in stability experiments
g	gravitational acceleration
h	see Fig. 47b; also denotes altitude in Fig. 36
H_{AW}	stagnation enthalpy at adiabatic wall
H_o	stagnation enthalpy
l	characteristic length
l_v	rolling moment due to sideslip
L	lift force
M_∞	freestream Mach number
n_v	yawing moment due to sideslip
p	static pressure (but see also Fig. 47c)
$\bar{p}$	see eq. (2)
p_b	see eq. (1), Fig. 5b and Fig. 47c
$\bar{p}_b$	see eq. (3)
p_s	see eq. (1) and Fig. 5b
p_{stag}	stagnation pressure
$\dot{q}$	local heat transfer rate
$\dot{q}_w$	heat transfer rate for a wedge
r_n	nose radius
r_o	Earth's radius
R_d	in Fig. 41, "effective" Reynolds number $\equiv \rho_\infty U_\infty d/(\mu_\infty$ at sea level)
Re	Reynolds number
s	semi-span
S	reference area
S_t	Stanton number $\equiv \dot{q}/(\rho_\infty U_\infty \times$ stagnation enthalpy in wind-tunnel nozzle)
T_{le}	leading edge temperature
T_o	freestream stagnation temperature
T_w	wall temperature
T_∞	ambient static temperature
$\bar{u}$	see Fig. 5a
$\bar{u}_b$	see eq. (3)
U_l	stream velocity on lower surface of wing
U_∞	freestream velocity
W	vehicle weight
x	see eq. (2) and Fig. 5b

x_o	measure of axis position in Figs 44-46
$\bar{x}$	see eq. (2) and Fig. 5a
y	see eq. (2) and Fig. 5b
$\bar{y}$	see eq. (2) and Fig. 5a
y_b	see Fig. 5b
y_s	see Fig. 5b
z	see eq. (2) and Fig. 5b

Greek letters

α	angle of attack (see eq. (2))
α_c	see Fig. 24
α_L	angle of attack measured at undersurface root chordline
α_o	mean angle of attack in experiments on pitch stability (see Section 2.3.)
α_u	angle of attack measured at uppersurface root chordline
Λ	leading edge sweep angle
β	angle of yaw
β'	see Fig. 23
γ	ratio of specific heats
ε	see eq. (2)
ζ	wave angle (=90^o for normal shock)
θ	for C_M derivatives, θ is angular displacement; for Section 2.2.2. see definition in Fig. 25
θ_c	see Fig. 23
ϕ	see Fig. 23
μ	viscosity
χ	see Fig. 47b
ρ	static density
$\bar{\rho}_\infty$	see eqs (2) and (3)
τ	see Fig. 47c
ω	design value of $(\zeta-\alpha_L)$ or, in Table 1, the frequency of pitching motion
Ω	see Fig. 47c

1. INTRODUCTION

With the roll-out of the Space Shuttle orbiter in September 1976 and the prospect of the first manned orbital operation by the Space Shuttle in late 1979, it is appropriate to examine the form which the Shuttle system is now intended to take (some ten years after NASA Space Shuttle Task Group issued their Summary Report in 1969) and to summarise some research in lifting reentry, which has been conducted in the intervening period and could influence future developments.

Figure 1 shows the main features of the Space Shuttle system (1-3) in launch configuration, together with principal dimensions and weights. The size of the orbiter is closely determined (3) by the call for a cargo bay measuring 18.3 m (60 ft) by 4.6 m (15 ft) diameter and for the ability to land horizontally at weights (3) exceeding 190,000 lb (86183 kg) and at approach speeds (1) of up to 200 kt (the limit on the tyres being (3) 225 kt). With the orbiter intended to fly at least 100 missions, the orbiter rocket engines replaceable after 55 launches and the solid-fuel boosters designed for 20 missions, the external liquid fuel tank is the only expendable item, but it nonetheless accounts for nearly one quarter of the cost of each launching (2). With a view to reducing future operating costs, Rockwell International has (3) already reexamined the possible use of liquid-fuelled recoverable boosters and also a fully reusable system (3) in which the external tank and liquid-fuel boosters would fly back to base (see Fig. 2 and also Ref. (4)).

Since the late 1960s, much American work in lifting reentry has been aimed specifically at easing the design of the Space Shuttle orbiter, either in straight wing or in flat delta form. It is therefore no surprise to find that some of the more generalised research results have originated in Europe and, thus, reflect conceptual attitudes, which differ from those which have determined the pattern of American research. In particular, the aerodynamics of lifting reentry is intrinsically hypersonic and much of the European effort in hypersonic research has been concerned (5), since the late 1950s, with the design and performance of so-called waveriders (6, 7); thus, it is equally no surprise that their special merits for lifting reentry were first recognised and have been subsequently demonstrated, in the main, by European research workers.

To recognise the merits of waveriders, it is first necessary to note, that at the high shock wave pressure ratios used for lifting reentry, convex- and flat-bottomed vehicles suffer from large cross-flows on their lower surfaces and from spillage of the lifting flow around their leading edges (see Fig. 3a); the spanwise pressure gradients, which account for these cross-flows, correspond to a progressive loss of lift across the span and so to a value of overall C_L which, at that particular angle of attack, is significantly less than for example that which a plane oblique shock would provide on a two dimensional wedge. Even local values of C_L along the root chord are below the wedge value and accordingly the shock angle in the plane of symmetry is less than that for a plane oblique shock.

In essence, the argument in favour of waveriders rests on the fact that deltas can be designed as by Nonweiler (6) to produce plane oblique shocks at angles of attack up to that for wave detachment (see Fig. 3b) so that a caret wing at design condition would be expected (8)

(a) to produce a stronger shock than would a convex- or flat-bottomed delta at the same angle of attack and Mach number, and
(b) to preserve the lifting pressure by preventing undersurface cross-flows.

More generally one might expect (8, 9) that significant anhedral on some or all of the vehicle heat shield would provide, again by "flow containment", but no longer with two-

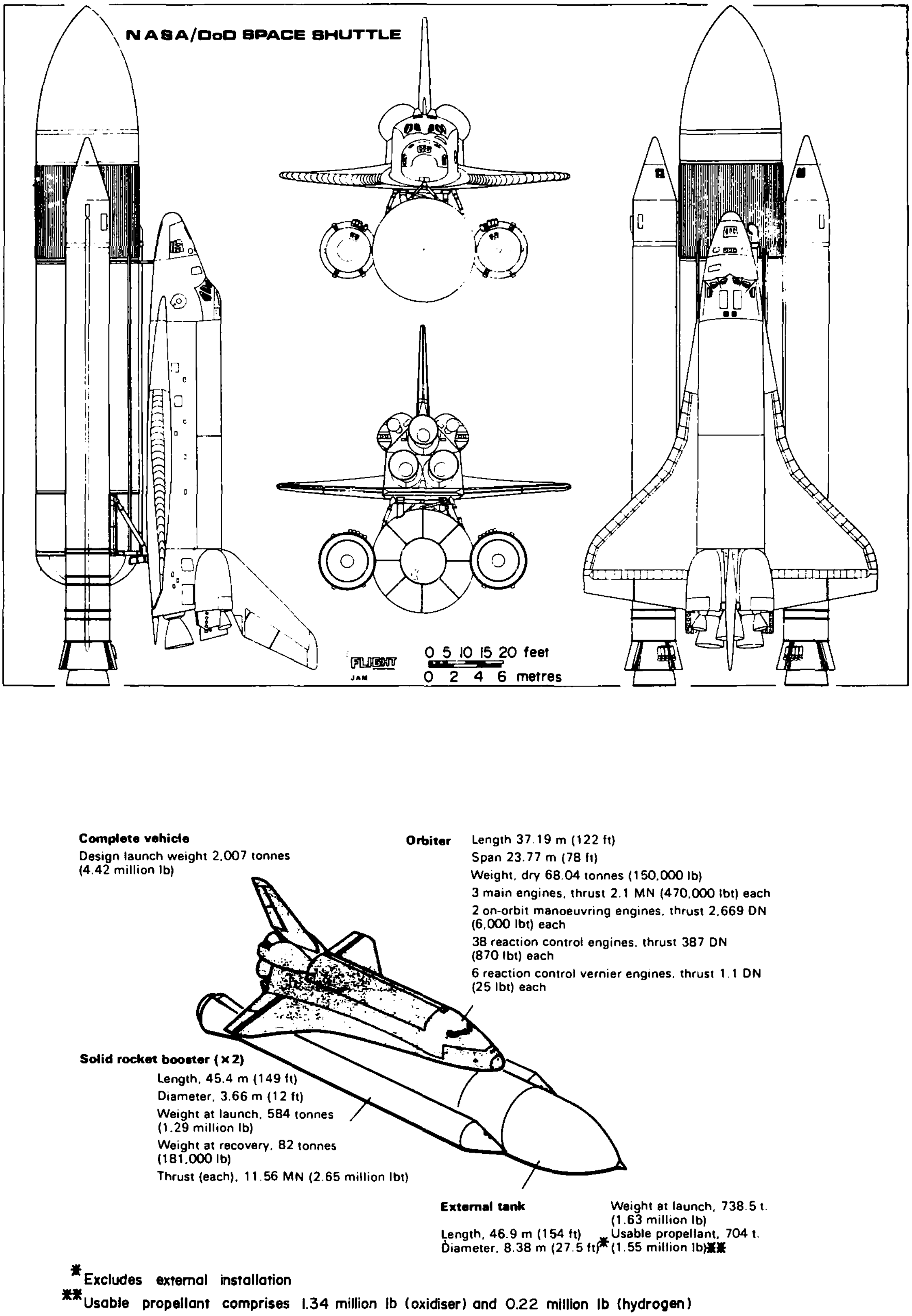


Fig. 1. The space shuttle — launch configuration and principal features.

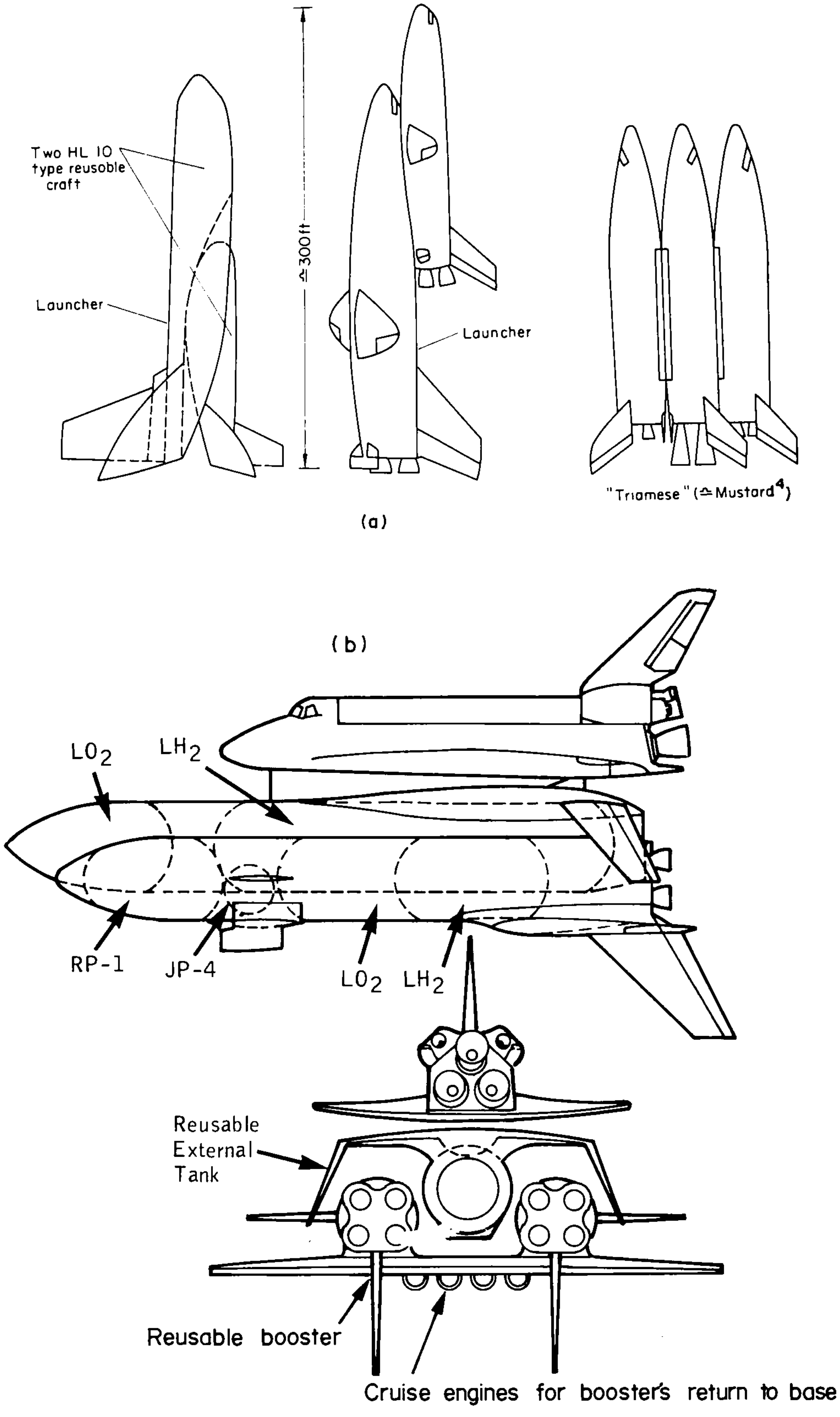


Fig. 2. Fully reusable space shuttle projects:
(a) early projects (e.g. Ref. (4)),
(b) Rockwell International (1976)(3).

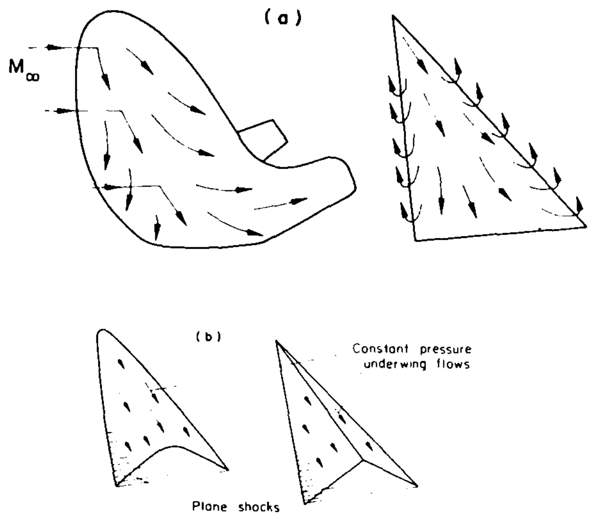


Fig. 3. Flows beneath lifting surfaces at high angle of attack and Mach number:
(a) flow spillage from convex- and flat-bottomed vehicles,
(b) flow containment by caret wings.

dimensional flow, an increased reentry C_L at conditions not far from design and possibly at angles of attack up to that for C_{Lmax}. Since 1970, these possibilities have been examined by analysis and experiment. The aerodynamic advantages of concavity have been shown to be significant and to remain so at all reentry angles of attack. In addition, it has been demonstrated that anhedral distributions need not be restricted to the simple inverted-V of the caret wing; in fact substantial convexity can be permitted in central regions and the practicality of designing realistic vehicles is greatly improved thereby.

To retain an attached shock at high angle of attack and hence to preserve zero spillage, may demand the use of "aerodynamically sharp" leading edges. The highly blunted edges conventionally assumed for certain types of reentry vehicle, would then have to be replaced by edges which are highly cooled, either by an active system (see, for example, Camarda (10)) or by the conductive system studied by Nonweiler (11, 12).

The most immediate advantage of increasing reentry C_L would be to reduce rates of heat transfer (a) at stagnation points (and on sides and upper surfaces) due to the increase in flight altitude and (b) under the entire heat shield due to reduced undersurface local velocities. Alternatively, an improved C_L per unit L/D would permit a reduction in wing area and, hence, in vehicle weight. The importance of reducing heat transfer rates and thereby the weight of the thermal protection system (TPS) can be gauged from the fact that the Shuttle orbiter TPS contributes more than 10 % to the orbiter's empty weight; this is some 25 % of the maximum launch payload and about 50 % of the maximum reentry payload. Significant savings in TPS weight would permit significant improvements in payload and in the economics of the Space Transportation System (15, 16).

The development of the research reported in this paper can be summarised as follows. Firstly, during the period 1969-1970 in which European collaboration in the Space Shuttle project was being considered, simple theoretical arguments and calculations were formulated (8), which suggested that a design conditions a carat wing would not only provide a simple flow, but should provide a higher lift coefficient than a flat delta of the same planform and at the same (high) values of angle of attack and Mach number. Secondly, in mid-

1970 specially conducted experiments confirmed (17, 18) that this was so, at least for the conditions tested (14, 19). First results obtained from thin shock-layer theory for caret wings at reentry conditions (20, 21) also began to show the extent to which they might offer improved aerodynamic performance, both at and away from design conditions. Thirdly, during the period 1970-1972, substantial theoretical and wind-tunnel data were obtained in the UK (9, 14) and in Belgium (22), which showed that several forms of concavity were aerodynamically superior to the inverted-V of the classic Nonweiler wing, that improved aerodynamic performance could be obtained with undersurfaces of more acceptable geometry and that some of these could actually include convex regions. Fourthly, the idea of reshaping a Space Shuttle orbiter to include large regions of undersurface concavity (23) was developed and (in 1972 and 1976) evaluated by testing, at reentry conditions (24, 25) and at subsonic speeds (26), two wind-tunnel models (one based on the MSC-NASA 040A orbiter and one a heavily modified version). Finally, consideration was given (14) to configurations for high performance lifting reentry gliders, designed ab initio to exploit the merits of concave or partly concave undersurfaces.

With regard to the design of realistic configurations, a longstanding and unresolved controversy surrounds the practicality or otherwise of applying the approach originally used by Nonweiler for the design of simple wings, to the design of complete vehicles. This controversy developed while both American and European research was aiming at the design of cruise vehicles (10, 27-29) rather than reentry gliders, and arguments, which may be valid for aircraft intended to take off conventionally and fly at low angles of attack and Mach numbers up to 12 (say), do not necessarily read across to vehicles which need no airbreathing propulsion, but must fly at widely varying angles of attack and at Mach numbers from 25 or so down to about 0.25.

Although Nonweiler first proposed the caret wing in the context of lifting reentry (6), it is only since 1970 that research has turned to such an application; as knowledge has advanced, so the need for flow simplicity has diminished and in the late 1970's concavity is not so much a consequence of striving for simple flows as a deliberate refinement of aerodynamic design for specific and very desirable engineering ends. In addition, largely since arguments over cruise vehicle design and the caret wing received their last major airing in 1968 (28), theoretical work has shown and experiment has confirmed that partial concavity may be at least as effective as simpler, but more extreme anhedral distributions.

The principal criticisms conventionally made of caret wings have been that the use of anhedral enforces longer undercarriages, increased wetted areas, and reduced volumetric efficiency. Whether or not these criticisms would prove valid for cruising aircraft (this is still an open question (28)) it is a fact that lifting reentry gliders bear only a superficial resemblance to supersonic or hypersonic cruise vehicles, one of the most important differences being that they can be substantially bulkier (8, 30). The main reasons for this difference are that, for lifting reentry gliders, supersonic and hypersonic lift/drag ratios need not exceed 2±1, and the demand for high volumetric efficiency and, hence, for minimum panel area militates (even more strongly than for cruising aircraft) against slender fuselages and large wing areas. A major result of the difference is that concavity on undersurfaces becomes easier to accommodate. Thus, in both context and content arguments regarding the practicality of waveriders must be reexamined together with those criticisms of concavity, which have become a part of conventional attitudes, but have sometimes been levelled at types of vehicle, which were not considered when the attitudes were formed.

2. AERODYNAMIC RESEARCH ON SIMPLE SHAPES

2.1. The significance of shock wave standoff angle

In Ref. (8) it was argued that for a lifting reentry vehicle at given Mach number and given angle of attack to provide as high a lift coefficient as possible, it must be shaped so as (a) to produce as strong a shock wave as possible and (b) to preserve the resulting high static pressure across the entire undersurface. It was then shown by physical arguments and simple calculations, that at or near design conditions the caret wing will provide a higher C_L at given L/D_p or a higher L/D_p at given C_L, than will a flat wing of the same planform and at the same Mach number. The analysis included the prediction that, at equal values of angle of attack, the caret wing would support a stronger shock than would a flat wing, i.e. that the shock wave standoff angle would be greater for the concave surface; from this prediction it was then deduced that heating should be less severe beneath the caret wing.

Thin shock-layer theory has since been used to elaborate the analysis, to include conditions far removed from the design point, to consider nonuniform anhedral distributions and to provide more general conclusions with regard to the relative performance of flat and concave delta wings. The advances in waverider design for lifting reentry owe much to the development of thin shock layer theory, principally in the UK, and to the rejection of Newtonian assumption which this has involved.

Thin shock layer theory deals with flows in which a large density change occurs across a shock wave and so, although restricted to inviscid conditions, it is well suited to the study of lifting reentry. In fact the theory successfully predicts standoff angles and distinguishes between various types of detached and attached flow (see Fig. 4). First applied to flows of relevance to hypersonic wings in 1962 (by Cole and Brainerd (31), by Kennet (32) and later by Antonov and Hayes (33), who effectively dealt with low aspect ratio wings at high angles of

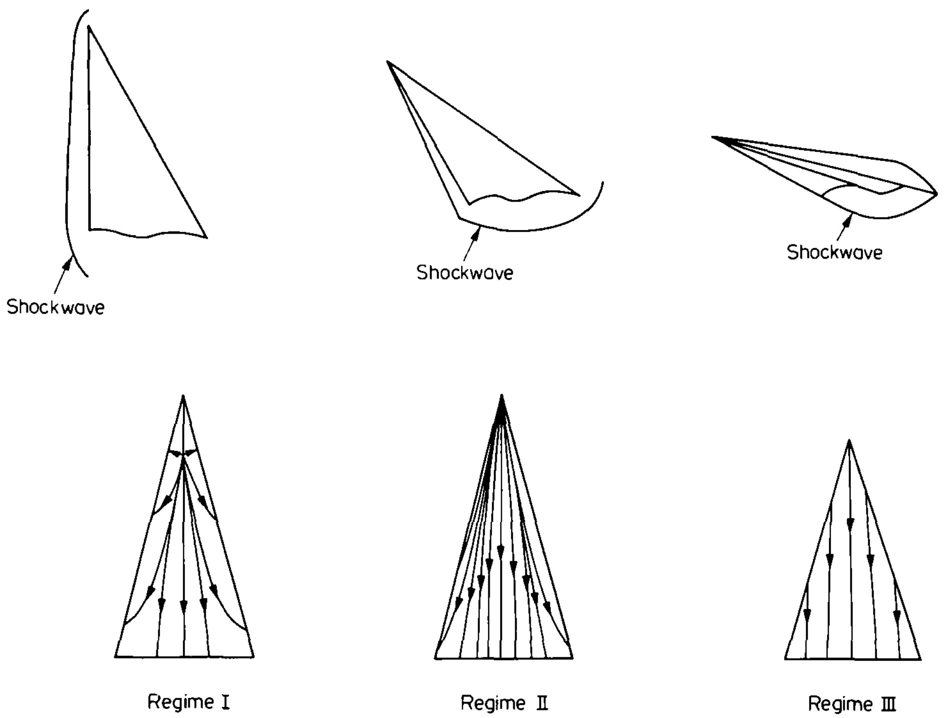


Fig. 4. Flow regimes under a conical wing in supersonic flow. (Data due to Roe (53) and Bashkin (54).)

attack), thin shock layer theory was recast in a particularly concise set of equations by Messiter (34) in 1963; solution was complex and was also restricted (34-36) to the detached shock condition (regime II in Fig. 4), but Squire (37,38) obtained numerical solutions for conical wings of flat, diamond and caret cross-section, and Hillier (39,40) included the effects of yaw. The theoretical results showed good agreement with experiment, and encouraged further work, especially since by this time (1970), Squire (41), Woods (42) and Roe (43,44) had successfully applied thin shock layer theory to the problem of flow beneath conical wings with shock waves attached to the leading edges* (regime III in Fig. 4).

Also in 1970, Roe (20) and Squire (9,21) started to investigate the claim (8), that at lifting reentry conditions concave undersurfaces should permit a delta wing to support a stronger shock at given angle of attack and so provide a higher C_L per unit L/D; in July 1970, the basic arguments in favour of flow containment (8) and the first results of thin shock layer theory for concave wings at reentry conditions (9,20,21) were presented at Euromech 19. Since then, Hillier (49) has extended thin shock layer theory to examine several nonconical shapes; this and further work on conical wings (50-53), was described in 1976 by Hillier *et al.* at Euromech 74. In particular, Roe has extended thin shock layer theory to include the treatment of heat transfer by deriving simple correlations to link the heating rates on the centre line of a conical body with the shock wave standoff angle. Roe's correlations (53) are obtained as now described.

In Fig. 4 three distinct flow regimes are shown, in one or other of which a sharp-edged conical body will operate at supersonic speeds (54). At hypersonic Mach number, thin shock-layer theory seems reasonably accurate in the second and third regime. For a body of prescribed conical shape, an important parameter is the curvature of the shockwave as it crosses a plane of symmetry, this single parameter being sufficient to determine the shockwave standoff angle and also the pressure, density and velocity anywhere in that plane of symmetry. If the initial shockwave curvature is given as $y_s''(0) = -a$, then the shock standoff distance and the post-shock pressure rise are given as

$$\left.\begin{aligned} y_s - y_b &= \frac{1-a+a(\ln a)}{(1-a)^2} \\ &\text{and} \\ p_s - p_b &= \frac{a^4-6a^3+3a^2+2a+6a^2(\ell n\ a)}{2(1-a)^4}\ , \end{aligned}\right\} \quad (1)$$

where the scaled quantities y and p are related to the physical quantities (i.e. actual distance $\bar{y}$ and actual static pressure $\bar{p}$) as follows:

$$\left.\begin{aligned} \bar{y} &= \bar{x}\ \varepsilon \tan\alpha .\ y, \\ \bar{p} &= \bar{p}_\infty + \bar{\rho}_\infty U_\infty^2 \sin^2\alpha(1+\varepsilon p)\ , \\ \varepsilon &= \frac{\gamma-1}{\gamma+1} + \frac{2}{(\sigma+1)M_\infty^2 \sin^2\alpha}\ , \end{aligned}\right\} \quad (2)$$

α = inclination of a reference plane lying close to the shocklayer.

*Since experimental data were scarce, Roe compared his results with exact solutions due to Babaev (45,46) and Voskresenskii (47); however, it is noted that the latter (and also South (48)) challenged the correctness of Babaev's results.

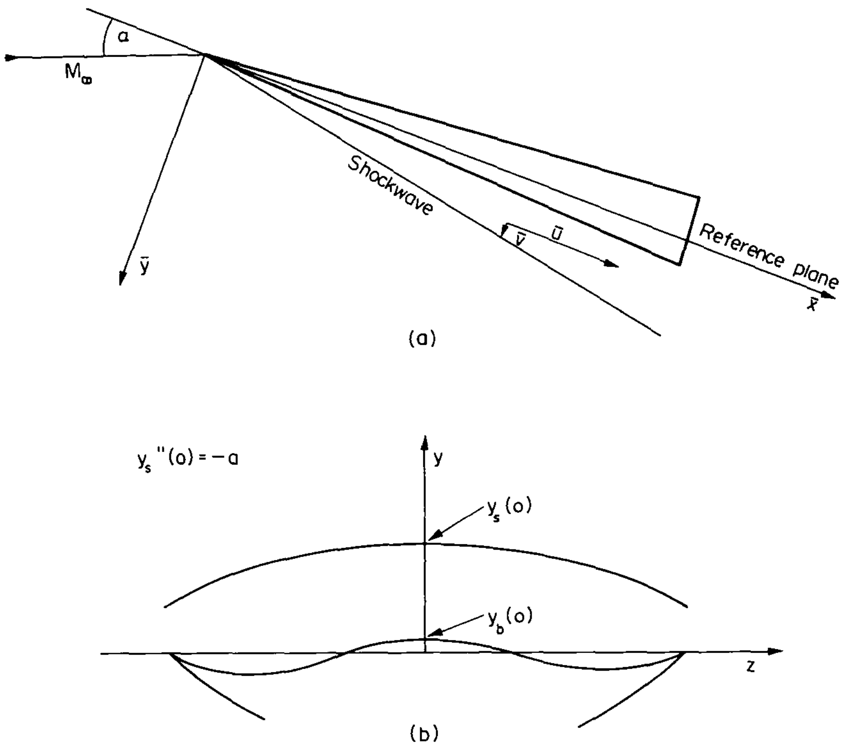


Fig. 5. Correlation between post-shock pressure rise and shock standoff distance. (Data due to Roe (53)):
(a) chordwise section, physical coordinates;
(b) spanwise section, scaled coordinates.

These equations have been shown (see Fig. 5c) to be especially accurate at those Mach numbers and shock strengths for which heating will be most severe, i.e. at reentry conditions.

To estimate the heating rates themselves, the parameter $\bar{p}^{1/8}\bar{u}^{1/2}$ is known (11) to indicate the rate of laminar heat transfer if cooling is by radiation only; in the above notation one then obtains

$$\bar{p}_b^{\,1/8}\bar{u}_b^{\,1/2} = \bar{p}_\infty^{\,1/8}U_\infty^{\,3/4}\sin^{1/4}\alpha\cos^{1/2}\alpha\left[1+\varepsilon\left(\left(\frac{3}{16}-\frac{\tan^2\alpha}{2}\right)y_s+\frac{y_b-1}{16}\right)\right] . \qquad (3)$$

This formula contains a Newtonian part plus a first order correction term. The Newtonian part gives the condition that $\bar{p}_b^{\,1/8}\bar{u}_b^{\,1/2}$ is proportional to the product $\sin^{1/4}\alpha\ \cos^{1/2}\alpha$ and Fig. 6a shows that as α increases this "temperature function" decreases quite strongly over most of th range for which the theory should be valid, namely $30^\circ < \alpha < 70^\circ$ or thereabouts. The correction terms produce effects, which are strongly related to angle of attack and, for the condition corresponding to C_{Lmax} (i.e. $\tan^2\alpha = 2$ or $\alpha \approx 55^\circ$ according to Newtonian theory), the correction factor becomes equal to $1 - \frac{\varepsilon}{16}(13y_s - y_b + 1)$; this expression depends much mor strongly on y_s than on y_b, so that for a given value of α the surface temperature will be reduced by any measure, which increases the shock wave standoff angle, for example by the introduction of undersurface concavity. In addition, if two vehicles are of the same wing loading*

*The term "loading" implies a force which should be measured in Newtons, but in the present paper (as in Ref. (8)) it is held that a much more useful parameter is the mass per unit wing area since this is independent of acceleration and flight attitude. The term "wing loading" is retained for its familiarity, but units are quoted in kg/m^2.

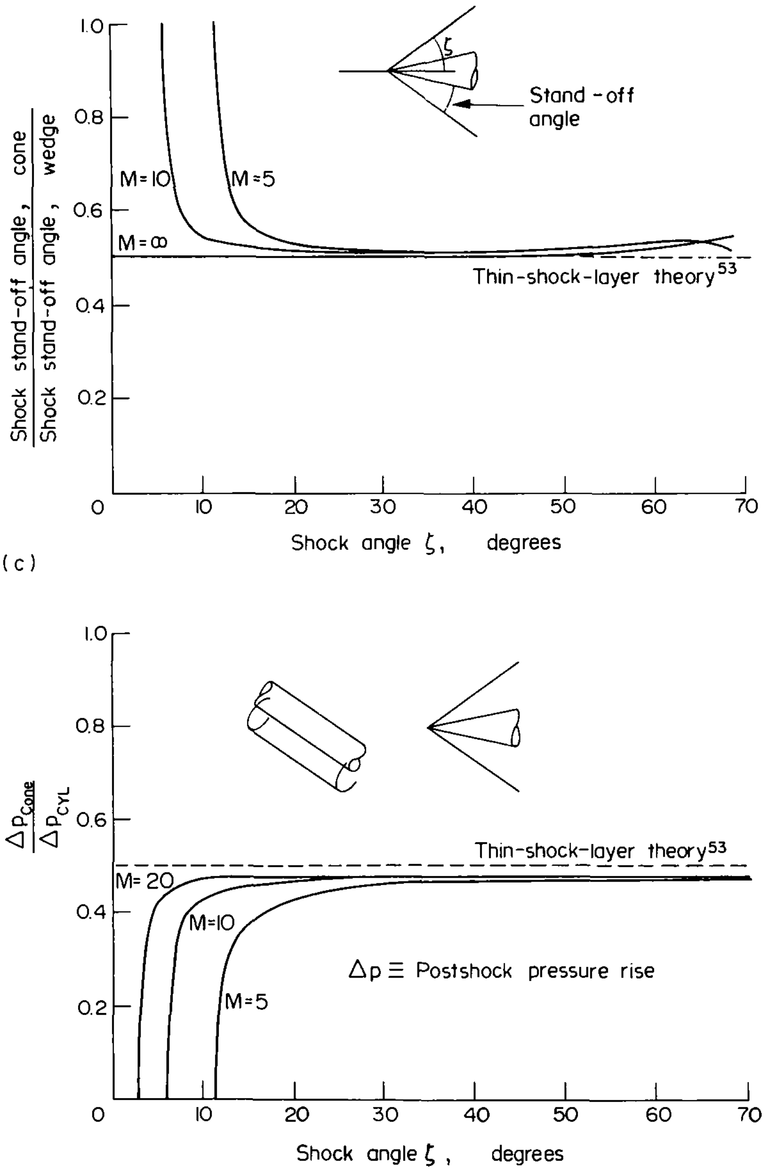


Fig. 5c. Confirmation of Roe's correlation by exact results.

but at given reentry L/D and cross-range performance they produce different C_L values (for example because one vehicle is concave), then in eq. (3) the value of $\bar{p}_\infty$ will be lower for the vehicle with the higher C_L.

One conclusion to be drawn from Fig. 6a is that, at least at Mach numbers where heating is a dominant consideration, reentry gliders should be flown at angles of attack far above 35^o; typically, those parts of the reentry for which heating is most severe ($M_\infty \approx 20 \pm 3$) may be flown at or beyond the angle of attack for C_{Lmax} (see also Fig. 6b) and the value of C_{Lmax} should itself be maximised by appropriate aerodynamic design provided that configurations so produced do not suffer unacceptable weight penalties due to special shaping. For the purposes of this paper, the significance of shockwave standoff angle is also that it is particularly simple to measure in wind-tunnel tests and is, therefore, convenient as a preliminary indicator of aerodynamic performance and of heating severity.

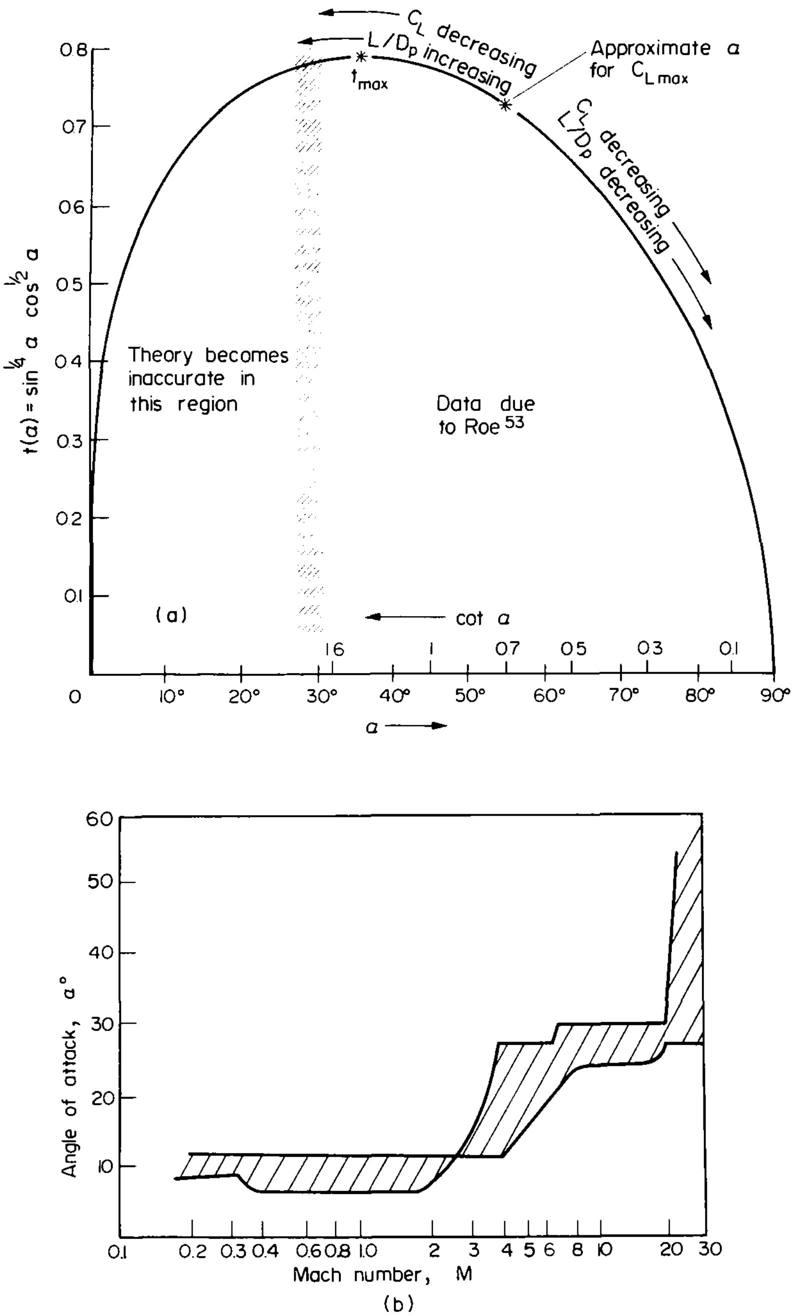


Fig. 6. Temperature function and orbiter trajectories:
(a) the function $t(\alpha) = \sin^{1/4}\alpha \cos^{1/2}\alpha$ (Roe(53));
(b) envelope of shuttle entry attitude histories. (Data due to Reding and Ericsson (55).)

In Fig. 7 a summary of data on shock standoff angle (14) confirms that flows beneath caret and other concave wings should provide higher static pressure ratios than do flat delta wings, so that higher lift and drag coefficients should occur. It suggests further that those with "root wedges" (and so of trapezoidal planform) promise especially high coefficients, because at off-design conditions they more closely approach pure wedge flows. These data on shock standoff angle also suggest that the heat transfer associated with concave undersurfaces should be lower than with such wings as the flat delta.

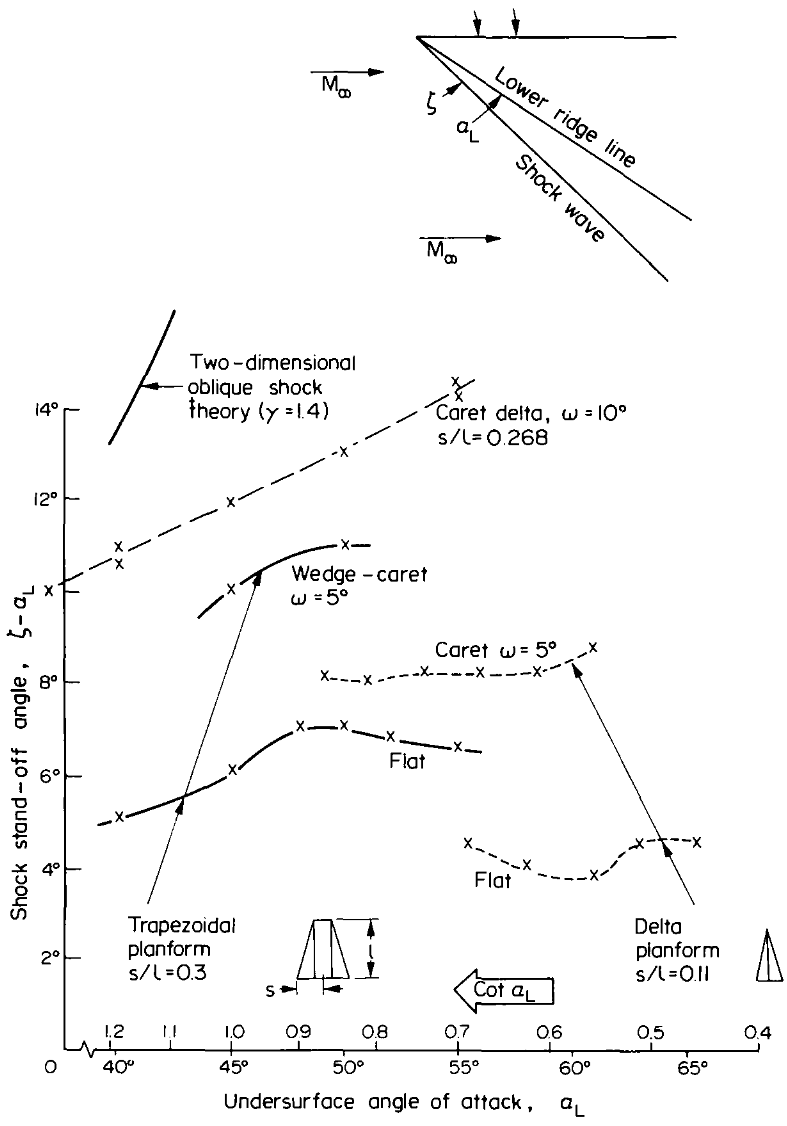


Fig. 7. Shockwave standoff angles in air and in nitrogen, M_∞ = 12.2 (Davies *et al.* (14)). For caret wings ω is the design value of ($\zeta-\alpha_L$), where α_L is the undersurface angle of attack.

2.2. Wind tunnel tests on lift and drag

Some of the earliest tests on the effects of concavity at lifting reentry conditions were performed by J. E. G. Townsend in the Helium Tunnel at NPL Teddington (1970) (14); caret wings were compared with a flat delta of the same planform, at a model Reynolds number of 625,000 and nominal Mach number 20. The results in Fig. 8 demonstrate that the caret wing with $\omega = 9^\circ$ was superior in C_L at the angles of attack tested (25-40°); the data for $\omega = 5^\circ$ were scattered, but all except two points supported the physical reasoning (and later computations by thin shock layer theory) that $\partial C_L/\partial\omega$ would be positive. Agreement between theory and these experimental data was poor, but their main interest lay in the correctly predicted trend of C_L with ω and all further tests were performed in nitrogen or air.

The earliest tests to measure forces on concave wings at high angle of attack and high Mach number in a flow of γ = 1.4 were those by Rao (19) and Carr (56) (1970 and 1971) in the

Imperial College No. 1 Gun Tunnel. For $\omega = 9.8^{o}$ an airflow of Mach number 12.2 and a model Reynolds number 0.75×10^6, a caret wing at design condition ($\alpha_L = 34^{o}$) provided over 10 % more lift than a flat delta and more than 20 % at an angle of attack of 40^{o}.

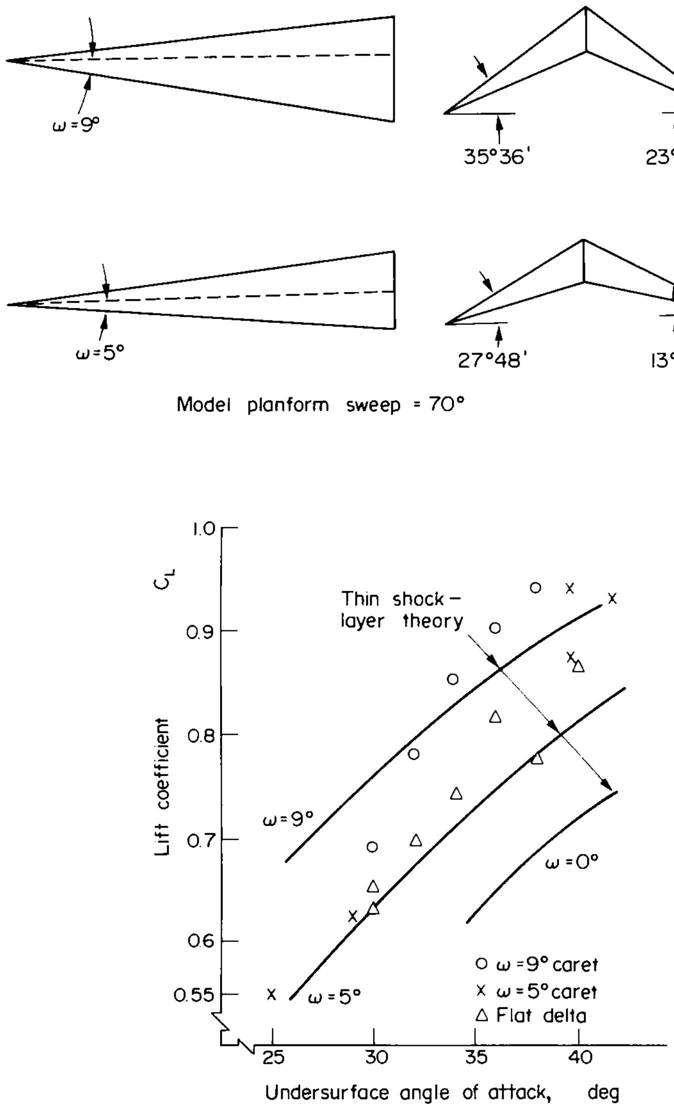


Fig. 8. Force measurements on delta wings in helium at $M_\infty \doteq 20$ (nominal) and $R_e \doteq 6 \times 10^5$. (Data due to J. E. G. Townsend (14,18).)

At angles of attack exceeding 45^{o} both tests suffered from tunnel blockage effects and measurements became unreliable. For this reason, the same models and balance were installed in the larger Imperial College No. 2 Gun Tunnel and tested by Coloman (57,58) in a nitrogen flow at Mach number 9, model Reynolds number 1.5×10^6 and angles of attack from 30 to 60^{o}. Even though the higher angle of attack limit was greater than in previous tests there was no blockage of the flow, and at C_{Lmax} (which occurred at about 50^{o} angle of attack for both shapes, at measured L/D ratios of 0.65) the caret wing produced a lift coefficient some 17 % higher than that of the flat delta. Superiority in C_L per unit L/D was retained over the angle of attack range tested, that is for $0.4 < L/D < 1.5$.

Further tests at angles of attack from 20 to 70^{o} have been reported by Jeffery (59) and

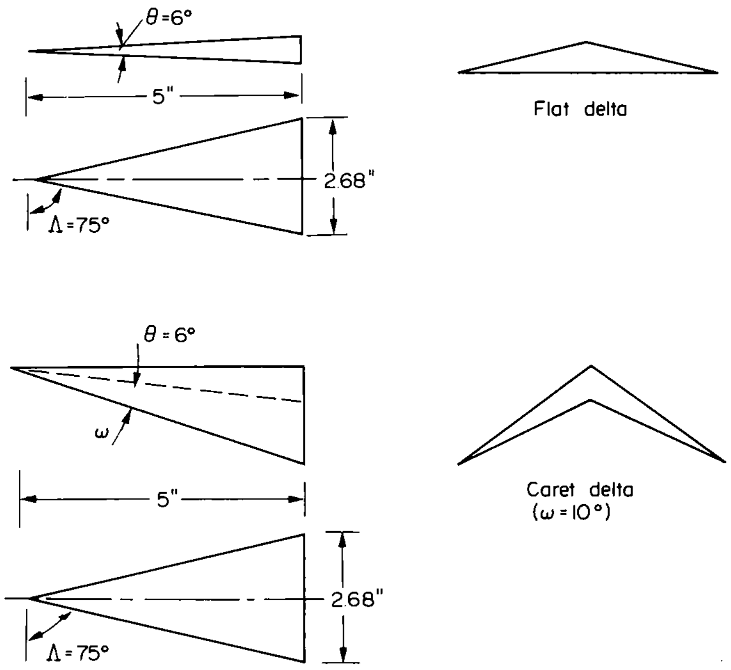


Fig. 9. Flat and caret delta models used to obtain data in Figs 10, 13, 38, 39 and 43.

by East and Lewis (60) at Mach numbers of 9 and 9.7, respectively, but for ω-values of about 5 and 8°, i.e. for caret models of more moderate concavity. In addition, Davies *et al.* (13) have tested a flat and a caret delta wing (ω = 0 and 5°), both having unusually low aspect ratio (s/ℓ = 0.11), at Mach number 12.2, model Reynolds number 10^6 and angles of attack between 50 and 65°. Shock standoff angle was roughly doubled by the use of concavity (see Fig. 7). Furthermore, in all these tests (13,59,60), the C_L values for the caret wings were superior to those for the equivalent flat delta throughout the test range, even though angles of attack were far removed from the design values. Also, it has been observed by Sykes (13) that increased pressures are retained by concave surfaces even when tested at Mach numbers far below the design value; these remarks however were made in the context of a Maikapar body rather than a Nonweiler wing.

Finally, Richards and Houwink (61) have evaluated flat and caret delta models (the latter having ω = 8.7°) by free flight tests in an airflow of Mach number 15 and at Reynolds number 2×10^6. The angles of attack varied from under 40° to just under 50 and 60° for the flat and caret delta, respectively. Once again, the caret wing provided a large margin of aerodynamic superiority, Richards and Houwink claiming (61) a C_L improvement of up to 23 %.

Figures 9-16 show most of the models used in the above tests and present the experimental results obtained in air or nitrogen in order of increasing Mach number. The major conclusion is that, in every case, C_L increases with concavity. Indeed, the extent to which C_L has been improved by simple anhedral distributions is encouragingly close to the maximum value of 30 % (for conical wings and γ = 1.4) anticipated (8) on the basis of thin shock layer theory (20). Modifications to planform, aspect ratio or anhedral distribution may, thus, be made as much for structural purposes (e.g. the reduction of wetted area per unit planform area) as for aerodynamic refinement, although the latter objective might still be achieved by exploring the effects of planform and anhedral modifications and the possible advantages of nonconicality (49)

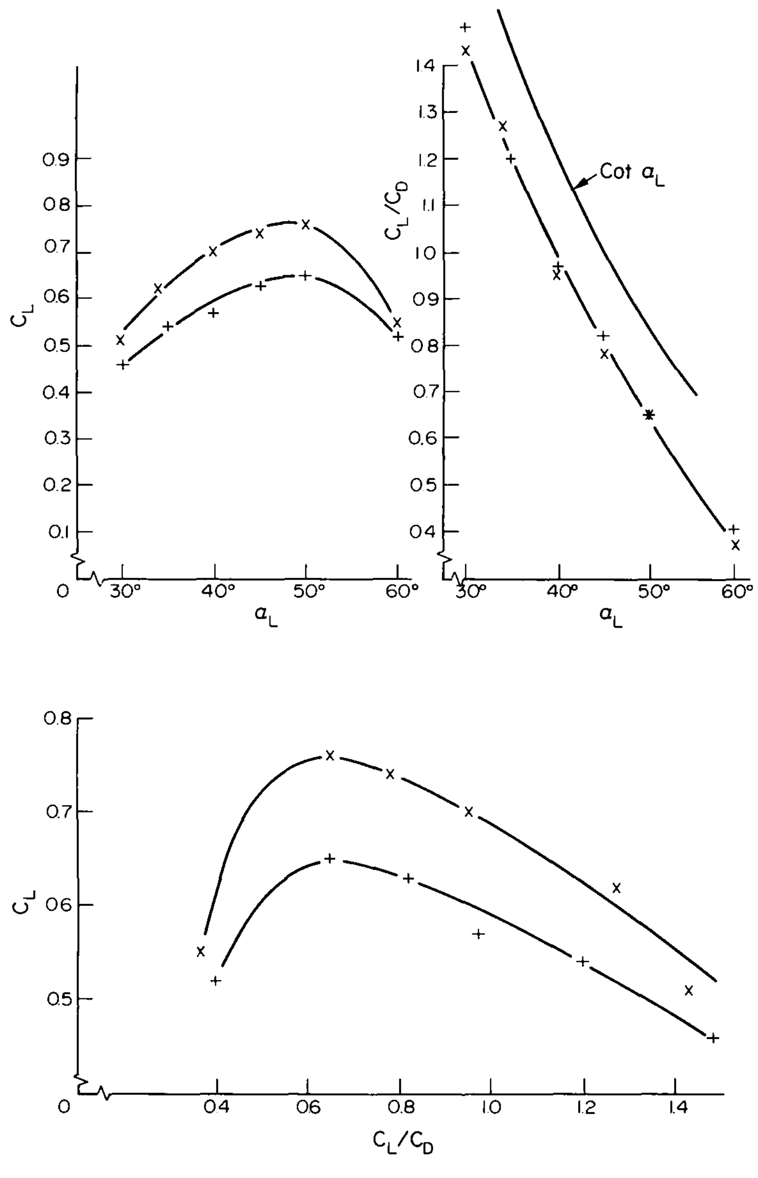


Fig. 10. Force measurements on delta wings at $M_\infty = 9$ and $R_e = 1.5 \times 10^6$ × caret delta ($\omega = 10°$) + flat delta. (Data due to Coleman (57,58).)

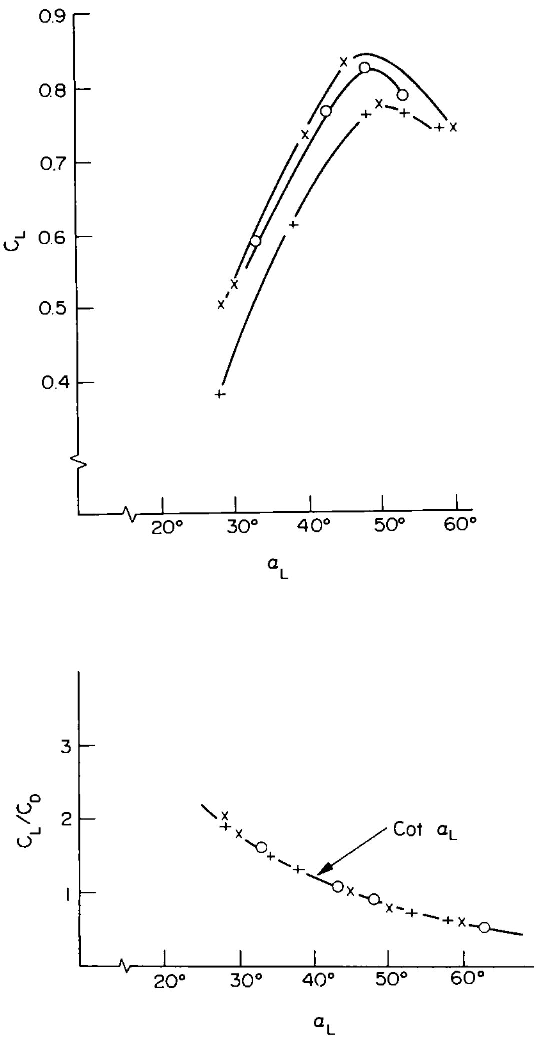


Fig. 11. Force measurements on delta wings at $M_\infty = 9$ and $R_e = 0.9 \times 10^6$. (Data due to Jeffery (59).) x caret delta ($\omega = 8^o$), O caret delta ($\omega = 5^o$), + flat delta.

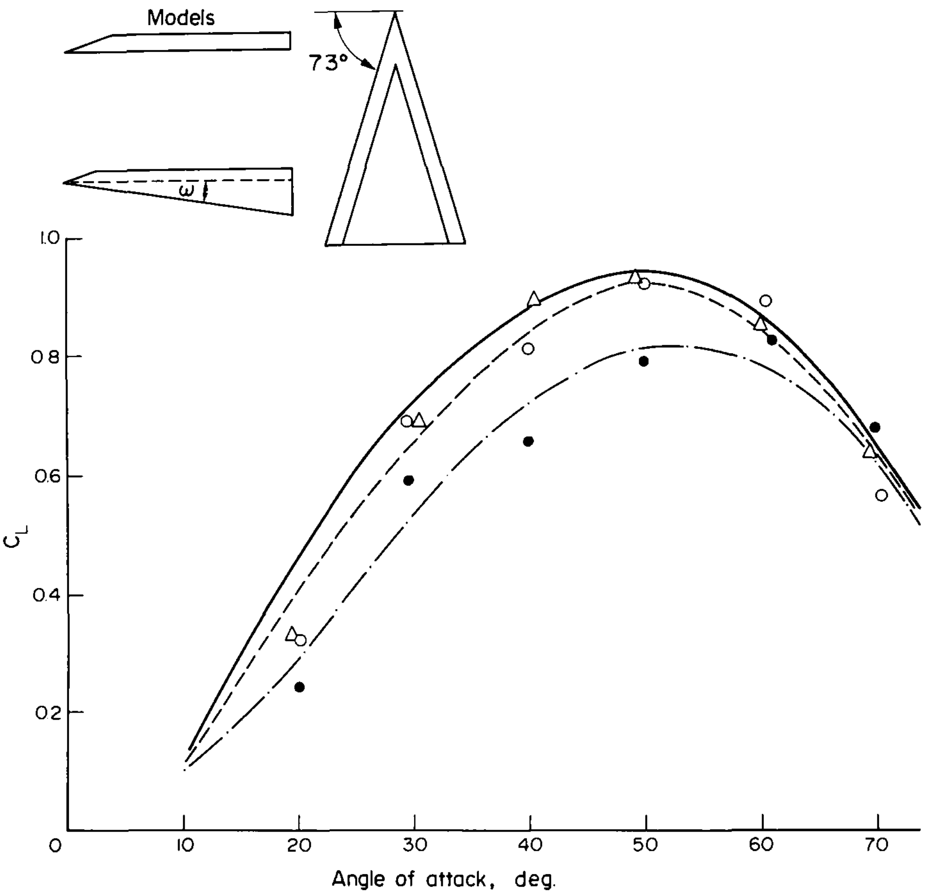
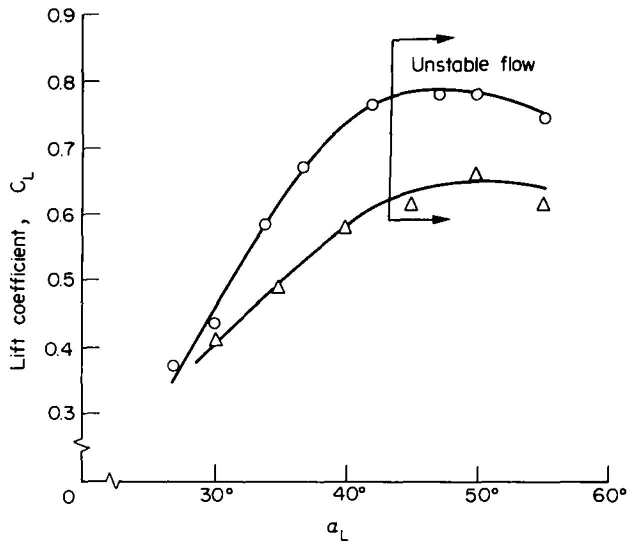


Fig. 12. Force measurements on delta wings at $M_\infty = 9.7$ and $R_e = 3 \times 10^5$. (Data due to East and Lewis (60).) Experiment and theory: o, —·– flat delta; o, - - - caret delta ($\omega = 4.6^o$); Δ, —— caret delta ($\omega = 7.96^o$).



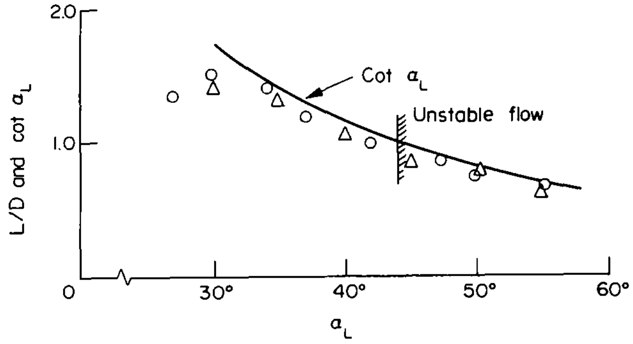


Fig. 13. Force measurements on delta wings at $M_\infty = 12.2$ and $R_e = 0.75 \times 10^6$. (Data due to Carr (56).) O caret delta ($\omega = 10^o$), Δ flat delta (see Fig. 9).

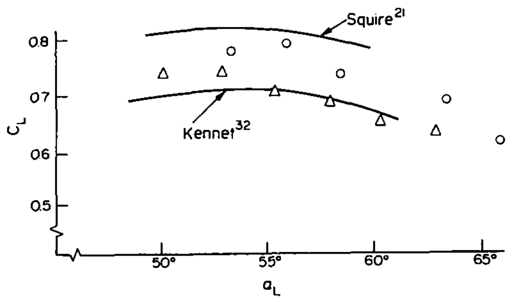


Fig. 14. Force measurements on delta wings at $M_\infty = 12.2$ and $R_e = 10^6$. (Data due to Davies *et al.* (13).) O caret delta ($\omega = 5^o$), Δ flat delta (N.B. $s/\ell = 0.11$, see Fig. 7).

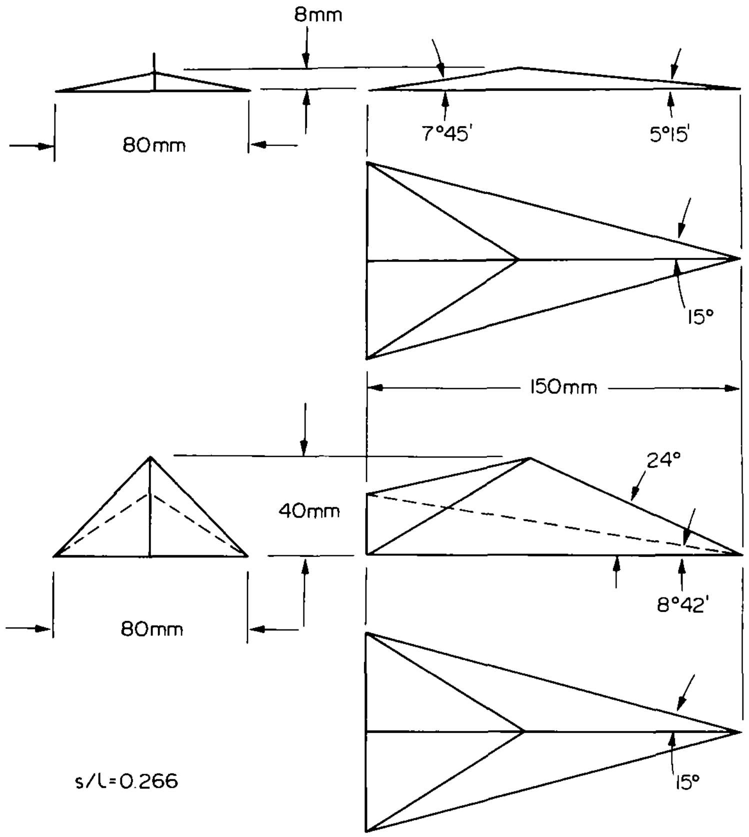


Fig. 15. Free-flight delta models tested in the VKI "Longshot" free piston tunnel (Fig. 16).

2.2.1. Some effects of planform redesign

The previous section concerned delta wings most of which were of similar aspect ratio (sweep angles of between 73 and 76^{o}) and so could demonstrate the effect of variations in ω. However, from the data on shock standoff angle (14) shown in Fig. 7, it is seen that at off-design conditions a concave wing of trapezoidal planform and s/ℓ of 0.3 can produce shock waves which are almost as strong as those from a caret delta having an ω-value twice as large and a planform only slightly more slender ($s/\ell \simeq 0.27$). These data increased the hope that "wedge-caret" wings would offer correspondingly large force coefficients, because at off-design conditions they would more closely approach pure wedge flows than would a delta wing of the same s/ℓ and ω. Since wedge-caret wings can also offer improved volumetric efficiency in related configurations, a range of such models was designed (24,62) to investigate their aerodynamic performance at high angle of attack and Mach number.

These trapezoidal "wedge-caret" wings were designed to support a plane shock at design conditions and so to produce the same design C_L as would a caret wing having the same ω value. The incorporation of the central wedge, however, was intended to preserve nearly two-dimensional flow (and so a stronger shock) over a wider range of off-design angle of attack. Preservation of two-dimensional flow up to the high angle of attack for shock detachment from a wedge would permit a wing to produce (8), at very high Mach number and $\gamma = 1.4$, a C_L of just over unity; at angles of attack beyond that for shock detachment (about 45^{o}) the C_L of a wedge with no lateral spillage might rise still further, since $C_{L\max}$ should not occur until $\alpha_L = 55^{o}$. The principal interest in the present tests was not merely to use concavity to improve upon the

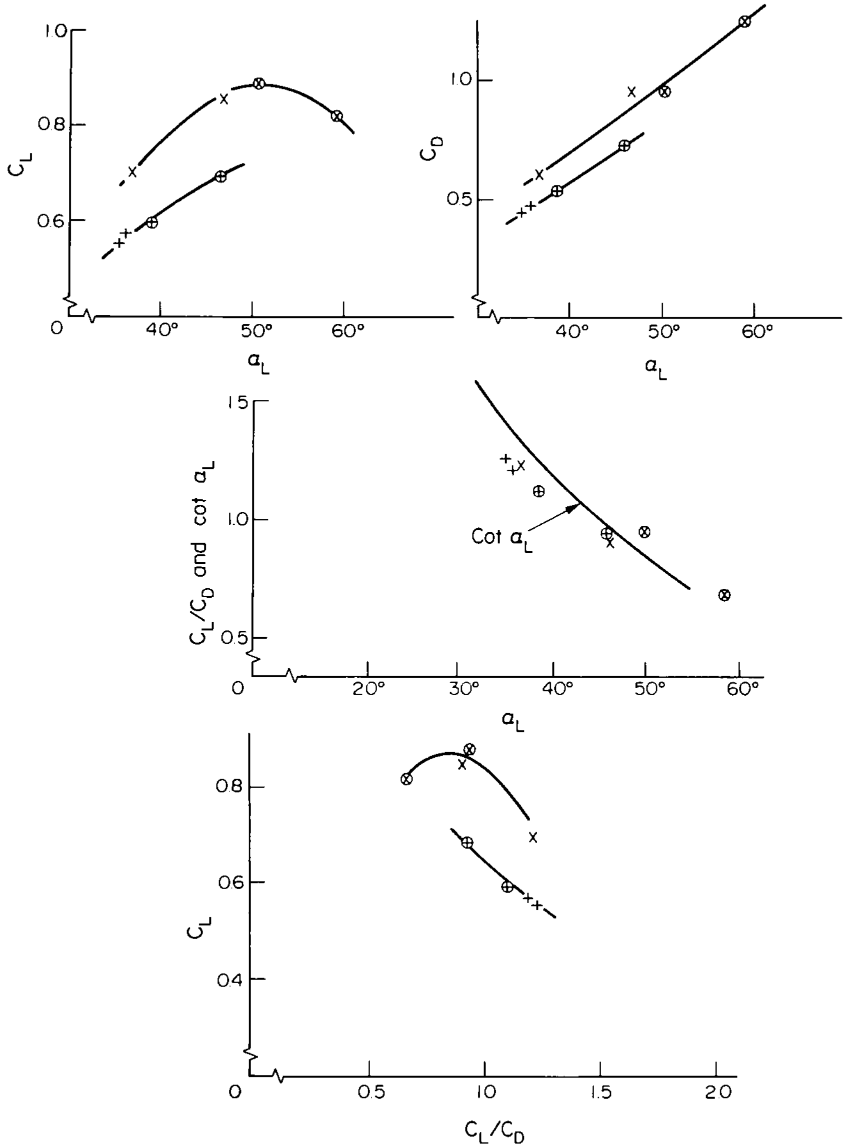


Fig. 16. Force measurements on delta wings at $M_\infty = 15$ and $R_e = 2 \times 10^6$. (Data due to Richards and Houwink (61).) x, o caret delta ($\omega = 8.7^o$), +, ⊕ flat-delta.

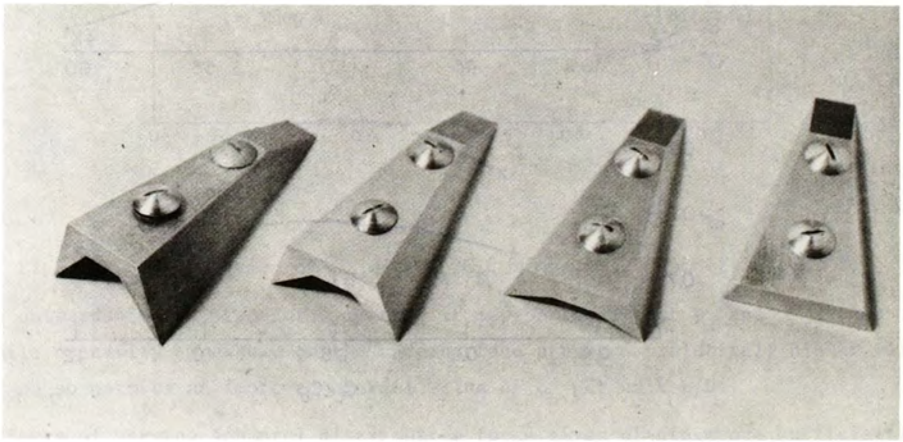


Fig. 17. Trapezoidal, flat and "wedge-caret" wings ($\omega = 0^o$, 4^o, 8^o and 12^o).

performance of a flat wing, but to find how nearly the trapezoidal wing could approach the theoretical "two-dimensional" performance and how large a value of ω would be required.

The models shown in Fig. 17 were tested (24,62) in the University of Southampton gun tunnel at a freestream Mach number of 8.4, a model Reynolds number of 5×10^5 and angles of attack from 40 to 60°. With the balance used, only lift forces could be measured, but for $\omega = 8°$ the wedge-caret wing produced some 20 % higher C_L than the flat trapezoidal wing, at the same angle of attack ($\alpha_L = 40°$). Further tests were therefore undertaken (25) using a balance which measured both lift and drag. From Fig. 18, it is seen that, at Mach number 9.7 and angles of attack from 40 to 55° the wedge-caret wing having $\omega = 8°$ produced about 20 % more lift coefficient than the flat wing; however, C_{Lmax} values were about 0.78 and 0.94, respectively, so that neither wing appeared to produce a specially high value, except by comparison with that which Newtonian aerodynamics would have predicted. These results therefore add new evidence that concavity improves lifting performance, but little on the best choice of planform or anhedral distribution. Since the wedge-caret wing produces a plane shock at design condition, the trapezoidal planform is intrinsically allied with anhedral distributions of the type shown in Fig. 17, that is a "partly filled in" inverted V. For plane-shock wave-riders it is therefore impossible to study the effects of planform and anhedral distribution separately. The anhedral distribution is apparently very important since it allows the central parts of the undersurface to be progressively filled in and, if this in turn forces the

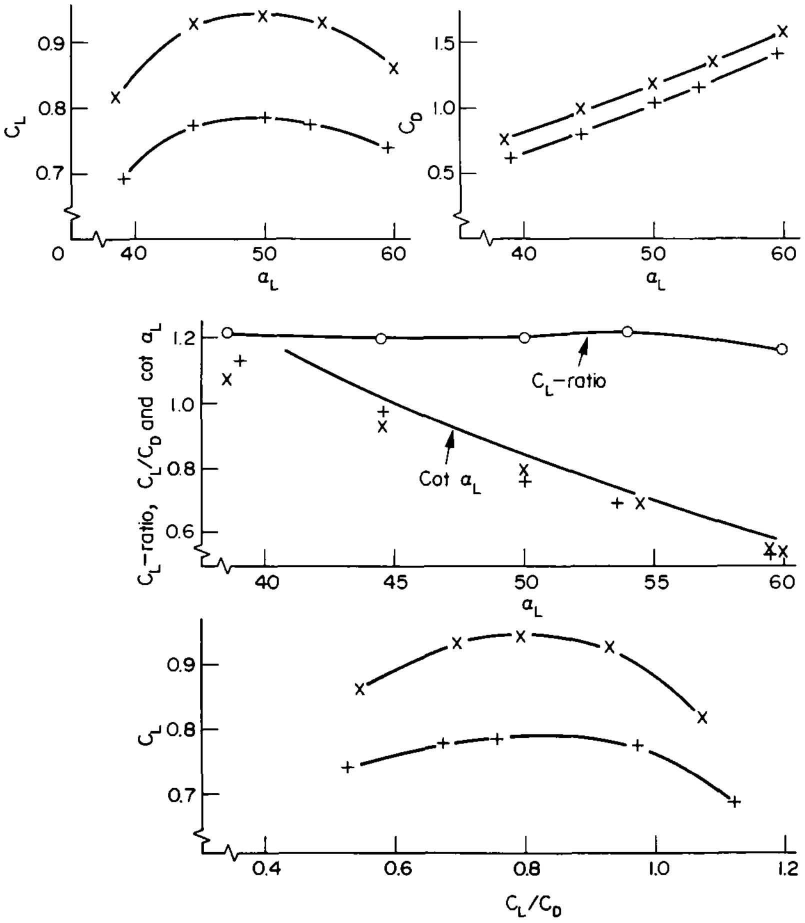
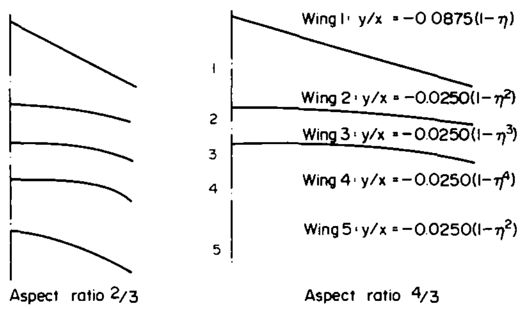
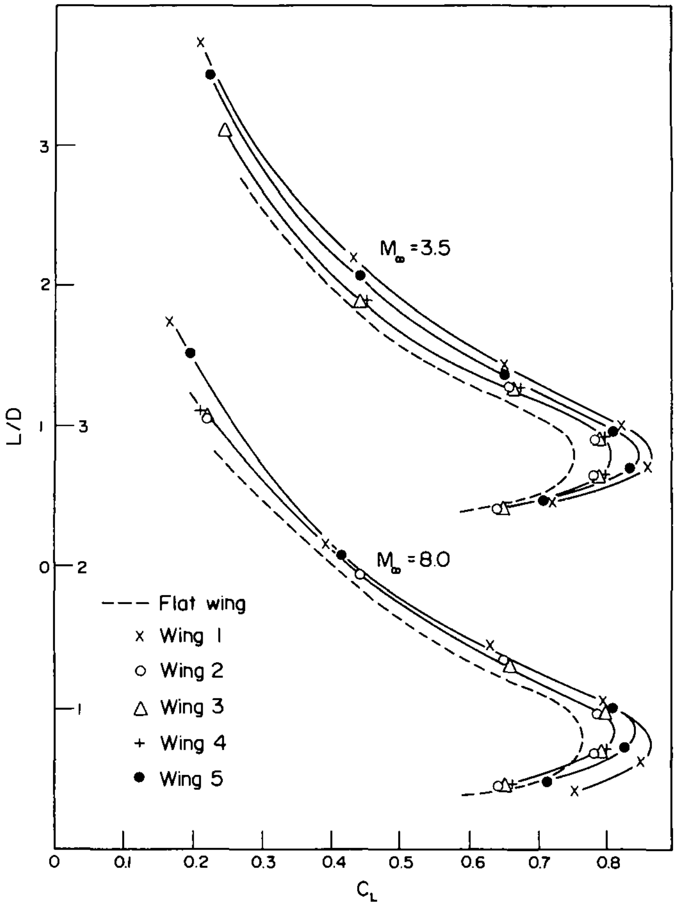


Fig. 18. Force measurements on trapezoidal wings at $M_\infty = 9.7$ and $R_e = 4 \times 10^5$. (Data due to East (25).) x wedge-caret wing ($\omega = 8°$), + flat trapezoidal wing.



(a)

Fig. 19. Delta wings with nonuniform anhedral distribution. (Data due to Squire (52).)
(a) Anhedral distributions of Squire's delta wings 1-5.



(b)

Fig. 19. (b) Variation of C_L with C_L/C_D, wings 1-5. Aspects ratio 2/3.

shock wave outwards in the plane of symmetry, it may be found that y_s and C_{Lmax} can be increased while concavity elsewhere on the undersurface preserves relatively higher values of local L/D and so permits an improved overall value of C_L per unit L/D.

The effects of varying anhedral distribution for a given planform are sufficiently complex to require elaborate theoretical methods and recourse to computer techniques. Such analyses have been performed by Roe, Squire and Hillier. The next section presents a short account of their results and experimental data due to Reggiori (63) and Houwink and Richards (22).

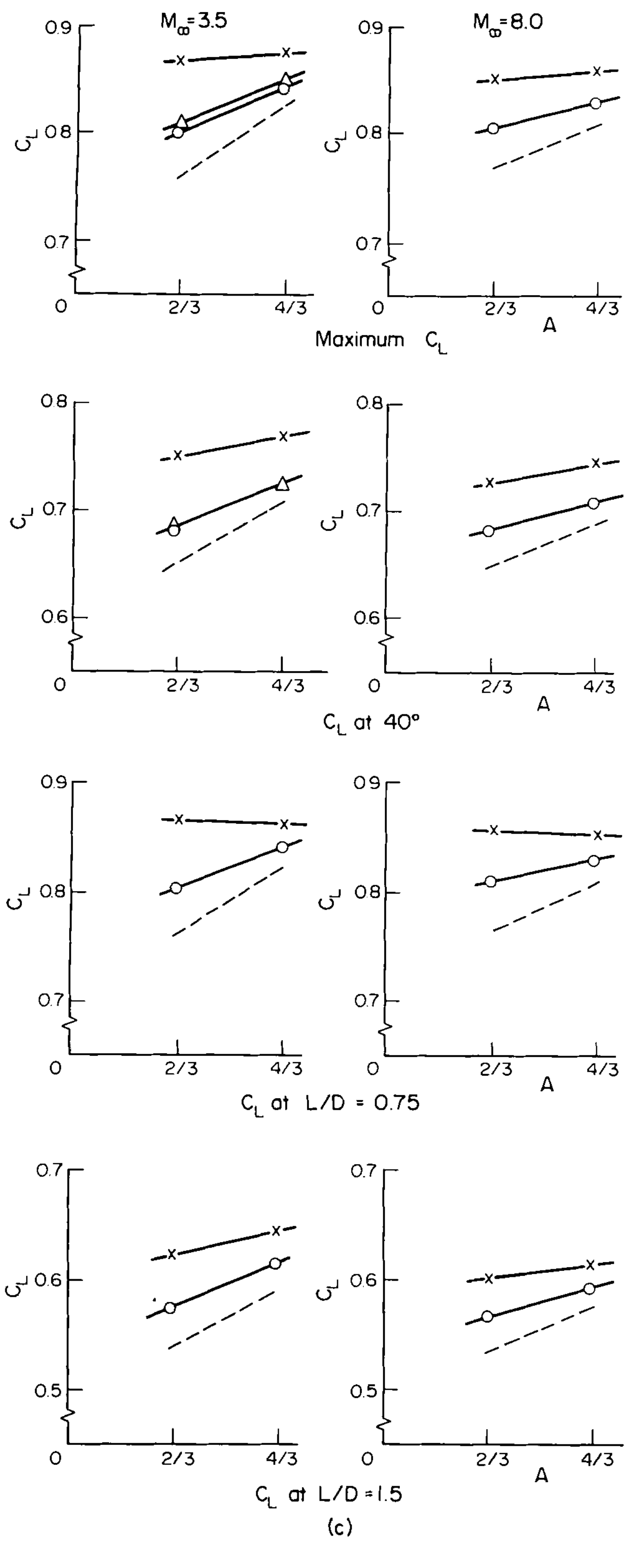


Fig. 19. (c) Effects of aspect ratio.

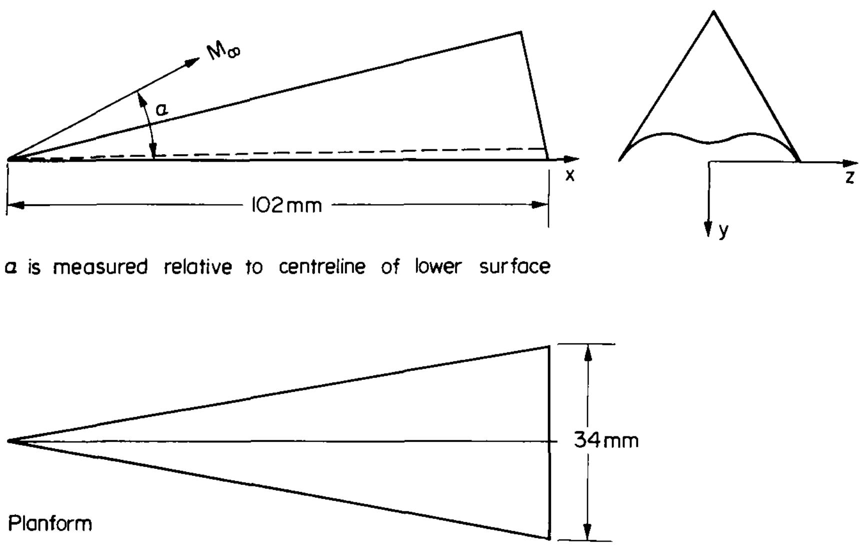


Fig. 20. Squire's "wavy wing" ($s/\ell \doteq 0.17$).

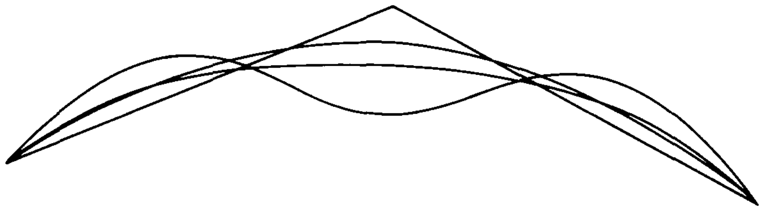


Fig. 21. Wings with similar performance at $M_\infty = 3.5$. $L/D \doteq 1.5$, $C_L = 0.62$. (Data due to Squire (52).)

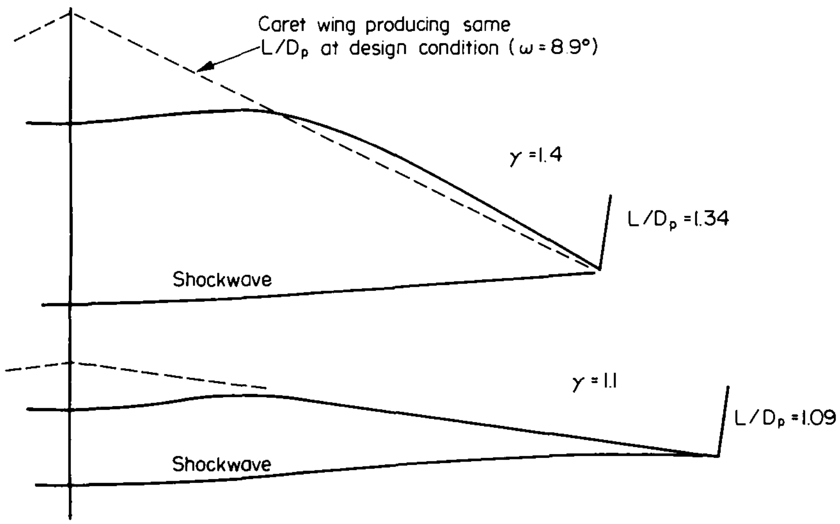


Fig. 22. Optimised wings for infinite M_∞ and $s/\ell = 1/3$. (Data due to Roe (14).)

2.2.2. The influence of anhedral distribution

For the design of lifting reentry vehicles with nonuniform anhedral the principal theoretical contributions are those of Squire (52) and Shanbhag (50) (especially for high α and high Mach number), Roe (14,43,44) (for moderate α at very high and moderate Mach numbers) and Hillier (39,40,49) who has considered the performance of wings with complex anhedral and wing-body combinations. Squire (52) studied the performance of delta wings as aspect ratio and anhedral distribution varied, for Mach numbers 3.5 and 8 and at angles of attack up to some 60^o. Wholly concave distributions ranged from "unrealistic" shapes (wings A to D (52)) to "realistic" shapes (wings 1-5 in Fig. 19a). Results for the "realistic" wings 1-5 are as shown in Fig. 19b and 19c; they confirm that aerodynamic gains are significant at both Mach numbers, and Fig. 19c shows that increasing the aspect ratio from 2/3 to 4/3 only partly reduces their advantage over the equivalent flat deltas.

Results were also obtained for the "wavy wing" of Fig. 20 in which very substantial "filling in" of central regions has greatly improved the configurational acceptability of the lifting surface. Data for the wavy wing were obtained for Mach number 3.5 only, but a typical

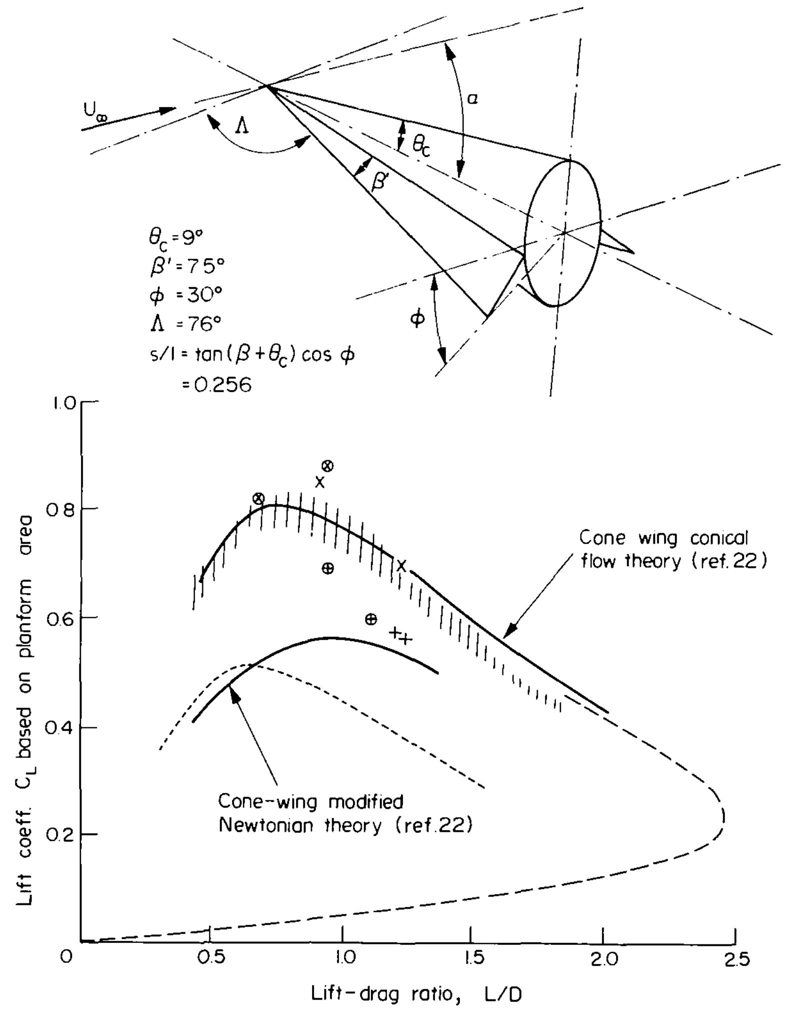


Fig. 23. Force measurements on delta wings and delta-cone wing-bodies at $M_\infty = 15$ and $R_e = 2 \times 10^6$. (Data due to Houwink and Richards (22).)
Experimental data: -------- cone, $\theta_c = 9^o$, $M_\infty = 15$ (22),
— — — cone-wing, $\theta_c = 10^o$, $\beta = 5^o$, $M_\infty = 5.8$ (63),
||||| cone-wing, $\theta_c = 9^\circ$, $\phi = 30^o$, $\beta = 7.5^o$ (22),
⊗ × caret delta, $\omega = 8.7^o$ } see Figs 15 and 16 (61).
⊕ + flat delta, $\omega = 0$

result was quoted by Squire as $C_L = 0.62$ and $L/D = 1.5$ which represents a gain in C_L of 0.08 when compared with a flat delta of aspect ratio 2/3. A comparison of shapes producing identical aerodynamic performances is presented in Fig. 21. The comparison shows the geometrical advantage offered by the wavy wing over the caret wing for Mach number 3.5, but prompts the question as to whether such advantages are retained at Mach numbers at which lifting reentry would occur.

Calculations by Roe (14) suggest that at infinite Mach number a form of wavy wing can offer the same L/D_p as a caret wing while presenting a substantial reduction in concavity (see Fig. 22), but a shock wave of increased y_s and hence a higher C_L. For $\gamma = 1.4$, the wavy wing and the caret wing produce C_L values of 0.72 and 0.71, while a flat delta would produce a smaller value (0.58) than either. The advantages remain at $\gamma = 1.1$ although the required geometries are significantly changed. Practical values of γ are unlikely to lie below 1.25.

In Roe's calculations, the form of the shock was specified as a Chebychev series, whose coefficients were determined so as to maximise lift coefficient at a given mean angle of attack (i.e. at given L/D). With the resulting wings, high pressures are generated on the most steeply inclined regions (i.e. the central regions) and are allowed to spread onto less inclined areas where they do not cause excessive drag. In the context of flow containment, these shapes apparently conform to desirable design principles, in that (a) high centreline pressures are associated with "pushing out" the shock wave, while (b) spillage is prevented by drooping the leading edges. They also permit the most deeply recessed part of the equivalent caret wing to be filled and yet produce slightly more C_L per unit L/D.

These theoretical results are far from complete and general conclusions cannot be drawn. However it is encouraging to find experimental evidence from Houwink and Richards (22), that partly concave undersurfaces provide substantially better aerodynamic performance at high

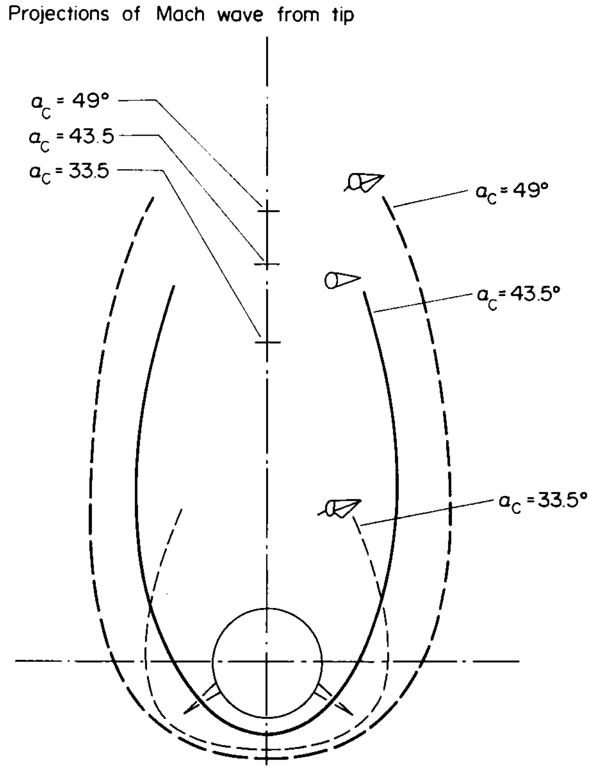


Fig. 24. Shock wave envelopes for a 9° semi-apex angled cone with and without wings. (Data due to Houwink and Richards (22).)

Mach number and high angles of attack than is obtainable with flat undersurfaces of the same planform. For Mach number 15 and $R_e = 2 \times 10^6$, it is seen (Fig. 23) that a cone of 9^o semi-angle, fitted with delta wings of pronounced anhedral (but with wing tips no lower than the belly of the cone) produces a C_{Lmax} of about 0.8, considerably more than that of a flat delta of similar planform. This relatively high performance is obtained even though, as seen in Fig. 24, the undersurface flow is only partly contained. However, oil flow photographs have confirmed (22) that flow spillage is not high for $\alpha < 40^o$ and, indeed, in Fig. 25, it is seen that undersurface static pressures hardly vary for $0 < \theta < 60^o$. These tests are particularly valuable since they were performed in the VKI Longshot Tunnel and conditions were, thus, fully representative of an orbiter trajectory (see Fig. 26).

It seems very likely that the design of partly concave undersurfaces as pursued by Roe, Squire, Richards and Houwink represents the most promising approach to the achievement of realistic undersurface shapes for lifting reentry vehicles intended to exploit the aerodynamic

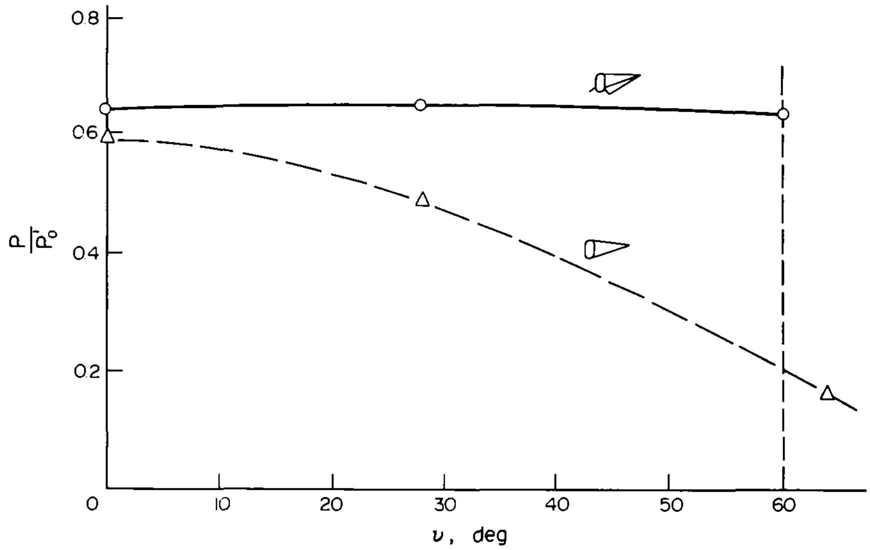


Fig. 25. The pressure distribution on a 9^o semi-apex angle cone with and without wings. (Data due to Houwink and Richards (22).)

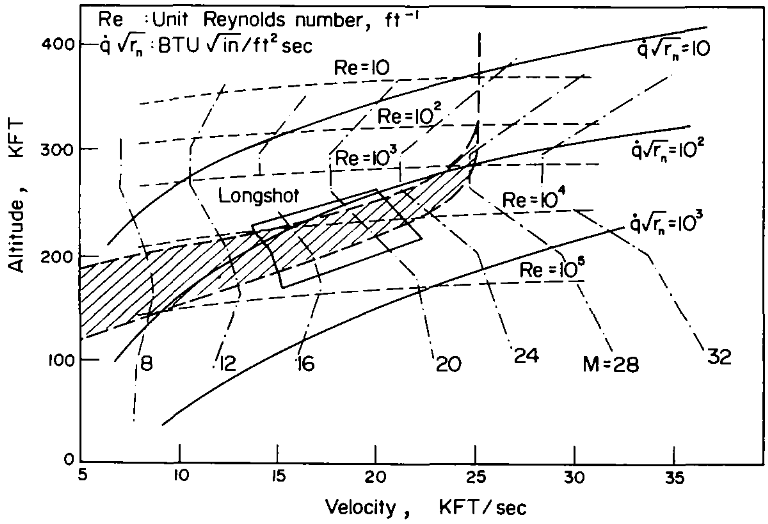
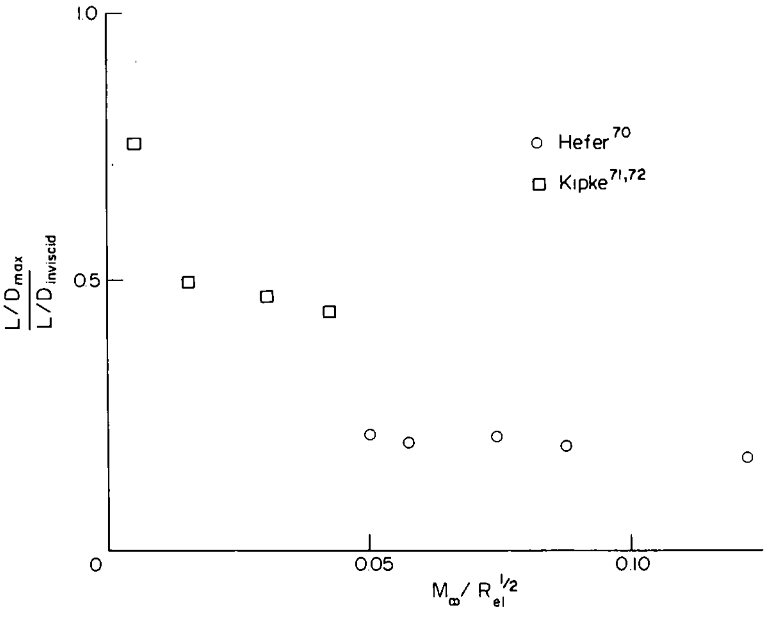
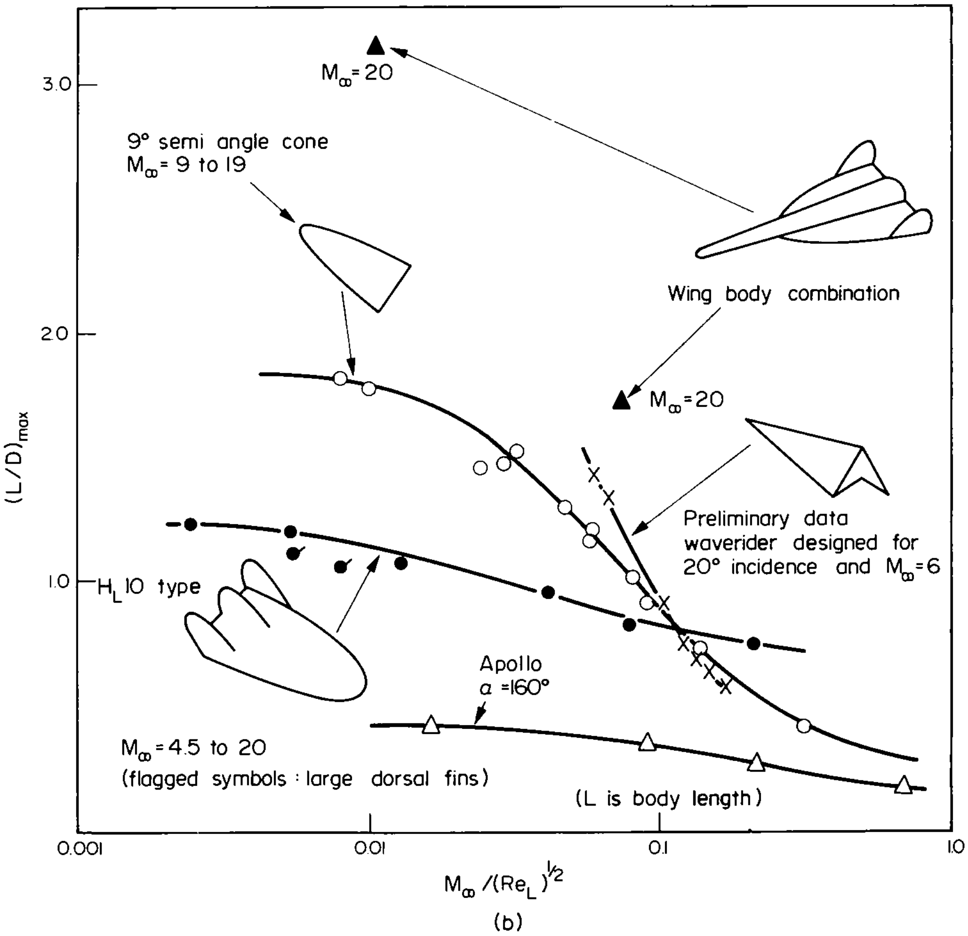


Fig. 26. "Longshot" tunnel simulation performance for 100 ft vehicle superimposed on orbiter trajectory.



(a)

Fig. 27. Potential effects of low Reynolds numbers on wings and bodies. (a) Viscous effects on hypersonic $(L/D)_{max}$ for caret wings. (Data due to Hefer (70), Kipke (71,72) and Küchemann (5).)



(b)

Fig. 27. (b) Viscous effects on hypersonic $(L/D)_{max}$ for 5 classes of vehicle. (Data due to Stollery (73).)

advantages of flow containment. It will be shown later that such undersurfaces can be readily installed both on modified versions of the Space Shuttle orbiter (23) and on vehicles offering higher cross-range. To date however there is a dearth of data, either theoretical or experimental, on the detailed effects of concavity on the distribution and severity of heat transfer at reentry conditions, the reduction of which is the principal objective of studying concave undersurfaces.

2.2.3. Results at low Reynolds number

At high altitudes and high Mach numbers, the flows surrounding reentry vehicles will at some conditions be influenced by viscous interaction, not merely near the leading edges of lifting surfaces, but beneath much of the vehicle and also in regions such as the sides and upper surfaces. Viscous interaction and flows with thick boundary layers may well be important during the launch phase, since launch configurations will frequently be complex and vehicle angles of attack will normally be low (64); fuel consumption can be considerably increased (1) by the associated viscous effects, and the use of aerodynamic lift forces to improve launch performance (65–68) may also be complicated by viscous considerations. During reentry, however, the forces due to low Reynolds number effects on a lifting reentry vehicle

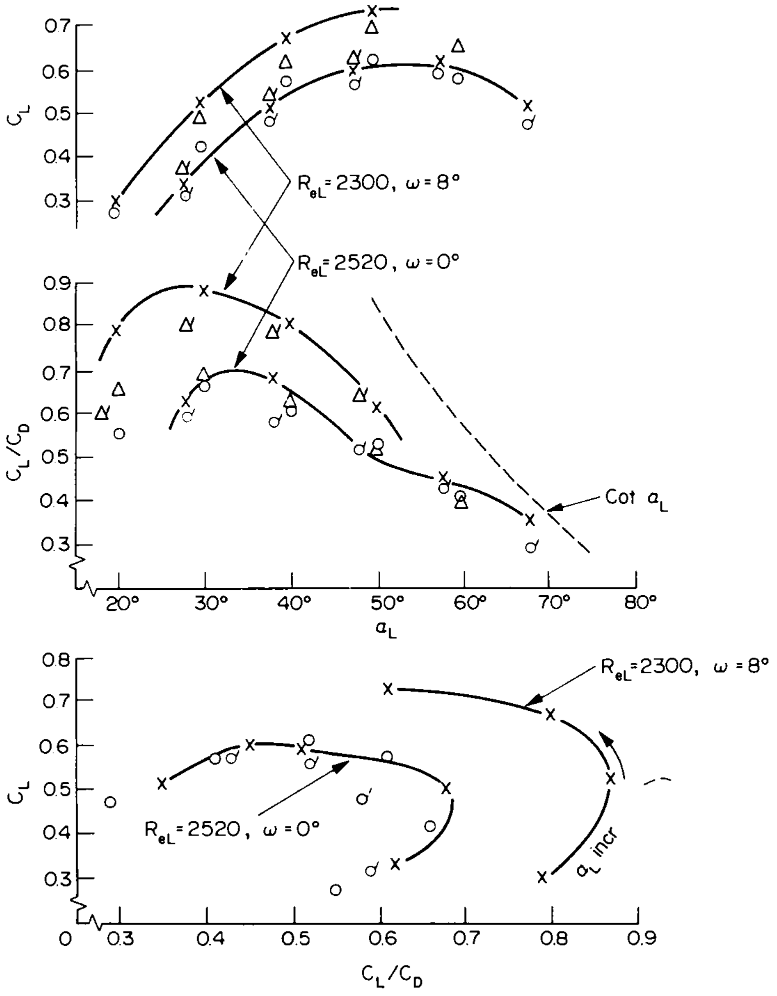


Fig. 28. Delta wings (s/ℓ = 0.268, $\omega = 0^{\circ}$ and 8°) at M_{∞} = 6 and low Reynolds numbers in air. (Data due to Metcalf and Berry (74).) o caret R_{eL} = 820, ơ flat R_{eL} = 800, Δ caret R_{eL} = 1400, Δ̸ flat R_{eL} = 1370.

need not necessarily be significant to reentry range and cross-range; this is because, for vehicles of realistic wing loading (50 lb/ft^2 (2.1kg/m^2) or more, say), the combinations of Mach number, Reynolds number and vehicle attitude occurring in a typical reentry trajectory seem likely to reduce the relevance of viscous effects except during the earliest stages of reentry, and at these conditions aerodynamic forces are small in comparison with centrifugal force and vehicle mass. Only when the designer is concerned to produce the maximum possible cross-range (and so will be dealing with fairly slender vehicles, which are to return high values of L/D throughout reentry and may be significantly smaller than the Shuttle) is it likely that low Reynolds number aerodynamics will significantly affect trajectory and vehicle design; this suggests that low Reynolds number aerodynamics will become of greater significance as performance specifications become more demanding.

To be specific, for $R_{e\ell} < 10^3$ (which implies altitudes exceeding 300,000 ft (91440 m) even for a small body of only 10 ft (3.1 m) length), forces are small and unimportant to trajectory performance (69). Low density effects become appreciable for slender bodies if $10^3 < R_{e_\ell} < 10^6$, and Hidalgo and Vaglio-Laurin (69) have shown that, for a sharp cone of 9^o semi-angle or for a body of more streamlined form, L/D is significantly affected only if $M_\infty R_{e_\ell}{}^{1/2}$ exceeds about 0.005. For a 10 ft (3.1 m) cone of low "wing" loading (30 lb/ft^2 (1.26kg/m^2) based on planform, say) in equilibrium glide at $M_\infty = 15$ and C_{Lmax} (about 0.5), altitude would not be higher than 200,000 ft (60960 m), but viscous effects on sides and upper surfaces would be important since $M_\infty/R_{e_\ell}{}^{1/2}$ would be about an order more than 0.005. However, for a vehicle of more practical size (e.g. man-and-cargo-carrying size), viscous effects would be of small importance.

An indication of the potential effects of low Reynolds number is shown in Fig. 27a (5). Results due to Hefer (70) and Kipke (71,72) are shown as the ratio of measured L/D_{max} to estimated $L/D_{inviscid}$ plotted against $M_\infty/R_{e_\ell}{}^{1/2}$. For values of the latter exceeding 0.005, it is confirmed that losses in L/D_{max} are significant and would seriously degrade the performance of small or lightweight vehicles operating at low angle of attack. Again, Fig. 27b shows that (73) large values of $M_\infty/R_{e_\ell}{}^{1/2}$ give rise to large losses in L/D_{max}; however, these values of $M_\infty/R_{e_\ell}{}^{1/2}$ will only occur when aerodynamic forces are insignificant unless the vehicles considered are of small size or wing loading.

However rare the circumstances in which forces due to low Reynolds number effects are significant to reentry range, the question will arise as to the relative susceptibility of different types of winged vehicle, although usually not at the condition for maximum L/D, but at the higher angles of attack typical of reentry. A detailed study of the relative performance of flat and caret deltas at low Reynolds numbers ($M_\infty/R_{e_\ell}{}^{1/2} > 0.10$) has been performed by Metcalf and Berry (74.) As shown in Fig. 28 the effects on C_L and L/D become significant only as L/D_{max} is approached or at Reynolds numbers which imply wing loadings well below 10 lb/ft^2 (0.42 kg/m^2). Since the latter are an order of magnitude too small, the effects of viscous interaction on aerodynamic forces during high-lift reentry seem likely to be significant only at the leading edges or in side and upper surface flows. However, with regard to the effects of concavity, it is worth noting that, at the highest Reynolds number (about 2500) and the attitude for C_{Lmax}, the C_L gain due to concavity is fairly large (about 20 %). Since C_D will also be affected, a better presentation of these data is to plot C_L against C_L/C_D (see Fig. 28), but even at the lowest Reynolds number (about 800) concavity gives gains in C_L (at given L/D), which can exceed 50 %. At the most realistic Reynolds number the L/D_{max} of the caret wing is 25 % above the flat delta value, and at L/D = 0.68, the caret wing gives nearly 45 % higher C_L.

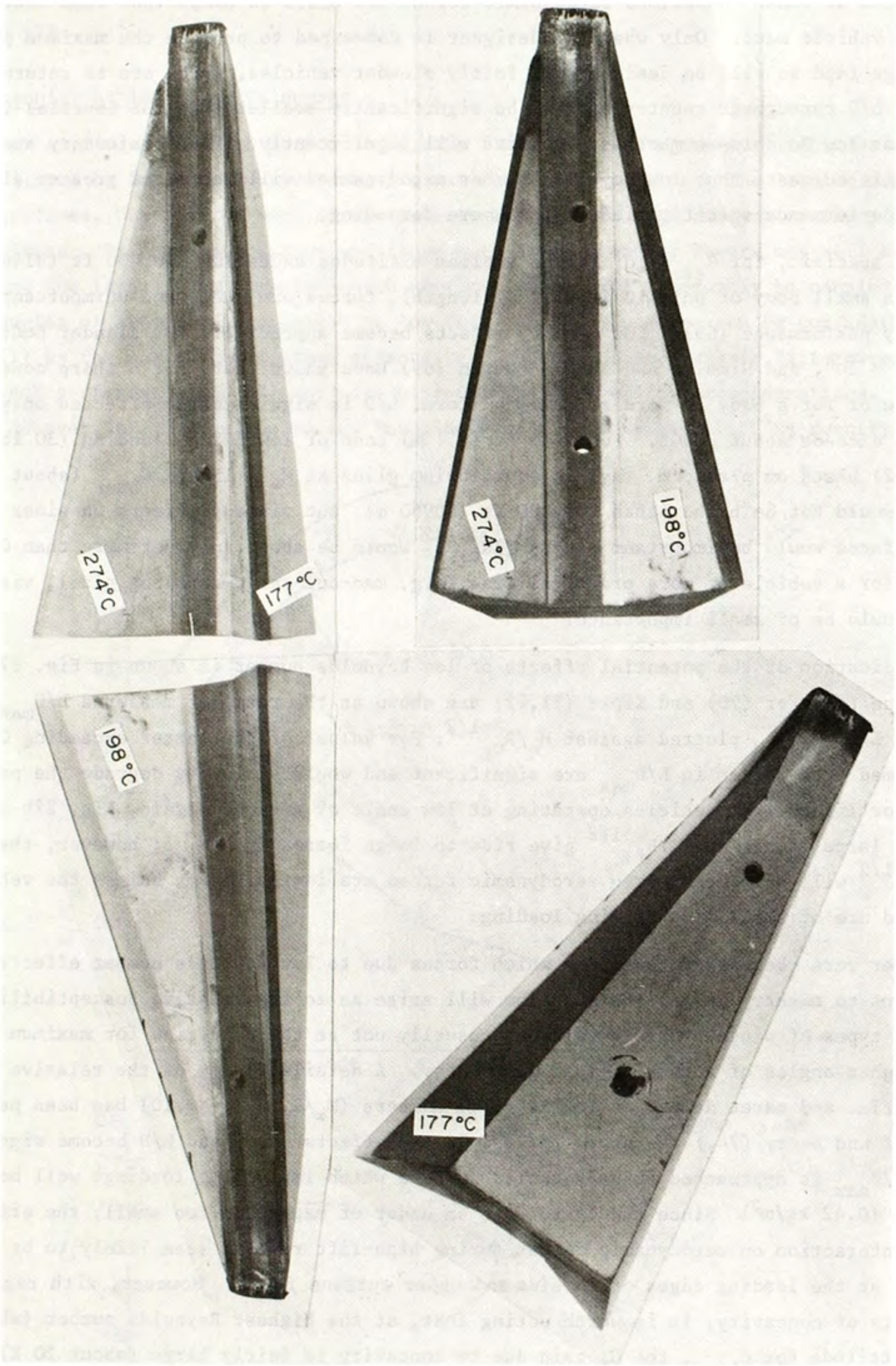


Fig. 29. Temperature distributions at low Reynolds number. (Data due to Davies and Berry (75).)

In practice, concavity may be used to increase C_L and altitude at given flight speed. Thus a better approach might be to compare data not at equal Reynolds numbers, but, for example, at equal implied wing loadings; the value to be kept constant, for given vehicle length, flightspeed and ambient viscosity, would then be $R_\infty C_L$.

With regard to heat transfer at low Reynolds numbers, Davies and Berry (75) have tested a trapezoidal wedge-caret wing (see section 2.2.1.), again at Mach number 6 and low Reynolds number, but with the model coated with heat-sensitive paint. The temperature distribution

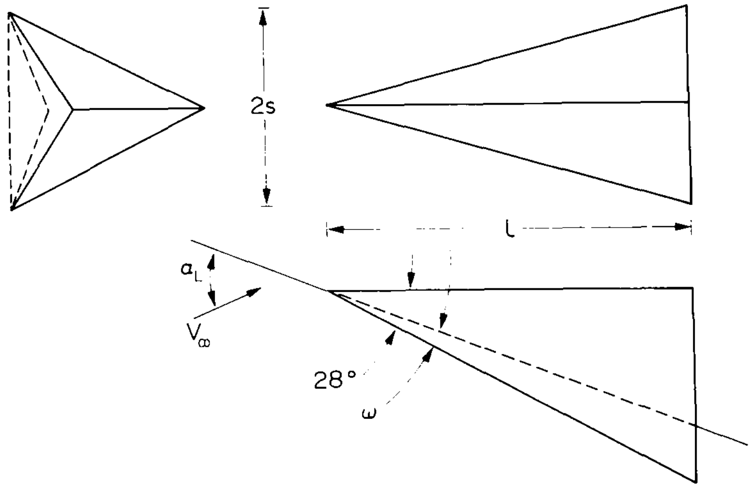


Fig. 30. Flat and caret delta wings for heat transfer tests at $M_\infty \doteq 20$–22 and $R_e \doteq 0.85 \times 10^5 - 1.2 \times 10^5 m^{-1}$ ($\ell = 76.2$ mm, $s/\ell = 0.267$, $\omega = 0^o, 5^o, 8^o$ (76).)

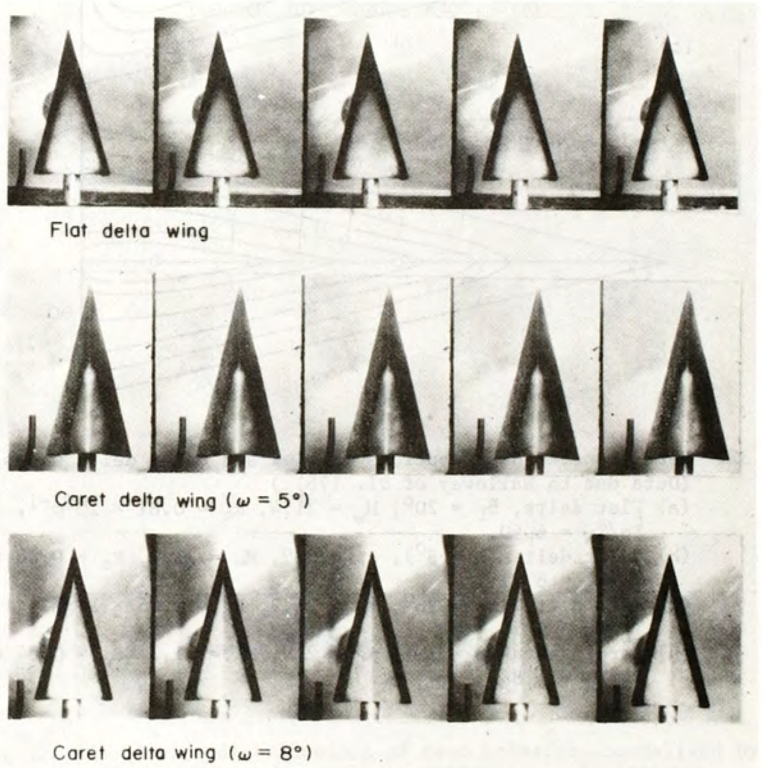


Fig. 31. Examples of phase-change paint recordings on models of Fig. 30. (Data due to Galloway *et al.* (76).)

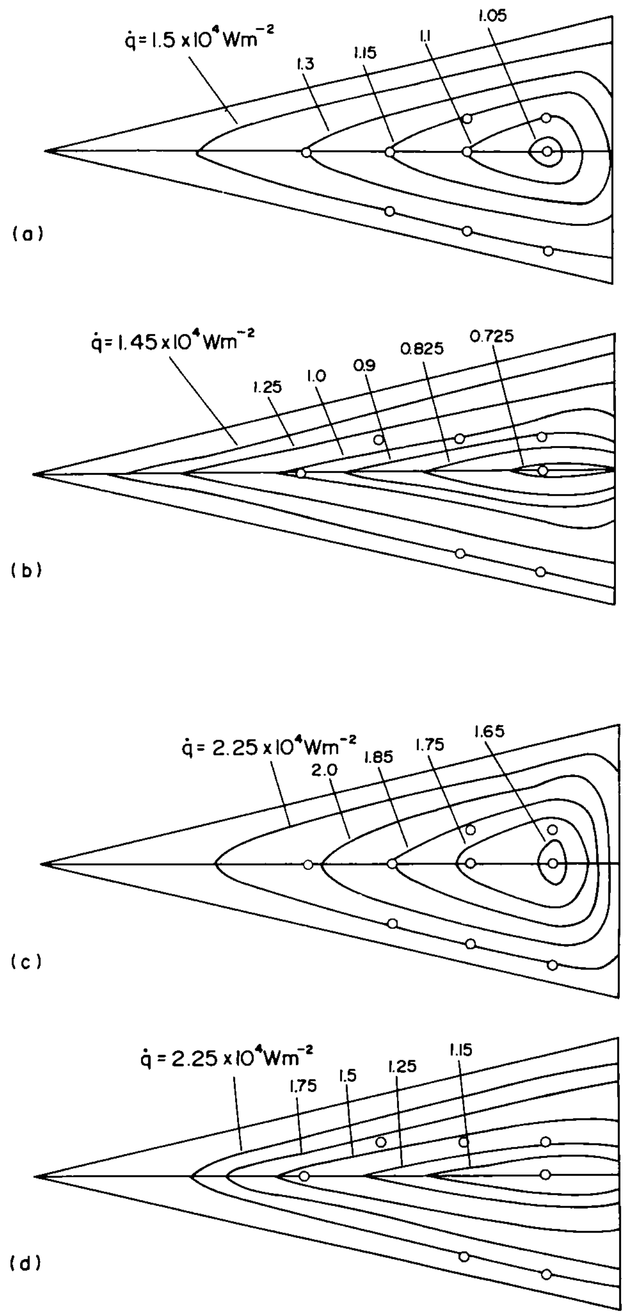


Fig. 32. Heat transfer distributions on flat and caret delta wings. (Data due to Galloway *et al.* (76).)
(a) Flat delta, $\alpha_L = 20^o$, $M_\infty = 21.4$, $R_e = 0.81 \times 10^5 m^{-1}$, $T_0/T_W = 6.90$.
(b) Caret delta ($\omega = 8^o$), $\alpha_L = 20^o$, $M_\infty = 21.2$, $R_e = 0.88 \times 10^5 m^{-1}$, $T_0/T_W = 6.71$.
(c) Flat delta, $\alpha_L = 28^o$, $M_\infty = 21.3$, $R_e = 0.85 \times 10^5 m^{-1}$, $T_0/T_W = 6.78$.
(d) Caret delta ($\omega = 8^o$), $\alpha_L = 27^o$, $M_\infty = 21.3$, $R_e = 0.84 \times 10^5 m^{-1}$, $T_0/T_W = 6.86$.

shown in Fig. 29 confirms that much of the undersurface flow remains two-dimensional at the conditions tested and indicates the extent to which severe heating extends inboard from the leading edges.

For detailed measurements of heat transfer distributions on flat and caret wings, results for low Reynolds numbers and Mach numbers up to 22 have been obtained in the Nitrogen Wind Tunnel at Imperial College (59,76).

Galloway *et al.* (76) investigated heat transfer on a set of three delta wings (see Fig. 30), by the use of phase change paint and of thermocouples attached to thin-shell regions of the model surface. Stagnation temperatures and pressures in the Nitrogen Tunnel varied between 1350 and 2000 K, and 3 and 5 MNm^{-2}, with Mach and Reynolds numbers varying from 20.54 to 22.20,

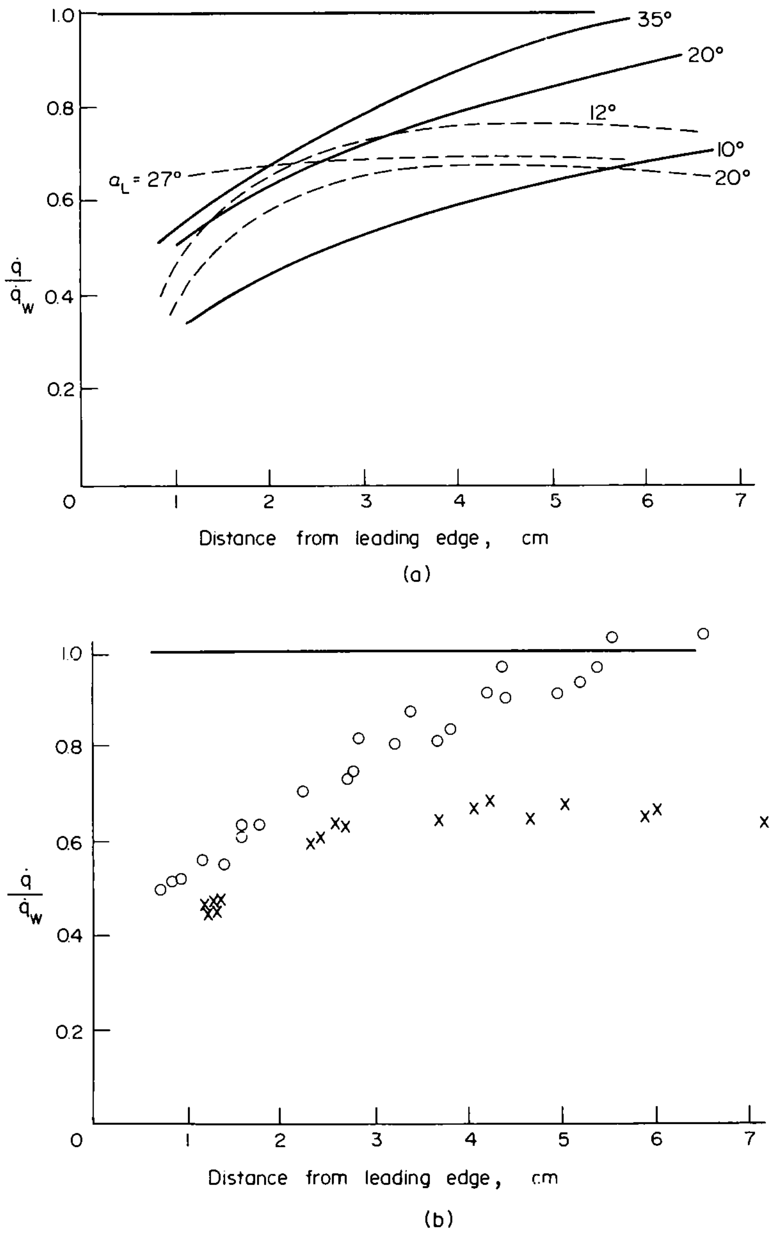


Fig. 33. Chord wise distributions of heat transfer, normalised by viscous wedge values. (Data due to Galloway *et al.* (76).)
(a) Mean curves averaged for several spanwise stations. $M_\infty(AV) = 21.2$, $R_{e(AV)} = 0.87 \times 10^5 m^{-1}$. —— Flat delta, – – – – caret delta, $\omega = 8^o$.
(b) Sample data sets for (a) above. o Flat delta, $\alpha_L = 35^o$, x caret delta, $\alpha_L = 20^o$, $\omega=8^o$.

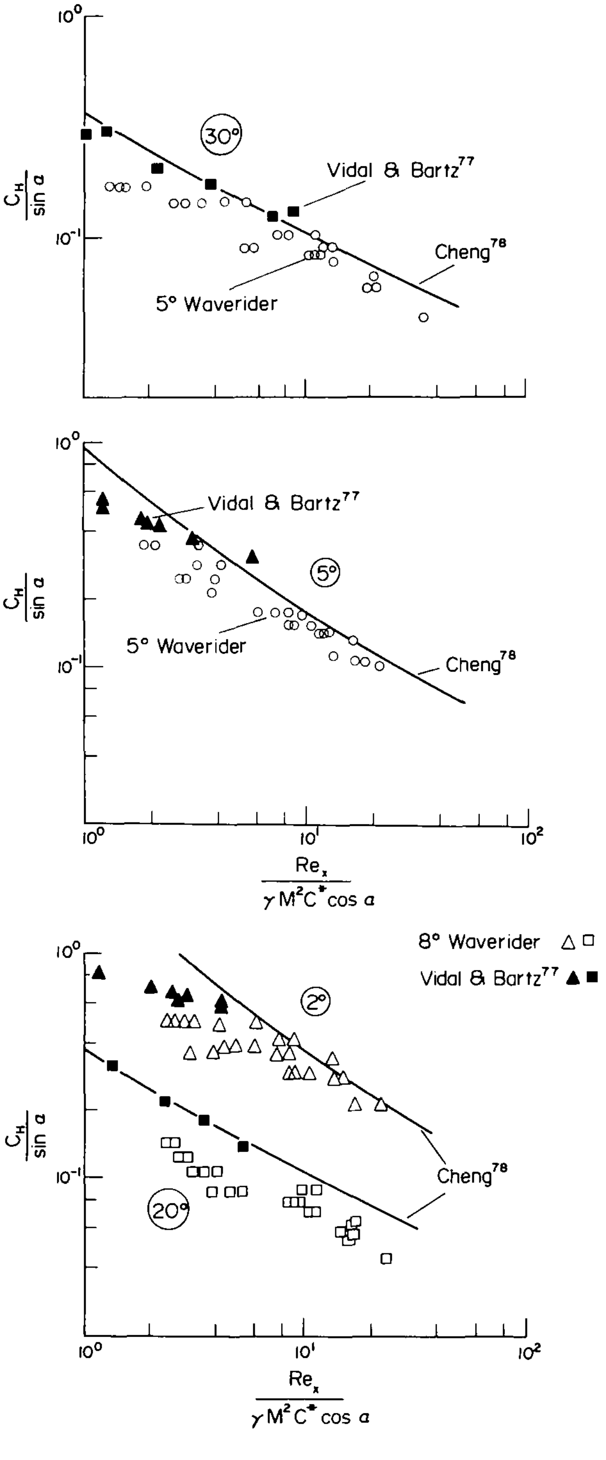


Fig. 34. Heat transfer correlations. (Data due to Galloway *et al.* (76) compared with wedge data due to Vidal and Bartz (77) and theoretical values due to Cheng *et al.* (78).)

and $0.78 \times 10^5 m^{-1}$ to $1.75 \times 10^5 m^{-1}$. Values of α_L varied from 2 to 35°, but only at one specific angle of attack ($\alpha_L = 20^o$) were data obtained on more than one model (the flat delta and the caret with $\omega = 8^o$); however, it is seen from Fig. 6b that the higher angles of attack tested are appropriate to the lower limit expected for Mach numbers of 20 and above. As with the model tested by Davies and Berry (75), heat transfer contours for the waverider models

(ω = 5 and 8^o) run nearly parallel to the leading edges (see Figs 31 and 32), whereas the flat delta exhibits well-rounded heat transfer contours. A more important difference between the flat and caret delta wings is, that, as seen in Fig. 32, the $\dot{q}$ values on the caret surface fall to significantly lower levels than those for the flat delta; in addition, the caret wing was probably producing a higher C_L at a given angle of attack and so, if compared with the delta wing at the same value of $R_\infty C_L$ and implied wing leading, it would be even less severely heated than Fig. 32 suggests.

A summary of much of the data obtained for two of the wings is shown in Fig. 33a as chordwise distributions of heat transfer, mean lines being drawn through data points for several spanwise stations; the amount of scatter which the mean lines conceal is shown for two cases in Fig. 33b. Finally, Fig. 34 compares caret wing data with the measurements for wedges made by Vidal and Bartz (77), and with the theoretical predictions of Cheng *et al.* (78,79).

These tests are among the very few, which attempt to quantify the effects of concavity on heat transfer (59,76,80).

2.2.4. Real gas effects

For lifting reentry gliders at full scale and in real air, Enkenhus (81) has shown that, almost independently of C_L and W/S, oxygen will start to dissociate ($O_2 \rightarrow O+O$) at a flight Mach number of about 7 and nitrogen will also dissociate ($N_2 \rightarrow N+N$) at Mach numbers above about 12.

The influence of real gas effects on the production of lift at high Mach number and high angles of attack is potentially of great importance since, as shown in Fig. 35, the best lifting performance from wedges (that is, the highest values of C_L per unit L/D_p) is obtained (14) for γ = 1.4 and, as the effective value of γ falls towards unity as a consequence of real gas

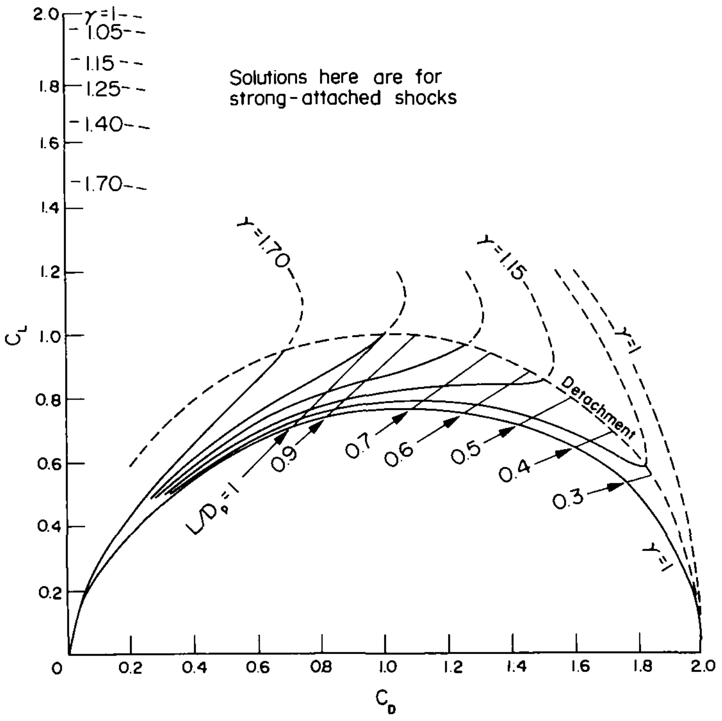


Fig. 35. Undersurface lift and drag for wedges in inviscid flow of infinite Mach number. (Data due to Roe (14).)

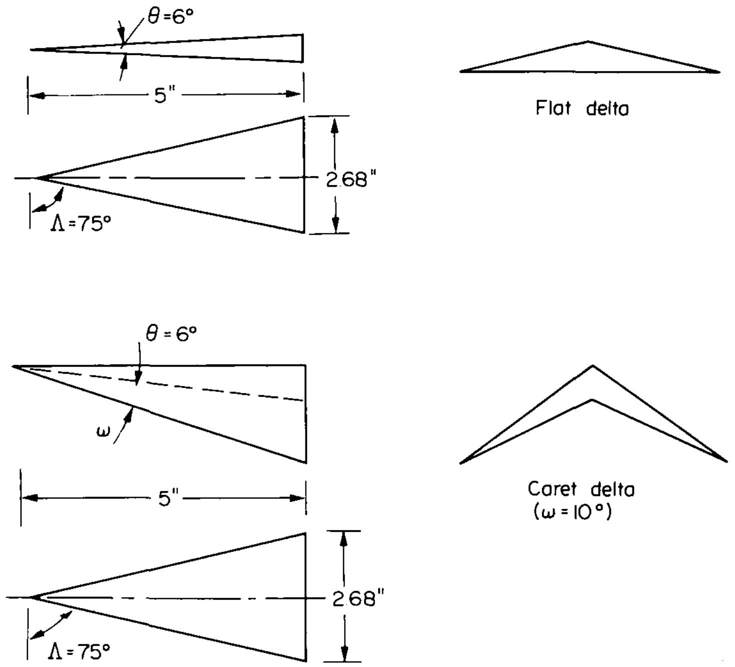


Fig. 9. Flat and caret delta models used to obtain data in Figs 10, 13, 38, 39 and 43.

by East and Lewis (60) at Mach numbers of 9 and 9.7, respectively, but for ω-values of about 5 and 8°, i.e. for caret models of more moderate concavity. In addition, Davies *et al.* (13) have tested a flat and a caret delta wing (ω = 0 and 5°), both having unusually low aspect ratio (s/ℓ = 0.11), at Mach number 12.2, model Reynolds number 10^6 and angles of attack between 50 and 65°. Shock standoff angle was roughly doubled by the use of concavity (see Fig. 7). Furthermore, in all these tests (13,59,60), the C_L values for the caret wings were superior to those for the equivalent flat delta throughout the test range, even though angles of attack were far removed from the design values. Also, it has been observed by Sykes (13) that increased pressures are retained by concave surfaces even when tested at Mach numbers far below the design value; these remarks however were made in the context of a Maikapar body rather than a Nonweiler wing.

Finally, Richards and Houwink (61) have evaluated flat and caret delta models (the latter having $\omega = 8.7^{\circ}$) by free flight tests in an airflow of Mach number 15 and at Reynolds number 2×10^6. The angles of attack varied from under 40° to just under 50 and 60° for the flat and caret delta, respectively. Once again, the caret wing provided a large margin of aerodynamic superiority, Richards and Houwink claiming (61) a C_L improvement of up to 23 %.

Figures 9-16 show most of the models used in the above tests and present the experimental results obtained in air or nitrogen in order of increasing Mach number. The major conclusion is that, in every case, C_L increases with concavity. Indeed, the extent to which C_L has been improved by simple anhedral distributions is encouragingly close to the maximum value of 30 % (for conical wings and γ = 1.4) anticipated (8) on the basis of thin shock layer theory (20). Modifications to planform, aspect ratio or anhedral distribution may, thus, be made as much for structural purposes (e.g. the reduction of wetted area per unit planform area) as for aerodynamic refinement, although the latter objective might still be achieved by exploring the effects of planform and anhedral modifications and the possible advantages of nonconicality (49) effects, serious losses in available C_L seem likely to occur. This fact was noted in a paper (8) in which wedge flow calculations by Bird (82) were quoted (see Fig. 36).

Since that time (1970), thin shocklayer theory has provided more general and more accurate assessments of the losses to be expected for flat and concave delta wings at lifting reentry conditions. For caret wings at design conditions, Squire (9) has shown that, for equilibrium flows (i.e. for γ = const. $\neq$ 1.4), plane attached shock waves are possible at angles of attack up to C_{Lmax} (55^{o} say) and, in addition, that the anhedral required for given aspect ratio becomes less as γ falls. This reduction in shock standoff angle ($\zeta - \alpha_L$) also results in a reduced C_L gain for concave wings but Squire has shown (9) that the gain remains significant.

Figure 37 shows curves for perfect gas (frozen flow) and real gas (equilibrium flow) for wedges and wings at Mach number 15 and 200,000 ft (60960 m) altitude. As noted by Bray (83), both these flows produce constant pressure beneath the wedge and so the percentage fall in C_L would be the same as the eventual percentage fall in surface pressure if an actual

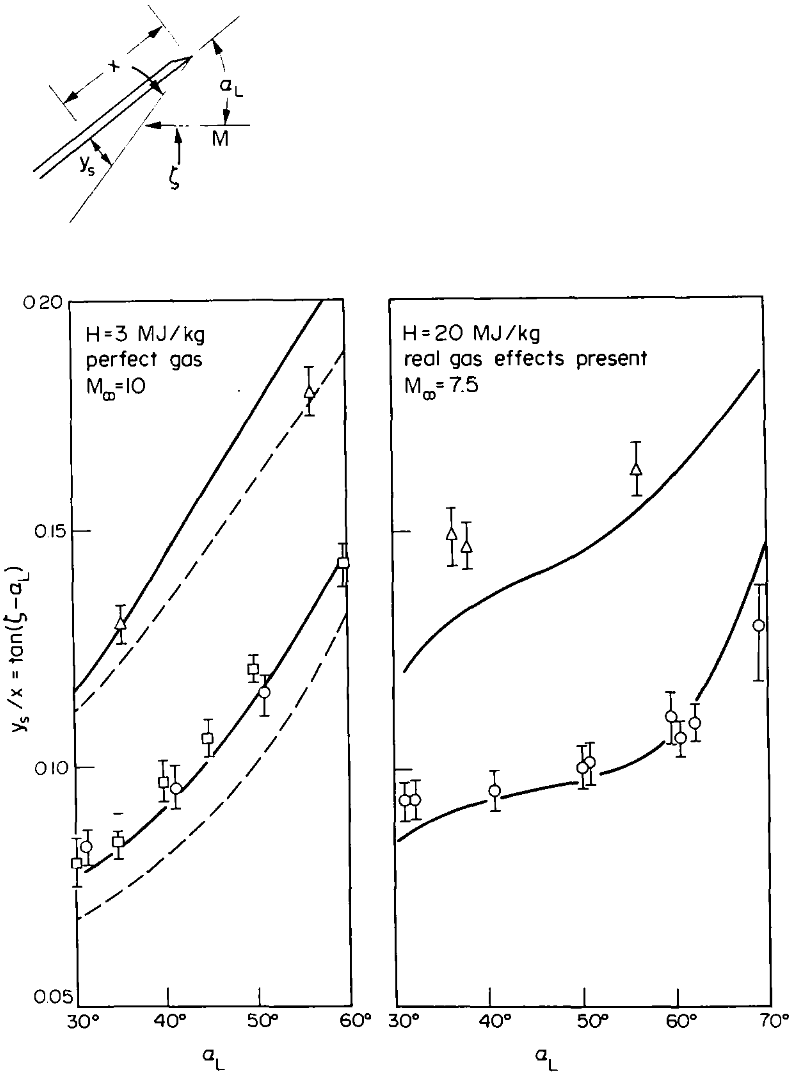


Fig. 38. Shock standoff angle beneath flat and caret delta wings in perfect and in real gas flows. (Data due to Stalker and Stollery (84).) —— Theory, with second order terms. - - - - Theory, second order terms omitted. Experiment: ▫ o flat delta, Δ caret delta.

nonequilibrium flow were to approach equilibrium; for the conditions chosen, the fall in pressure along the surface of a wedge in nonequilibrium flow and at 40^o angle of attack is 10 % of the pressure at the planform apex. This percentage increases rapidly with angle of attack, and also with flightspeed and altitude; possible losses can exceed 20 % and vibrational relaxation can also involve a fall of up to some 5 %.

Since the caret wing at near-design conditions produces a flow resembling that beneath a wedge nonequilibrium effects of similar magnitude may be expected. However, Fig. 37 shows that although a caret wing (or more generally perhaps a concave wing) will suffer greater losses in C_L than will a flat delta of the same planform, the caret wing remains superior in terms of both C_{Lmax} and C_L per unit L/D. This theoretical conclusion is very important and its confirmation by experiment is essential.

Experimental research on real gas effects in the flow beneath flat and caret delta wings has been reported by Stalker and Stollery (84). The models shown in Fig. 9 were tested in the T3 free piston shock tunnel at the Australian National University, in which airflows were produced at stagnation enthalpies of 20 MJ/kg and 3 MJ/kg, at flow Mach numbers of 7.5 and 10, respectively. At the lower enthalpy the flow behaved as a perfect gas, whereas at the higher value real gas effects were present. Measurements of shock standoff angle are shown in Fig. 38, from which it is seen that for both wings shock standoff angle increases strongly with angle of attack at perfect gas conditions, but that at high enthalpy, this effect is much reduced (though less so for the caret wing). The major conclusion, however is that at a

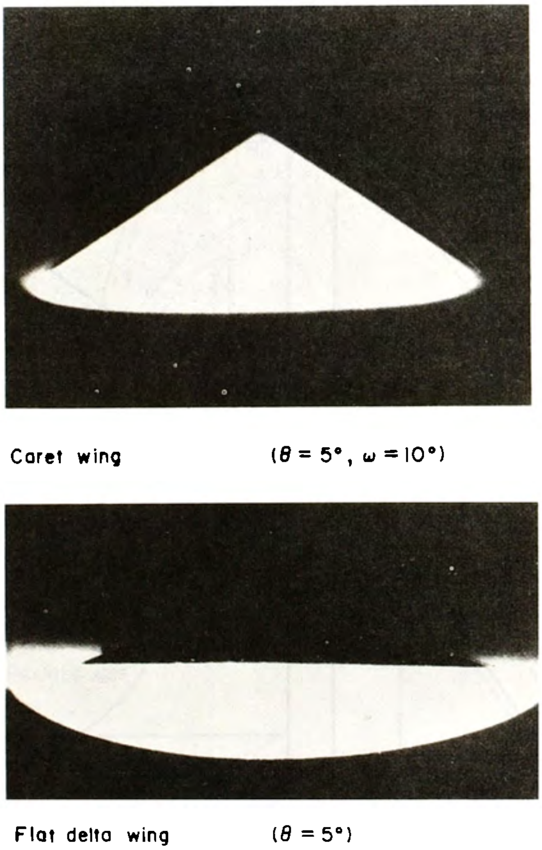


Fig. 39. Rearview luminosity pictures for caret and flat delta wings at $\alpha_L = 70^o$. (Data due to Stalker and Stollery (84).)

given angle of attack in a given real gas flow the caret wing still produces a substantially stronger shock than does the flat delta. Furthermore, Fig. 39 demonstrates that flow containment by the caret wing is still achieved.

Clearly, further experiments should study the extent to which these data may imply the retention of superior lifting performance by the caret wing and also should measure relative levels of heat transfer to flat and concave deltas. The effectiveness of the T3 tunnel in measuring surface pressures and local heat transfer rates has already been demonstrated (85) in tests on a reentry glider model supplied by MBB (see Figs 40 and 41) and in other more recent test programmes. In interpreting heat transfer results obtained on models of different shape, however, it will be necessary to allow for the fact that where models produce different values of C_L at a given angle of attack and wind tunnel condition, the implied wing loadings will be different or, for equal wing loadings, the model producing the higher C_L corresponds to full scale flight at greater altitude and relatively lower ambient density.

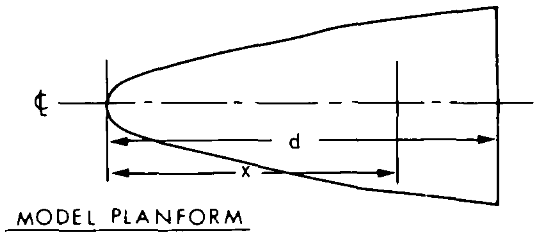
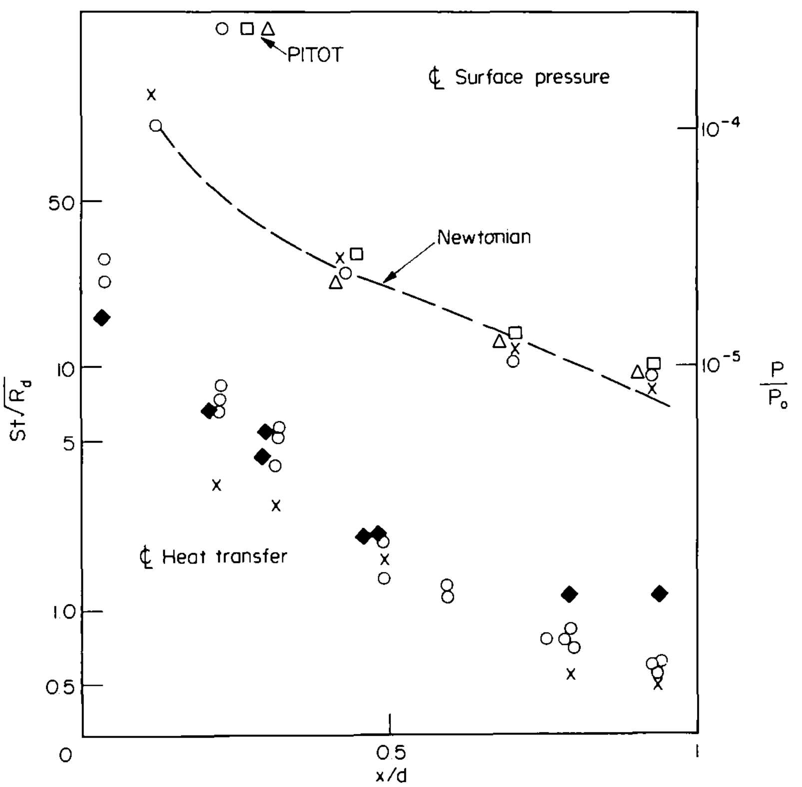


Fig. 40. Testing of reentry glider model. (Data due to Stalker (85).)

Finally, at angles of attack, for which a caret wing will support shock patterns which lie within the recessed region between the lower surface and the leading edge plane (see Fig. 42), Pike (86) has shown that substantial losses in lift will still arise; also, shock wave-boundary layer interactions will not only complicate thermal stresses, but will themselves be complicated by the presence of non-equilibrium flows. It is known that for $\gamma = 1.4$ such shock patterns will arise (8) in certain types of trajectory; they will probably not occur until after reentry is complete and certainly not at very high angles of attack, but they could enforce a change from high to low angle of attack at a lower Mach number than that for optimum cross-range. Against this, real gas effects reduce the concavity required by a delta wing operating at high Mach number; thus if enough concavity is provided to improve C_L per unit L/D at conditions, where real gas effects are small, the shocks produced at higher Mach numbers may lie within the recess (see Fig. 42) or, alternatively, the vehicle should be able to operate at higher angles of attack before the shock wave becomes detached.



	$R_d \times 10^{-5}$	M
○	1.8	7.7
△	1.5	7.2
□	1.2	7.6
◆	6.2	5.8
x	2.0	9.6

$P_o \equiv$ nozzle reservoir pressure

Fig. 41. Test results for reentry glider model. (Data due to Stalker (85).)

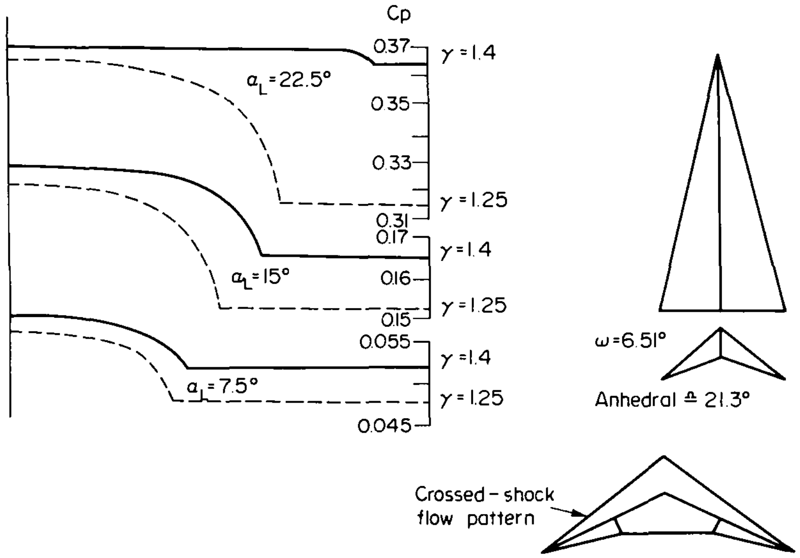


Fig. 42. Spanwise distribution of undersurface pressure coefficients at $M_\infty = 10$. (Data due to Pike (86).)

2.3. Research on aerodynamic stability and control

As reentry proceeds progressively larger aerodynamic forces become available from reaction controls or aerodynamic surfaces of the usual type ; thus, the possibility increases that a re-entry vehicle may be made not merely controllable, but aerodynamically stable and, hence, less dependent for its survival upon electronic and/or mechanical reliability.

Since early work on lifting reentry (largely concerned with the design of the Dynasoar X-20) was terminated in the early 1960s, much research has been aimed at understanding the problems of axisymmetric bodies rather than those intended to support large lift forces; although the advent of the Space Shuttle changed this situation, much of the ensuing work has concerned the specific problems of controlling particular orbiter designs and the basic configurations (large fuselage plus almost flat delta) have sought to be controllable rather than stable. For the orbiter as it now exists simulator studies are stated (1) to have shown, that given good weather and a reasonably straightforward approach the orbiter could be flown without stability augmentation since it has neutral stability over a very wide speed range in all three axes. Nonetheless, with neutral stability a vehicle may need to be controlled during much of its reentry process and is at greater risk than if it were inherently stable.

In the context of concave delta wings, Fig. 43 suggests that at least for Mach numbers of the order of 10 stability and control of flat and caret delta wings will not be complicated by significant shifts in the aerodynamic centre, while for trapezoidal planforms East (25) found that the aerodynamic centre moved by only small amounts.

In the absence of more generalised experimental results and flight experience, substantial theoretical contributions are all the more important. Following earlier work on the pitching oscillation of wedges with attached shock waves, in 1968 Hui (87,88) presented an exact treatment of both static and dynamic pitching characteristics of caret wings at high and low angles of attack; some of this work was described by East at Euromech 20 (1970), since when Liu (89,90) and Hui (90) have perturbed Hui's steady flow solution for delta wings with attached shock waves (91,92) and so predicted pitching stability derivatives for flat delta wings with shocks attached to the leading edges. Further analysis by Hui and Hemdan (93) has been concerned with the aerodynamic stability of the slender delta wing performing small amplitude pitching oscillations in hypersonic flow, with the shock wave attached to the wing apex, but detached from the leading edges. Hui and Hemdan show that flat deltas at these conditions will always be dynamically stable, whereas with shocks attached to the leading edges, the wing could become dynamically unstable as Mach number falls. If a similar conclusion can be shown to apply for caret wings then the use of high angles of attack during periods of most intense heating would also impart pitch stability to a lifting reentry vehicle; the study of waveriders with shock waves detached from the leading edges would then become even more profitable.

East and Ghosh (94) have undertaken experiments on flat and caret delta wings at Mach number 9.5 and angles of attack of about 15 and 26.5°. This work was described at Euromech 74 in 1976 and Fig. 44 shows the rigs used together with typical oscillation records. In experiments to measure $-C_{M_\theta}$ for double delta wing models using the gas bearing pivots, good agreement was observed for the 26.5° angle of attack experiments with inviscid theoretical predictions based on Liu's theory. However, poor agreement was observed for the lower angle of attack. Thus to check the double model technique, the half model was used to measure the stiffness derivative $-C_{M_\theta}$ on a half wing of chord and sweepback equal to those of the double models. The relevant experimental details are shown in Table 1.

The results for $-C_{M_\theta}$ are shown in Fig. 45 and within the experimental scatter no large deviations from Liu's theoretical predictions are observed at either 15 or 26.5°. At this

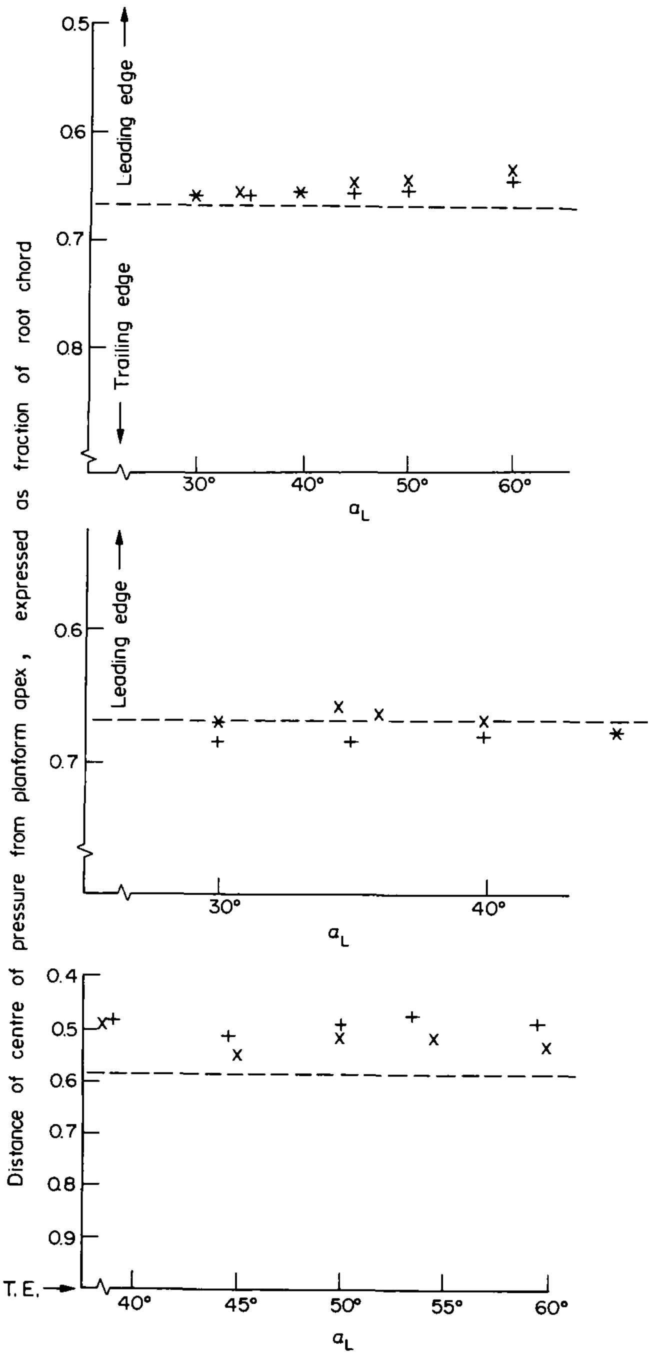


Fig. 43. Location of centre of pressure on delta and trapezoidal wings
(a) $M_\infty = 9$, $R_e = 1.5 \times 10^6$. (Data due to Coleman (57,58).)
(b) $M_\infty = 12.2$, $R_e = 0.75 \times 10^6$. (Data due to Carr (56).)
(c) $M_\infty = 9.7$, $R_e = 4 \times 10^5$. (Data due to East (25).)
– – – (a, b and c) Centre of area, x (a, b) caret delta, $\omega = 10^o$ (see Fig. 9), x (c) trapezoidal wedge-caret wing, $\omega = 8^o$ (see Fig. 17), + (a, b) flat delta (see Fig. 9), (c) flat trapezoidal wing (see Fig. 17).

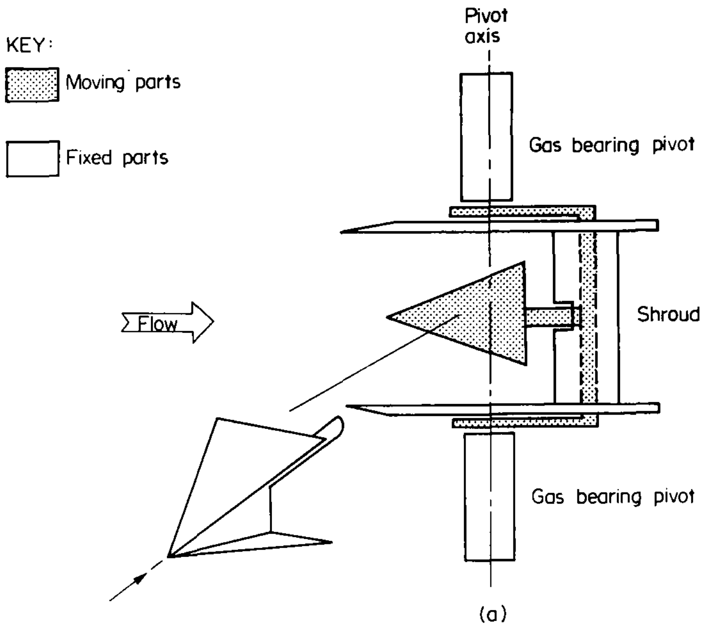


Fig. 44. Dynamic pitch stability measurements at hypersonic speeds using oscillitory rigs (AASU).
(a) Gas bearing pivot for double-wing technique.

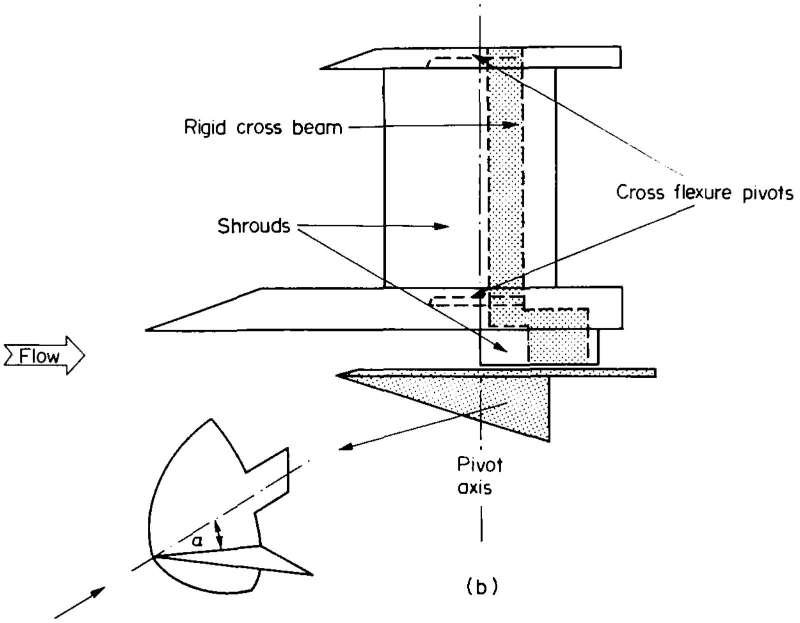


Fig. 44. (b) Cross flexure pivot for half-wing technique.

Table 1

Sweepback Λ	α_o	M_∞	Chord c (cm)	Area S (cm^2)	$R_{e_{\bar{c}}}$	f_o (Hz)	$\frac{\omega c}{U_\infty}$
62^o	15^o	9.5	5.5	8.01	1.8×10^6	100-145	0.029-0.042
62^o	26.5^o	9.5	5.5	8.01	0.84×10^6	100-145	0.029-0.042

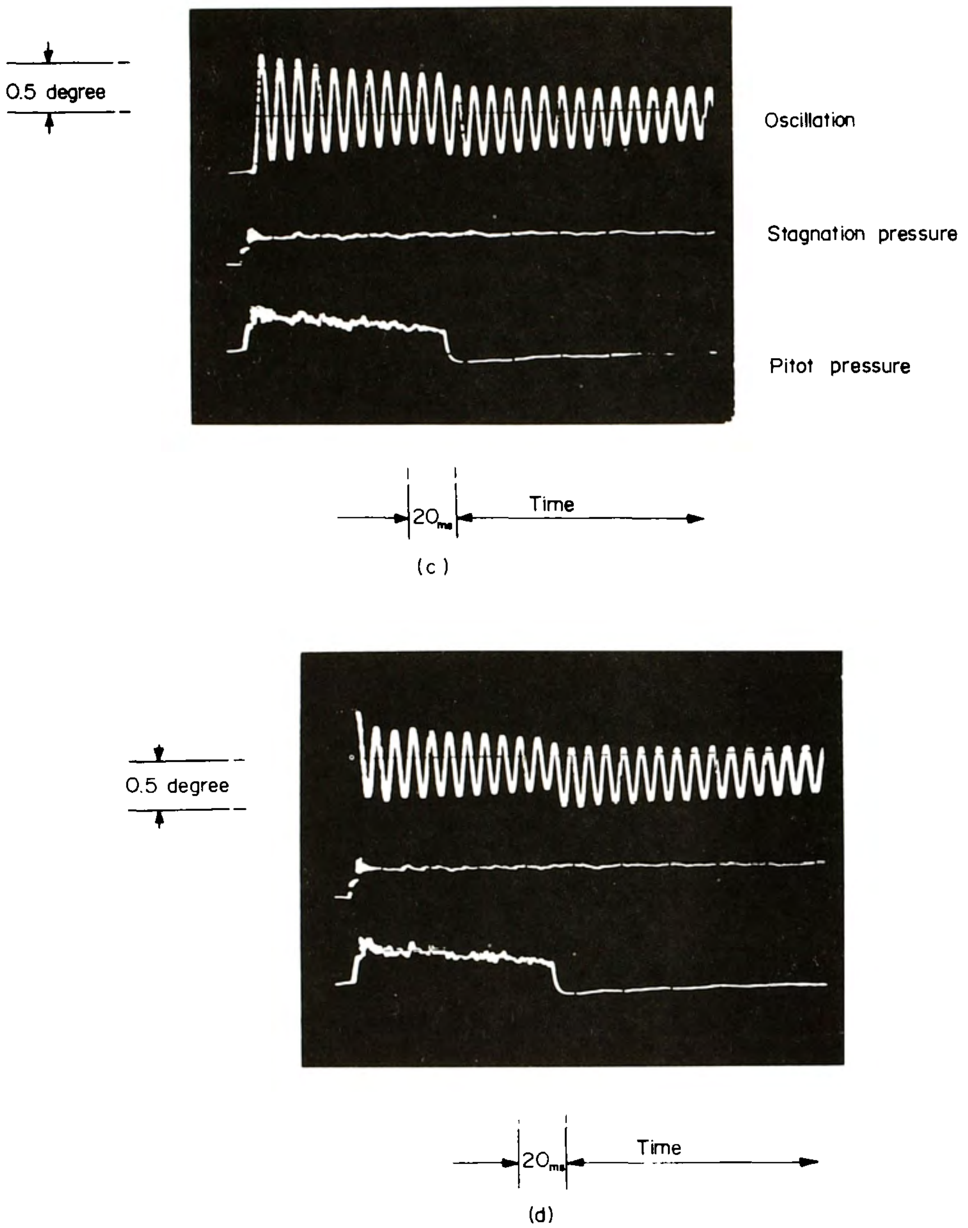


Fig. 44. (c, d) Oscillation records for half-delta wing. $\Lambda = 62^{\circ}$, $M_{\infty} = 9.5$, $\alpha_L = 26.5^{\circ}$, $P_{stag} = 135$ bar, $x_0/C = 0.81$. Initial displacement nose down (c), nose up (d).

stage no explanation for the anomalous results has been proved, but it seems possible that some dynamic interaction between the flow over the double model and the flow over the shroud may be the cause. Static pressure measurements on both windward and leeward surfaces of a static double wing arrangement showed that no significant static interaction occured.

The additional pitching stiffness of the flexure supported models resulted in higher frequencies than for the air bearing system. From 6 to 8 complete cycles were available for analysis during a typical running time. It therefore proved possible to obtain some preliminary measurements of the damping derivative $-C_{m_{\dot{\theta}}}$ using the difference between results from initially nose up and nose down displacements. The results for 15 and 26.5° angle of attack and half delta wings are shown in Fig. 46. High values of the damping derivative $-C_{M_{\dot{\theta}}}$ are observed for

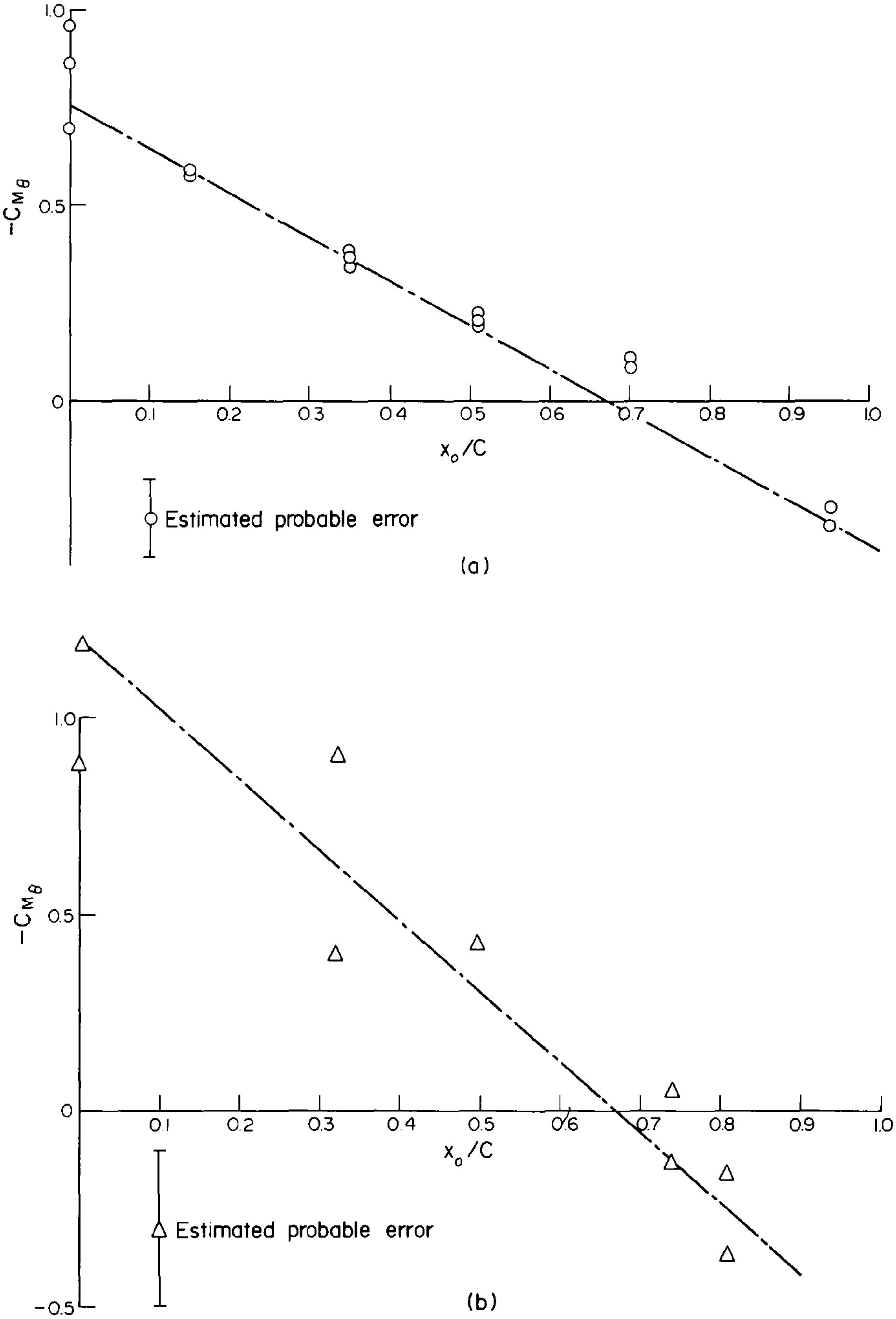


Fig. 45. Delta wing stiffness derivative – C_{M_θ} vs. axis position x_o/C (half-model). (Data due to East (94).) $M_\infty = 9.5$, $\Lambda = 62^o$.
(a) $\alpha_L = 15^o$, $R_{ec} = 1.8 \times 10^6$.
(b) $\alpha_L = 26.5^o$, $R_{ec} = 0.84 \times 10^6$.
Δ Experiment. — - — Theory (Liu (89)). $M_\infty = 10$, $\Lambda = 62^o$.
(Theory is approximate for (b) since shock is detached.)

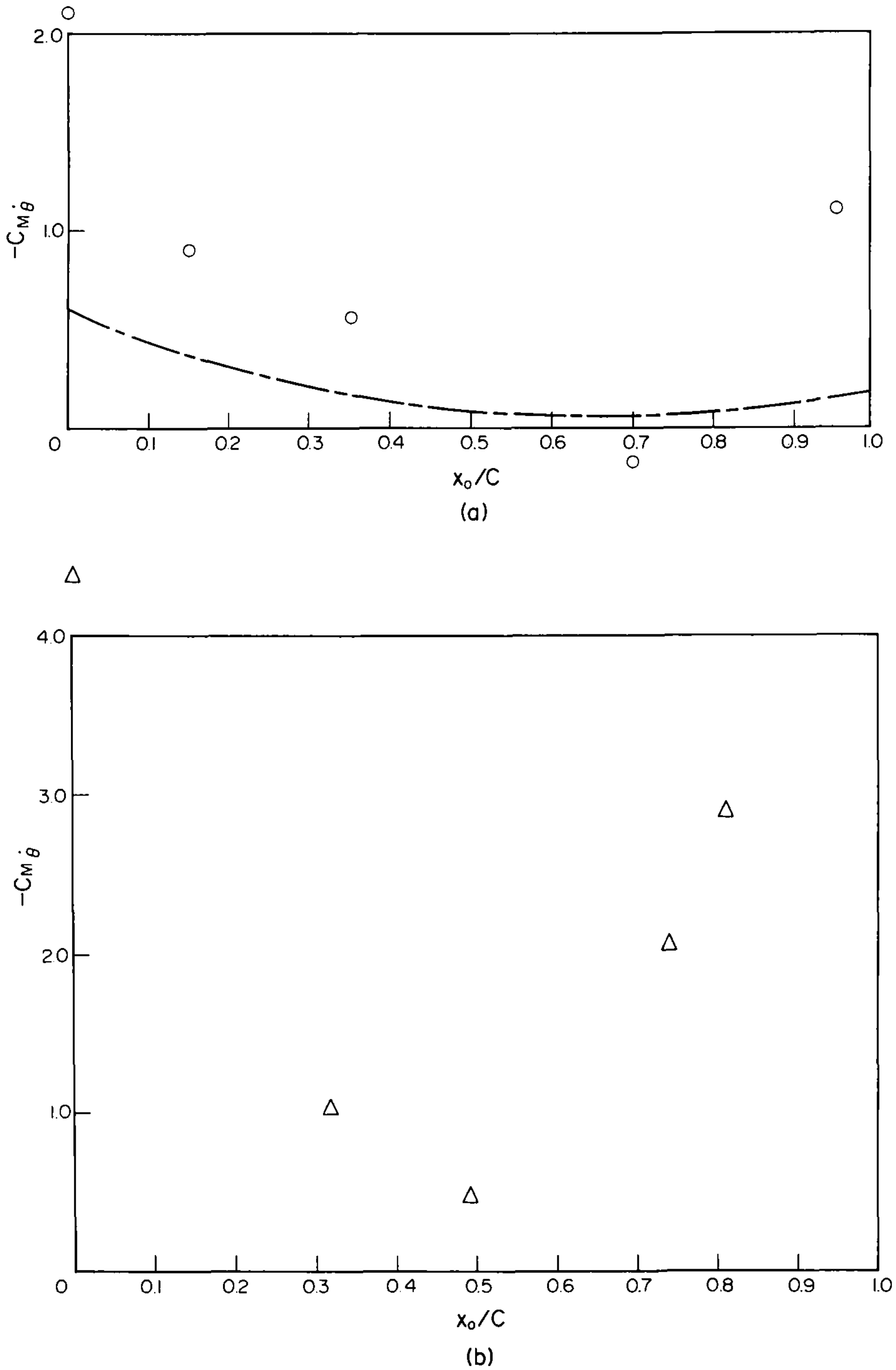


Fig. 46. Delta wing damping derivative $-C_{M_{\dot{\theta}}}$ vs. axis position x_0/C (half-model). (Data due to East (94).)
(a) o Experiment: $M_\infty = 9.5$, $\alpha_L = 15°$, $\Lambda = 62°$, $R_{ec} = 1.8 \times 10^6$. Frequency parameter: $k = \omega c/U_\infty = 0.026 - 0.039$.
— - — - — Theory (Liu (89)), $M_\infty = 10$, $\alpha_L = 15°$.
(b) o Experiment: $M_\infty = 9.5$, $\alpha_L = 26.5°$, $\Lambda = 62°$, $R_{ec} = 0.84 \times 10^6$, $k = 0.029 - 0.042$.

leading and trailing edge axis positions. Small negative values were measured for mid-axis positions for the 15° incidence model, whereas small positive values resulted for the 26.5° wing. Comparison with Liu's theoretical inviscid prediction was made for the 15° case, but was not possible for the 26.5° wing since the theory is invalid for wings with detached shock waves. The measured derivatives for forward and rearward axis positions are greatly in excess of predictions, whereas for mid-chord positions the theory predicts small positive damping compared with the small negative damping observed. The large departure between theory and experiment may be due to viscous effects. Considerable departure from the predictions of inviscid theory has been both predicted and observed by Orlik-Ruckemann (95) for low Reynolds number flows past cones. The results presented by East could be due to the dynamic behaviour of the expected laminar/transitional boundary layer at the prevailing Reynolds numbers. It is worth noting their free-flight tests on static and dynamic stability have been undertaken in the VKI Longshot Tunnel at Mach numbers of 15 and 19 and high Reynolds number, but that winged shapes were not included (96).

In America, data have been obtained on the dynamic stability of a Space Shuttle orbiter (a modified 089B orbiter). In addition to results at subsonic, transonic and supersonic speeds (97-99), wind tunnel tests have been conducted (100) at Mach number 8, model Reynolds number 1.2×10^6 to 4.8×10^6 and angles of attack between -4.9 and 26.5°. Taken together, the data in these four papers are stated (100) to be the only measured damping data for the Shuttle development programme and may indicate the order of acceptable pitching characteristics where augmentation is available.

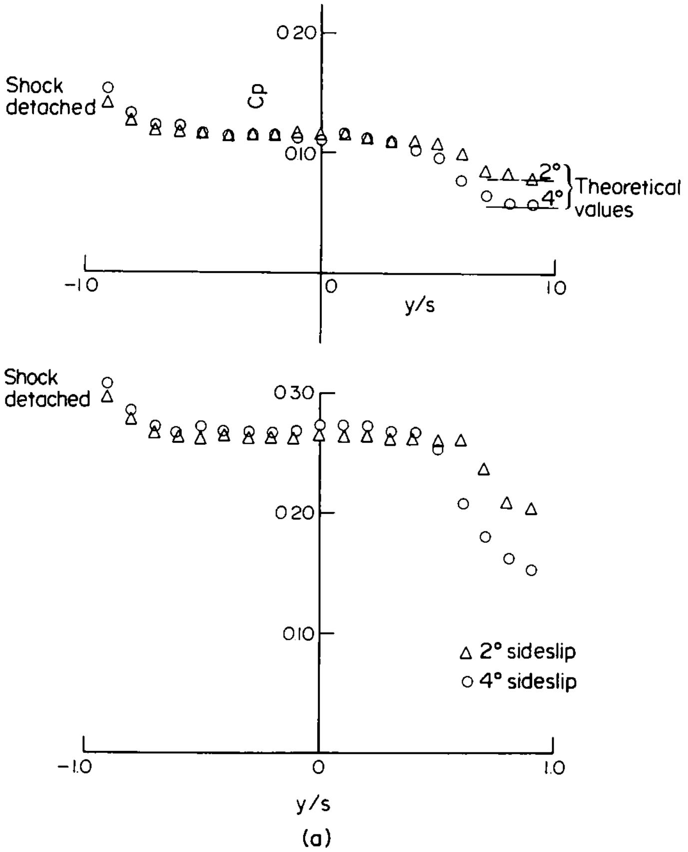
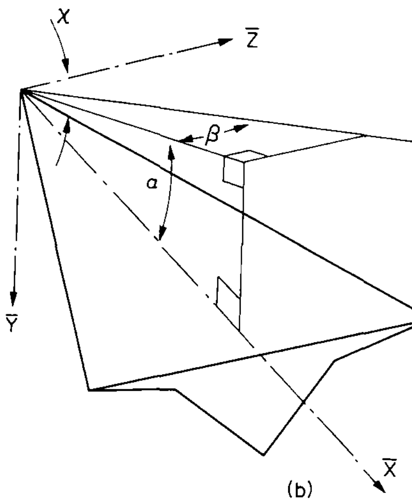
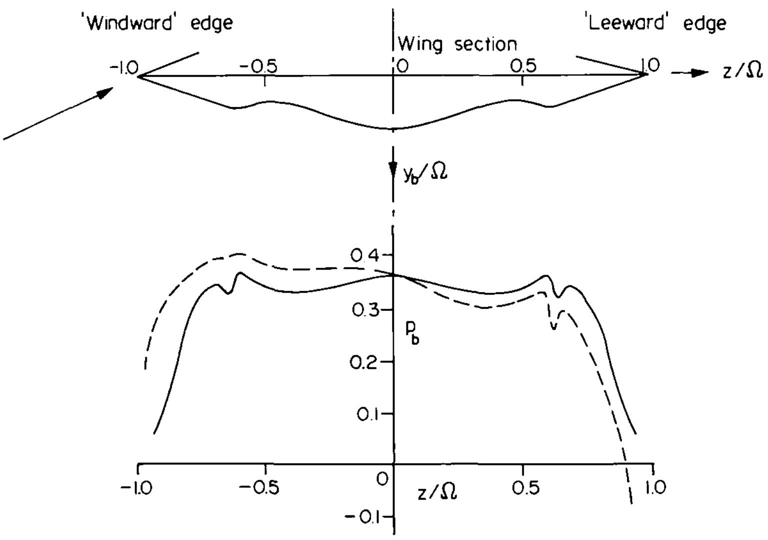


Fig. 47. Effects of side slip on flows beneath delta wings.
(a) Experiment at low angles of attack. Caret delta wings at 2 and 4° sideslip. Both wings tested at design Mach number (4.3) and design angle of attack.
Upper $\omega = 11.4°$, $s/\ell = 0.25$, $\alpha_L = 8.4°$.
Lower $\omega = 11.2°$, $s/\ell = 0.25$, $\alpha_L = 16.1°$.
(Data due to Crabtree and Treadgold (102).)

There is thought to have been no reported research on the stability and control in yaw of concave wings at reentry conditions and, indeed, a given yaw angle may become progressively less influential as angle of attack is increased. However, Bagley (101), Crabtree and Treadgold (102) demonstrated in the 1960s, that, at design conditions, even small angles of sideslip disturb the undersurface flow of caret wings and that this effect is concentrated near the leading edges (see Fig. 47a); thus, the pressure changes produced by yaw will simultaneously produce powerful effects on rolling characteristics. For the conditions tested by Treadgold (moderate Mach number and angle of attack) the yawing moment due to sideslip seems likely to be stabilising, at least for undersurfaces considered in isolation. This conclusion agrees with theoretical predictions due to Hillier (39) although again for "low" Mach number (about 3.5) and low angles of attack. For complete waverider configurations, however, Kipke (103)



(b)



(c)

Fig. 47. (b) Co-ordinate system used in thin shock layer theory (Hillier (104)).
(c) Results as calculated by Hillier (104). —— unyawed (Ω = 0.5), - - - yawed (Ω = 0.5, τ = 0.1). Symbols defined as: $\Omega = \epsilon^{-\frac{1}{2}}\cot\alpha.\cot\chi$, $C_p = 2\cos^2\beta\sin^2\alpha(1+\epsilon\, p_b)$, $\tau = \tan\beta/(\epsilon^{\frac{1}{2}}\sin\alpha)$.

has shown that both rolling and yawing moments due to sideslip can be statically unstable, although Kipke's tests were again at quite low angles of attack. It is therefore possible that to obtain even static stability for a given vehicle over a wide range of high and low speed conditions, the consideration of configuration design may be required at an early stage and that one possibility would be to include the stabilising influence of appropriately chosen sideflows as part of the undersurface design exercise.

The design of complete waverider configurations is often based on combining component bodies and flows which are individually understood and which, being based on aerodynamically sharp leading edges with attached shock waves, operate in aerodynamic isolation from each other, at least, when near design conditions. On the other hand, waveriders at lifting reentry conditions should sometimes operate, on both heating and stability considerations with shock waves detached from their leading edges (93). Hillier (104) has shown that the associated flows may, nonetheless, be predicted (see Figs 47b and 47c) and has confirmed (104) that the sideflows can contribute to yaw and roll stability (14). This situation is of value to the considerations of Section 3.2.

2.4. Resumé of Section 2.

From the above data it is concluded that

(a) the basic contention that concave undersurfaces allow increased reentry C_L on delta and related wings is correct,

(b) on structural and aerodynamic grounds, concave wings should be examined by theory and experiment to indicate planforms and anhedral distributions, which will further improve C_L per unit L/D, and maximum C_L,

(c) on more general grounds, the promise of reduced rates of heat transfer at stagnation points, beneath the entire heat shield and probably elsewhere as well, justifies detailed examination of concave components or configurations for reentry gliders (such as the Space Shuttle Orbiter), for reusable boosters (3,4) or for vehicles derived from the NASA/USAF Hyper 3, the X-24B or the X-24C.

In the context of this last conclusion, possible applications are outlined in Section 3.

3. CONFIGURATION DESIGN

A feature of lifting reentry gliders is likely to be that, in comparison with hypersonic cruise vehicles, they can be substantially bulkier. This is because (a) the supersonic and hypersonic L/D need not exceed 2±1 (rather than 5±1, say), (b) a bulky cargo and conventional rocket engines together enforce relatively large cross-sectional and base areas and (c) the demand for high volumetric efficiency and, hence, for minimum panel area militates against slender fuselages and large wings. Some NASA designs for reentry gliders are shown in Fig. 48a; even the Hyper 3 is seen to be reasonably bulky and, thus, to allow the introduction of concavity or partial concavity as shown in Fig. 48b. In that figure, vehicles X and Y have the same enclosed volume, base area and cross-sectional area distribution, the same span, length and planform area, the same combinations of planform sweep angles, the same length of main undercarriage legs and the same ground clearance, provided that landing attitudes are identical. On the other hand, the caret version uses a body undersurface of $\omega = 5^o$, which is known from experiment (see Section 2.1 and Fig. 7) to produce at 55^o angle of attack in an airstream of $M_\infty = 12.2$, a lift coefficient some 10 % higher than that of the flat-bottomed body; the heat transfer per unit wetted area of the caret heatshield will, thus, be reduced due to the lower value of V_ℓ/V_∞, which will accompany the increase in shock wave static pressure ratio. Subject

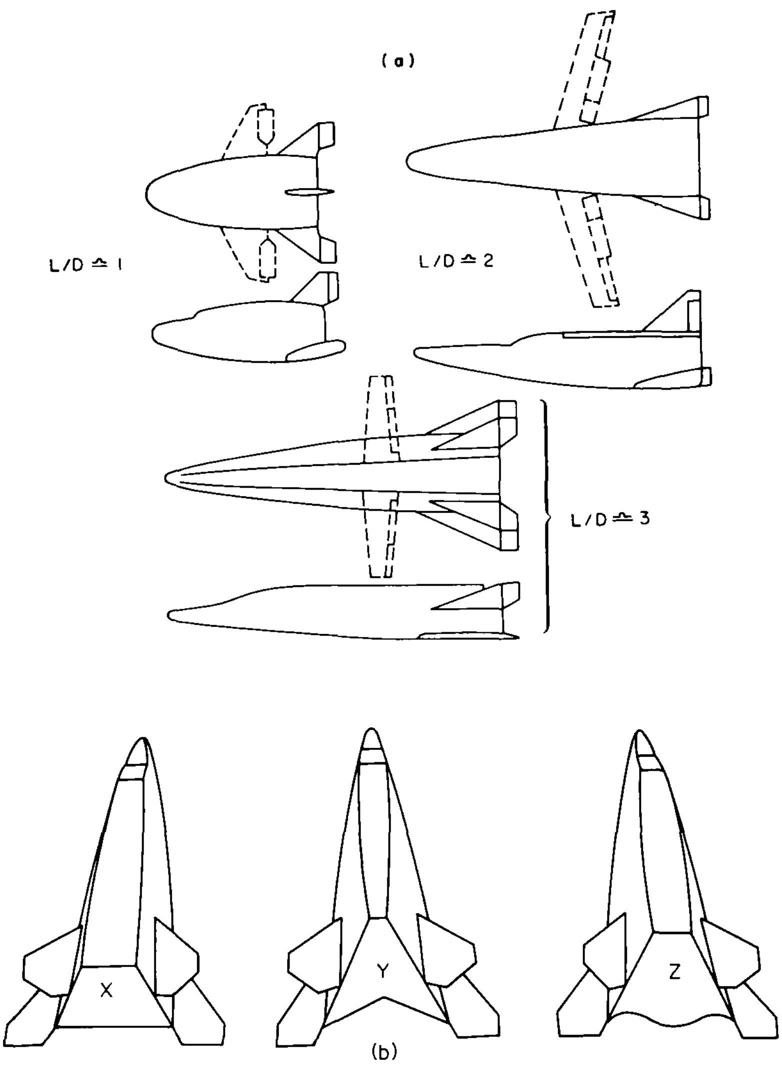


Fig. 48. Lifting reentry gliders (a) NASA designs
(b) introduction of concave undersurfaces.

to radiation between one side and the other, the caret heatshield should, thus, be of lower weight per unit wetted area and, in contrast to one conventional criticism of the caret wing, it is *not* obvious that its greater wetted area necessarily implies a greater total heatshield weight. However, it is clear that for identical wing loadings, heating at stagnation points (and probably on sides and upper surfaces) would be substantially less for the caret vehicle.

It is true that, if vehicle X were designed to accommodate a cargo of given maximum dimensions, then vehicle Y might fail to accommodate that cargo due to the point of the inverted V; the use of partial concavity as on vehicle Z seems likely to rectify this situation.

In Fig. 49 some typical cross-sections of designs for the Space Shuttle orbiter are compared with that of the Hyper 3 and again, if partly concave undersurfaces are incorporated into these basically bulky shapes, the modifications produce smaller changes in geometry than would be needed for shallower vehicles.

3.1. Vehicles derived from the Space Shuttle orbiter

For the Space Shuttle orbiter the design has been dominated by the call for large cargo

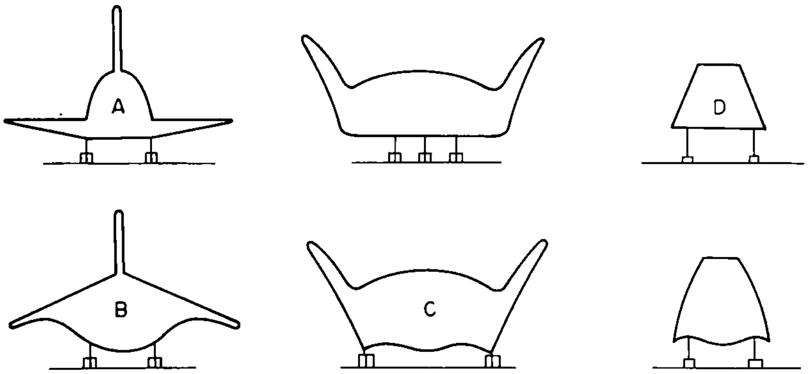


Fig. 49. Orbiter cross-sections.

dimensions and this has led to the use of a large diameter fuselage mounted on a low wing (see vehicle A in Fig. 49). However, a high-wing configuration (see vehicle B) can still contain the cargo and it also permits substantial anhedral without increasing the undercarriage length or reducing ground clearances at given landing attitude (23). In comparison with low-wing configurations the total panel area can actually be reduced and although the proportion allotted to heatshield area will have risen, an improved reentry C_L would moderate heat transfer rates. It was therefore thought worthwhile to consider the design and to conduct wind-tunnel tests on a model of a "high-wing orbiter".

The orbiter selected for modification was the NASA-MSC/040A, which at that time (1972) represented one of the most recent American designs (105), was based on the use of external, expendable tankage and quite closely resembles the NAR orbiter shown in Fig. 1. For this orbiter, the undercarriage was mounted just outboard of the fuselage side and for the high-wing orbiter it was decided (23), that for the given leg length and given wheel positions, the skin and major load-bearing members should remain in the same position so that leg length would not be changed. Also it was decided, that for given landing attitude (17°) and for given

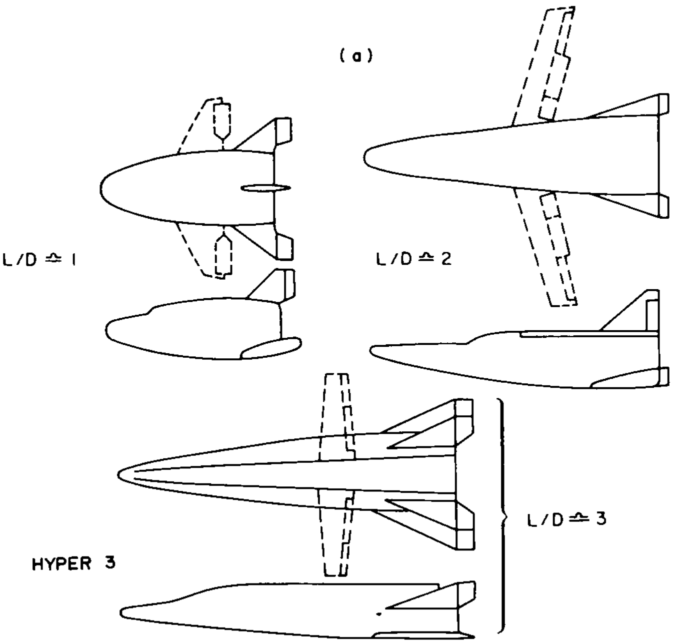


Fig. 50. Orbiter with anhedral.

clearance between the wing tips and the runway, the wing tips of the modified orbiter should be at the same position as with the unmodified 040A. Between the wing-tip and the undercarriage leg concavity was introduced, permitting the partly concave contour shown diagrammatically in Fig. 50. Two principal features emerge (23). Firstly, for given tip clearances substantial concavity is permissible for the high-wing orbiter. Secondly, although the convex belly of the new vehicle will have a reduced ground clearance, stone-damage to the heat shield would only occur from the runway at the end of a flight and direct contact with the ground would only occur in the event of undercarriage failure; thus, the vehicle, as redesigned to accommodate concavity, also provides greater structural depth beneath the cargo and a greater clearance around it. In point of fact, the cross-section is now substantially oversize for the given cargo and the vehicle could be reduced in span and cross-sectional area. Thus, concavity allows a reduction in configuration size at constant cargo size. If equipment were relocated in the volume newly available on either side of the cargo bay, the vehicle might also be reduced in length.

If high- and low-wing orbiters are designed for equal spans, it is found that the total panel areas are approximately the same. Thus in reducing the vehicle size, partial concavity

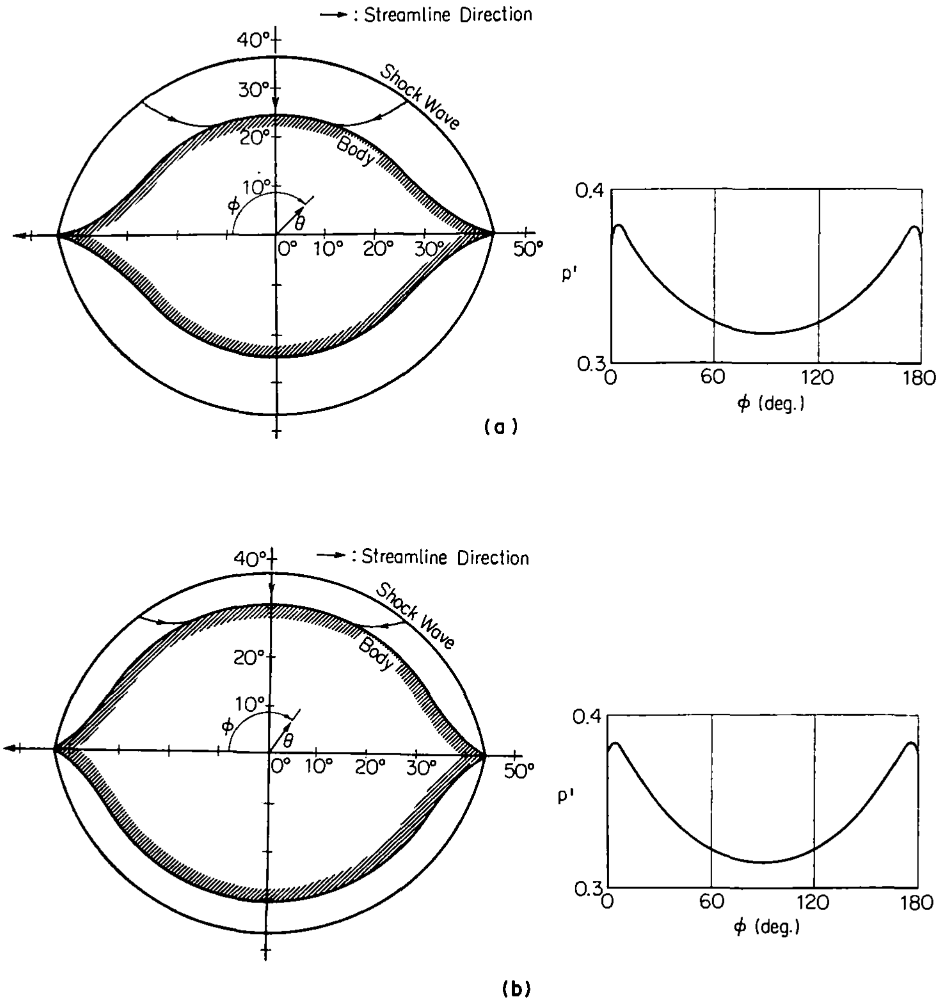


Fig. 51. Examples of flows on "lip-like" cones. (Data due to Tamaki *et al.* (106).) Flow pattern and static pressure distribution for (a) $M_\infty = 2.8$, (b) $M_\infty = 5.0$. Scaled static pressure $p' = (\text{surface static pressure})/\rho_\infty U_\infty^2$.

would permit reductions in total wetted area and in undercarriage length. The vehicle structure cost and weight might be reduced as a result, and the payload fraction would then increase It is possible that for the high-wing orbiter, wing loading may rise due to the fact that the ratio of heatshield wetted area to total wetted area is greater. One of the principal points of Refs (8) and (11), however, is that (at given wing loading and, hence, at given local static pressures) heat transfer per unit wetted area of a concave heatshield should be less since, at given flightspeed, local flow velocities will be lower; thus, in raising C_L, concavity can be permitted to raise W/S without necessarily destroying all the advantage.

Further possibilities suggested by Fig. 50 are, that (a) flow containment may be more pronounced at the rear end of the vehicle so that a far-aft centre of pressure may occur during reentry and (b) the proposed "flared" cross-sections at the forward end may contribute a stabilising rolling moment during reentry; also (c) the theoretical work of Tamaki *et al.* (106) suggests that the flared cross-sections might be formed as the "lip-like" cones of Fig. 51 and so might support an attached shock wave at lower angles of attack.

In view of the above a high- and low-wing orbiter (HWO and LWO) were designed; firstly, to determine the extent to which reentry $C_{L_{max}}$ and C_L per unit L/D might be improved by partly concave undersurfaces and secondly, to check that at subsonic speeds well-behaved vortices will

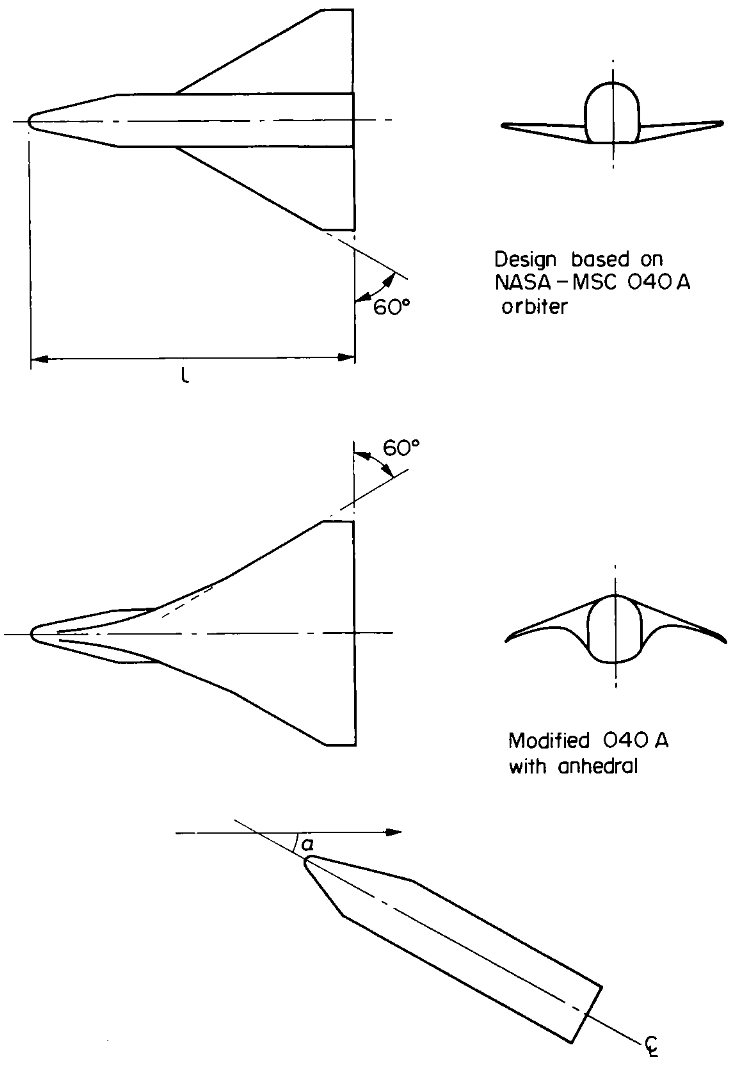


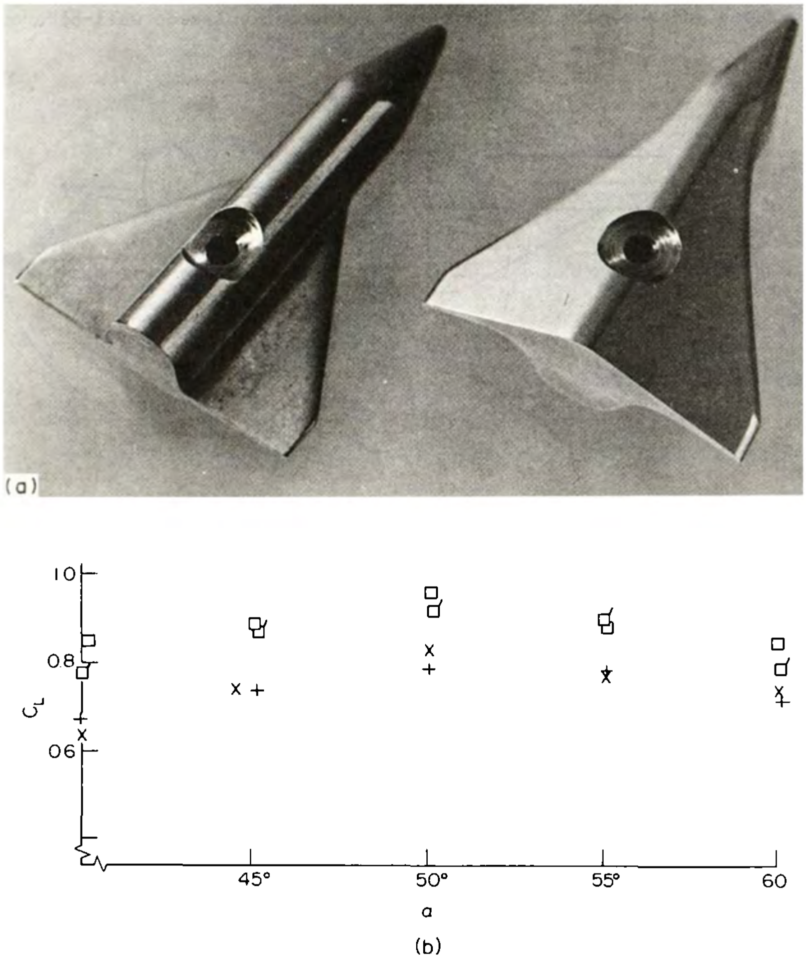
Fig. 52. Design details for orbiter models.

form at the leading edges and so will allow controllable operation up to higher angles of attack than would be permissible for a wing with attached flow (107).

The models were shaped as shown in Fig. 52, the value of ℓ being 4 cm for the hypersonic tests (see Section 3.1.1.) and 27.9 cm and 90 cm for the low speed tests (see Section 3.1.2.).

3.1.1. Performance at reentry conditions

Figure 53a shows the two models tested at hypersonic speeds by Davies and Townsend (24) (M_∞ = 8.4, R_e = 3 × 10^5) and by East (25) (M_∞ = 9.7, R_e = 4 × 10^5). Both sets of tests were performed in the AASU Gun Tunnel at Southampton University, but at different times (1972 and 1976) and using different balances. Davies and Townsend measured C_L values only and their data are compared with those of East in Fig. 53b. It is seen that quite close agreement exist and that the HWO produces significantly greater lift than the LWO at given angle of attack. Figure 54 presents all the data obtained by East and shows that for angles of attack between 40 and 50^o, and at Mach and Reynolds numbers of 9.7 and 4 × 10^5 the HWO produces some 20 % more lift than the LWO.



(b)

Fig. 53. Tests on orbiter models. (Data due to Townsend and Davies (24) and East (25).)
(a) Low-wing (LWO) and high-wing (HWO) orbiter models.
(b) Variation of lift-coefficient with angle of attack.
□ HWO, + LWO, M_∞ = 9.7, R_e = 4 × 10^5. □ HWO, x LWO, M_∞ = 8.4, R_e = 3 × 10^5.

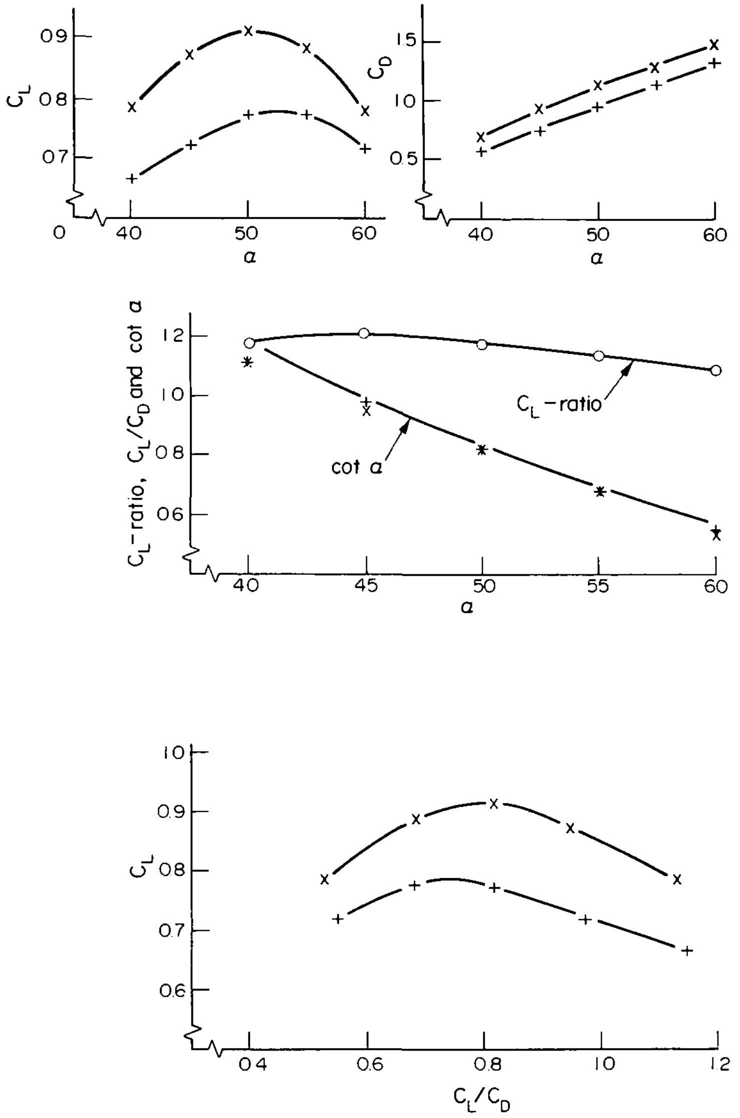


Fig. 54. Force measurements and C_L ~ ratio for orbiters at $M_\infty = 9.7$ and $R_e = 4 \times 10^5$. (Data due to East (25).) x high wing orbiter, + low wing orbiter, O ratio of HWO C_L to LWO C_L.

Flow photographs are shown in Figs 55-57. In the first the planview flows for $\alpha = 40^o$ demonstrate, that the wing shock of the HWO lies perceptibly closer to the wing leading edge and so confirms that spillage is probably reduced. Also, the flows for $\alpha = 50^o$ (see Fig. 56) show a sudden increase in the HWO shock angle, which must result from flow containment and explains the improvement in C_L shown in Fig. 53.

In Fig. 57 ($M_\infty = 9.7$) the sudden change in shock strength is still observed for the HWO and is seen to move upstream as angle of attack is increased. This shockwave behaviour is consistent with that on simpler shapes (see Fig. 7), but since the orbiter derives substantial lift from the forward fuselage, the downstream flow containment due to anhedral on the wing could produce rearward shift in aerodynamic centre. The comparative performance of the LWO and HWO has been examined by East; the aerodynamic centre on the HWO was solve 10 % of total length further aft than on the LWO, but for both models and for angles of attack between 40 and 60^o the position of the aerodynamic centre was almost invariant with attitude.

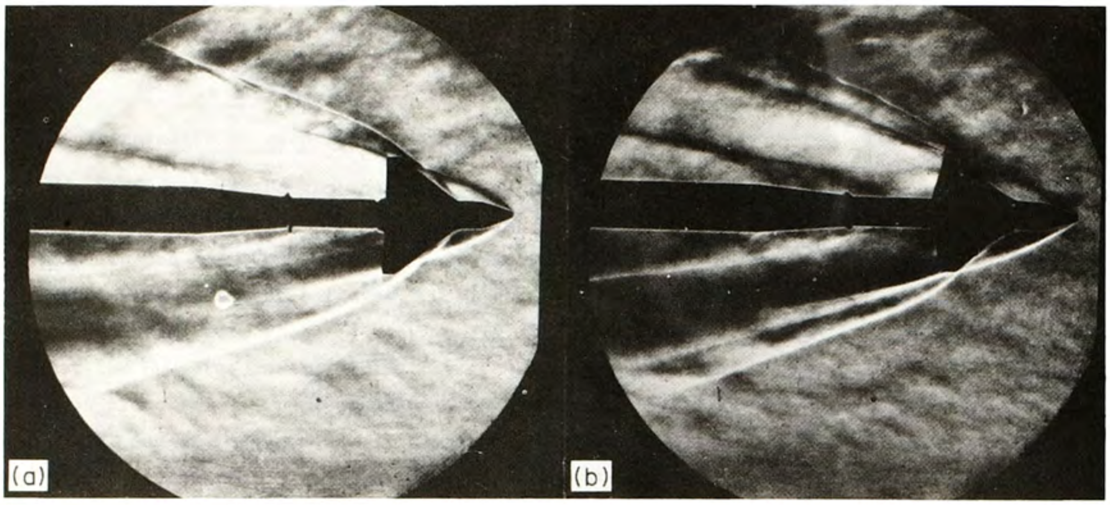


Fig. 55. Schlieren photographs of hypersonic flow over (a) low- and (b) high-wing orbiters. $M_\infty = 8.3$, $\alpha = 40^o$. (Data due to Townsend and Davies (24).)

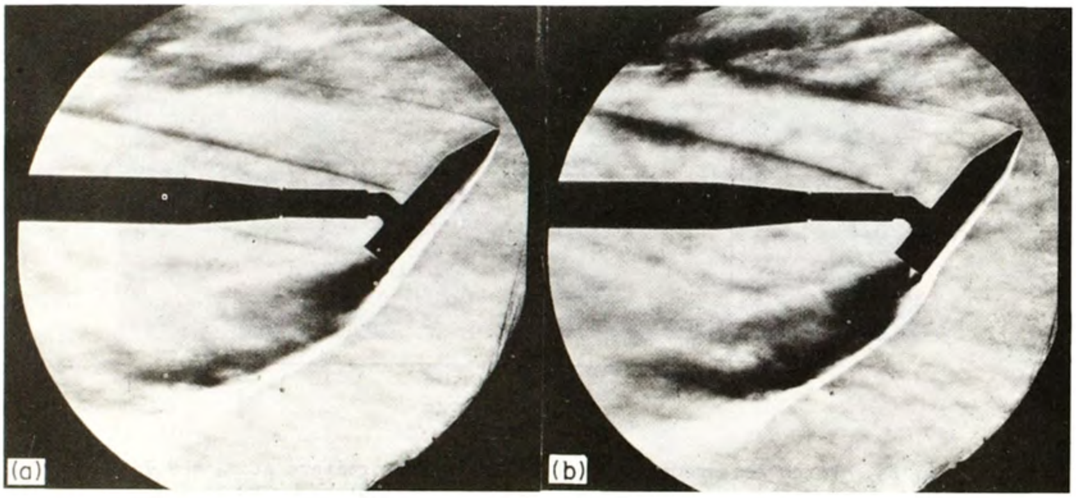


Fig. 56. As Fig. 55, but $\alpha = 50^o$.

3.1.2. Performance at subsonic speeds

The models used in low speed tests at small scale (ℓ = 27.9 cm) are shown in Fig. 58; of these the shape selected for tests at higher Reynolds number (ℓ = 90 cm) and to investigate performance in ground effect was geometrically identical with the HWO model in the hypersonic tests. The main purpose of these experiments was to establish (a) that vortex flows would form in a well behaved manner above the wing and (b) that lift, drag and pitching moment characteristics would demonstrate stall-free operation and no aerodynamic irregularities at angles of attack up to 30^o or more in free air and about 15^o in ground effect. Figure 59 presents data obtained at Surrey University (26) and demonstrates that vortex flows were retained up to a fairly high angle of attack. Figure 60 shows that aerodynamic characteristics seem conventional (108,109) and offer a basis upon which detailed aerodynamic refinement should allow

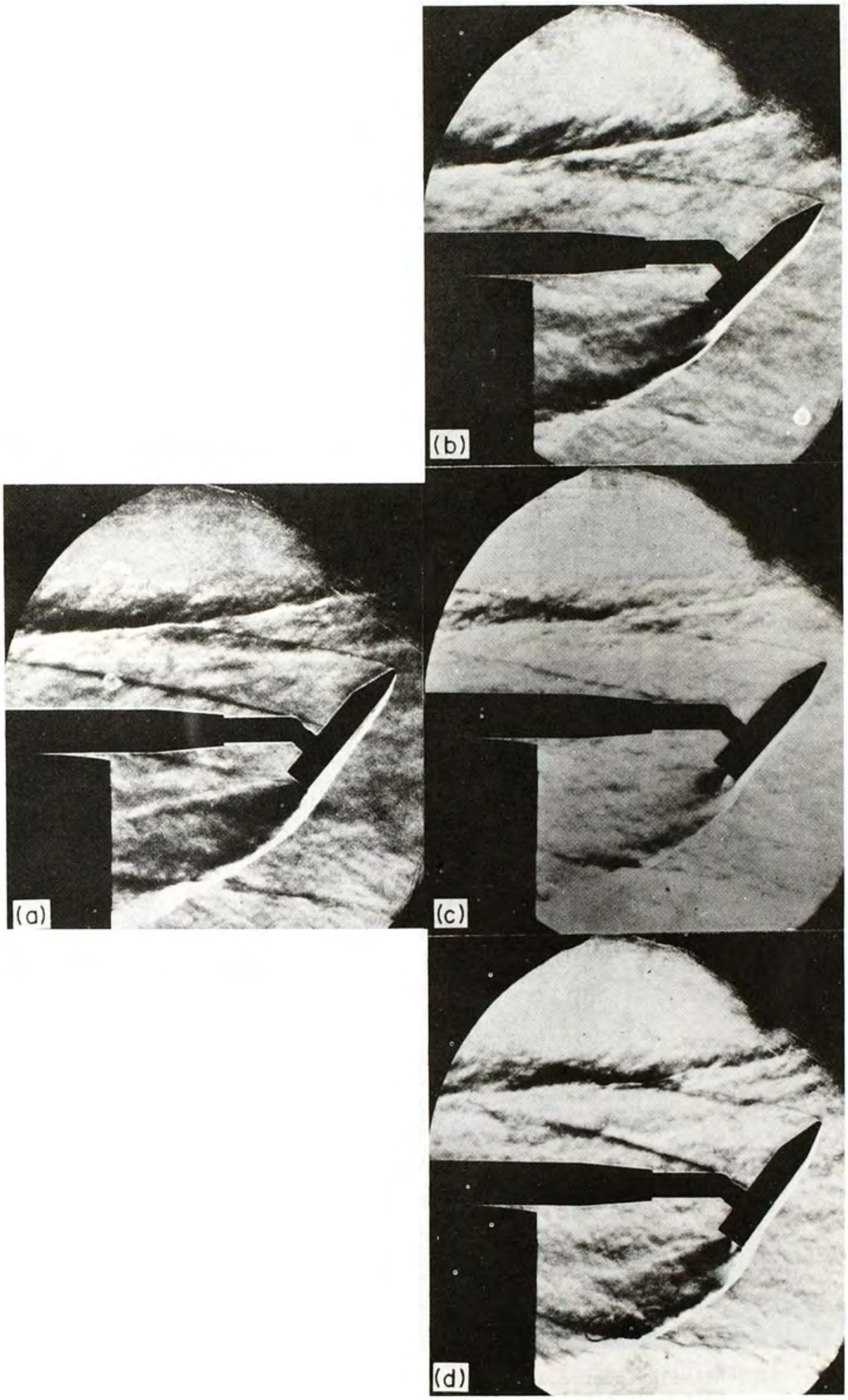


Fig. 57. Schlieren photographs of hypersonic flow over low- and high-wing orbiters at $M_\infty = 9.7$. (Data due to East (25).)
(a) LWO, $\alpha = 50^o$. (b) HWO, $\alpha = 45^o$, (c) HWO, $\alpha = 50^o$, (d) HWO, $\alpha = 55^o$.

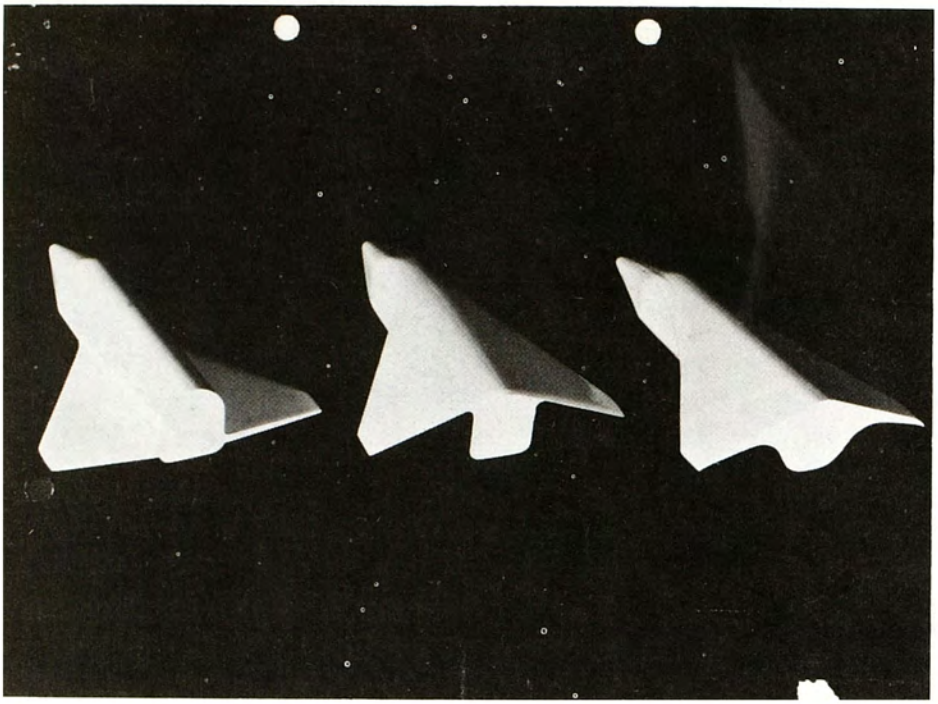


Fig. 58. Orbiter models for low-speed tests.

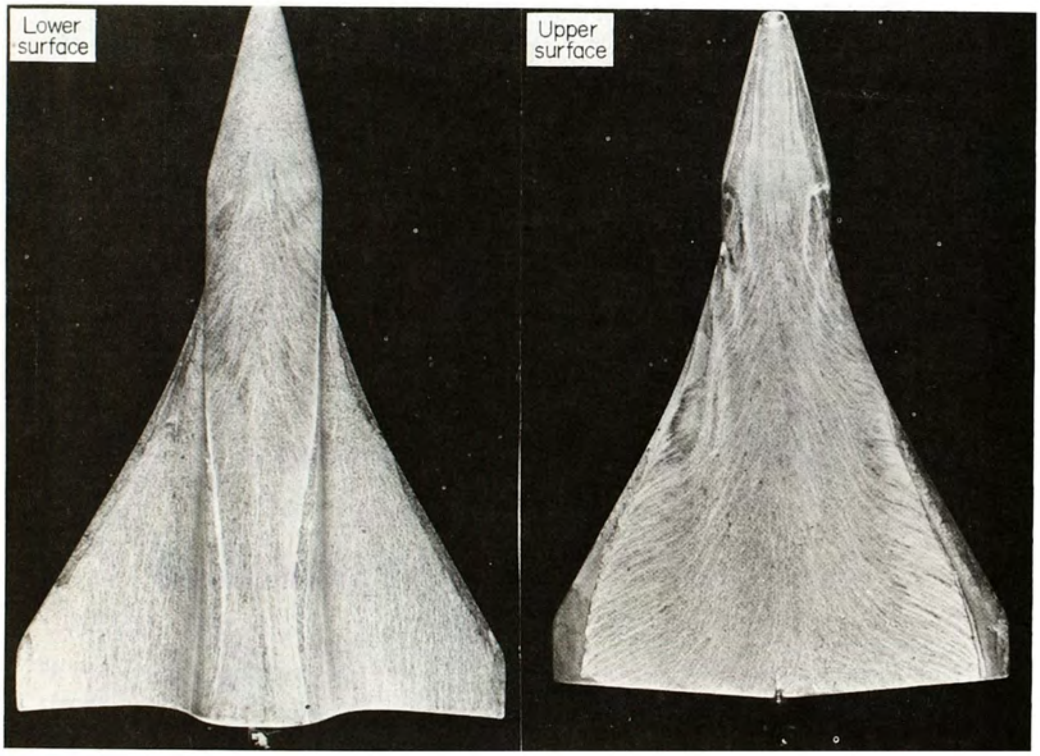


Fig. 59. Oil flow photographs at $\alpha = 27.5^{\circ}$ for model ($\ell = 27.9$ cm) of high-wing orbiter. (Data due to Edwards and Davies (26).)

more than acceptable low speed performance. In particular there is a gradual increase in C_L with angle of attack up to some 34°, with the prospect that high g, stall-free terminal manouevres may be permissible prior to a conventional landing.

In Fig. 61, data from tests at Southampton University (26) (uncorrected for the effect

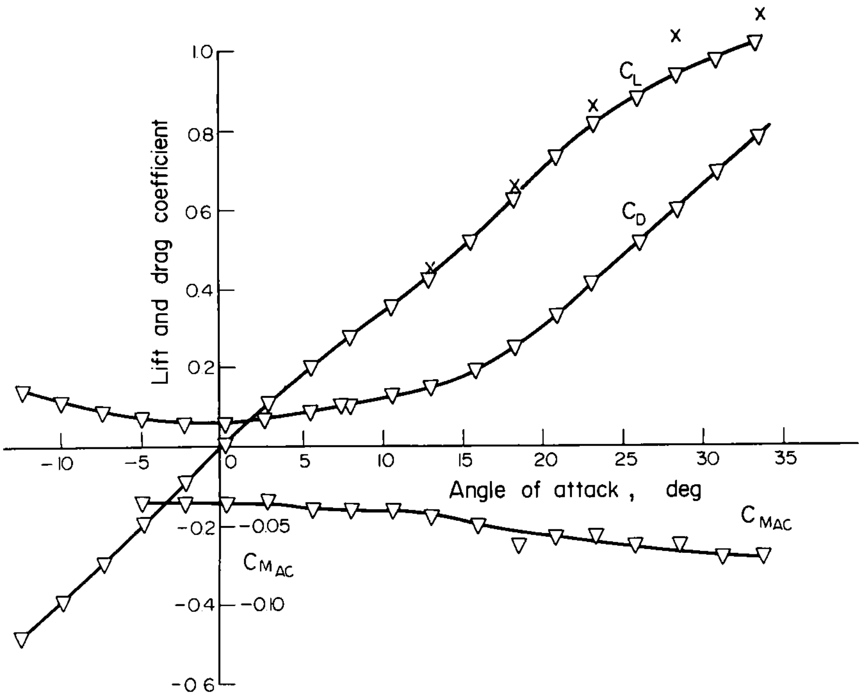


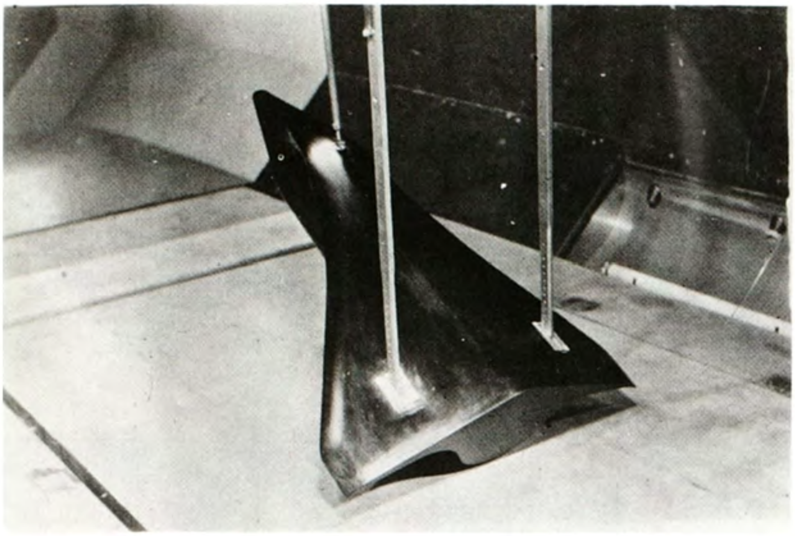
Fig. 60. Lift, drag and pitching moment about the aerodynamic centre for high-wing orbiter model (ℓ = 27.9 cm) at subsonic speeds. (Data due to Edwards and Davies (26).) ∇ Smooth model. x Model with sand roughened leading edge.

of wake constraint) show that ground effect on the HWO substantially increases the lift generated at angles of attack up to about 15°. At 12½° the HWO sustains a lift increase of about 30 %; wind-tunnel data and flight test data obtained on the Concorde prototypes have shown (110), that at 12½° angle of attack the Concorde gains nearly 50 % in lift as a consequence of ground effect and it may be that aerodynamic refinement of the HWO should aim at improving this aspect of its performance.

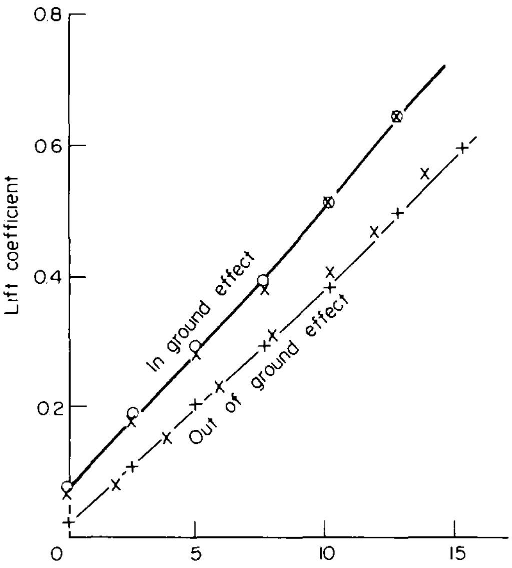
The measured values of C_L/C_D are shown in Fig. 62. Those obtained at the smaller scale peak at a lower value than those for the large HWO model, for which the influence of ground effect is slightly deleterious at angles of attack which correspond to landing conditions.

It is emphasised that the above tests were performed to demonstrate that desirable types of flow could be supported by the HWO, rather than to attempt the achievement of competitive performance levels. However, data on the 040A orbiter at low speeds and in and out of ground effect are available (111-113); as shown in Fig. 61b, the HWO produces about the same variation of C_L (with angle of attack and out of ground effect) as does the 040A orbiter, as tested by Romero and Chambliss (112).

Finally, tests on the lateral characteristics of the HWO were not conducted, but the behaviour of several delta and near-delta wings, all with pronounced anhedral on the upper surface (see Fig. 63a), have been investigated by Keating and Mayne (114). For wing E measurements of yawing and rolling moment are plotted as coefficients in Fig. 63b; the corresponding lateral stability derivatives ℓ_v and n_v are also shown and suggest no great handling difficulties for typical landing conditions (102). Indeed, the stability and fin effectiveness of all the wings tested were judged (114) quite adequate by the standards of slender wing designs of the late 1960s. It seems reasonable to suppose that this would also be true for the HWO as tested here.



(a)

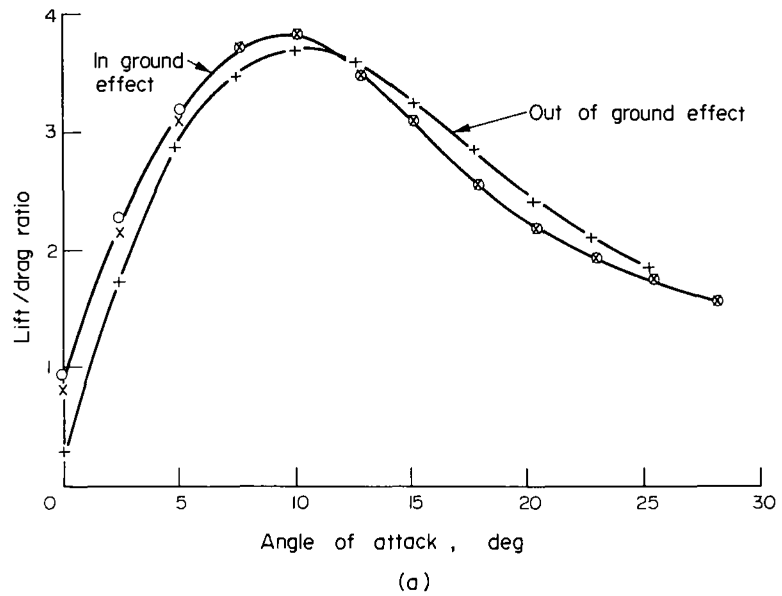


(b)

Fig. 61. Lift characteristics of high-wing orbiter model (ℓ = 90 cm) in and out of ground effect at subsonic speeds. (Data due to Edwards *et al.* (26).)
(a) Model mounted in AASU 7 ft × 5.5 ft (2.1 × 1.7 m) low speed tunnel.
(b) Orbiter lift in and out of ground effect.
Upper Curve: HWO in ground effect, O zero belt speed, x belt moving.
Lower curve: + HWO out of ground effect (26), x 040A (see Fig. 52) out of ground effect (112).

3.2. Orbiters of more advanced aerodynamic performance

The application of concave undersurfaces and aerodynamically sharp leading edges to variants of the Space Shuttle orbiter has been shown to offer aerodynamic merits at hypersonic and subsonic speeds, but the resulting shape does not produce known flows; it would, therefore, require elaborate theory and much experimental work before it could be proved viable throughout



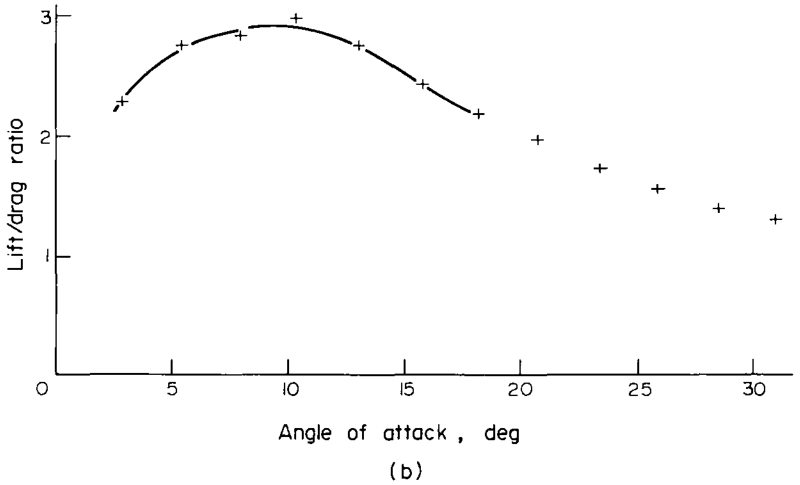


Fig. 62. Lift-drag ratio of high-wing orbiter models at subsonic speeds. (Data due to Edwards *et al.* (26).)
(a) 90 cm (3 ft) model in and out of ground effect. Ground effect: O zero belt speed, x belt moving, + out of ground effect.
(b) + 27.9 cm (11 in.) model, out of ground effect.

the speed range. In an attempt to use the shapes which, as already described in Section 2., have been designed by existing theoretical methods and proved by experiment to be aerodynamically attractive, the present Section considers the design of vehicles derived from such shapes as those of Fig. 48; these lend themselves more easily to the "known flow" design method and incidentally provide a basis for higher performance at reentry conditions and probably during the post-reentry glide.

Two major design objectives are added to the requirements that reentry aerodynamics should be relatively simpler and reentry performance at least as high as that of the Space Shuttle orbiter. The first is that considerations of high speed stability should be permited to influence the basic body shape, preferably allowing a reduction in the wetted area of fins and tailplanes and possibly an increase in useful volume, as compared with the Hyper 3.

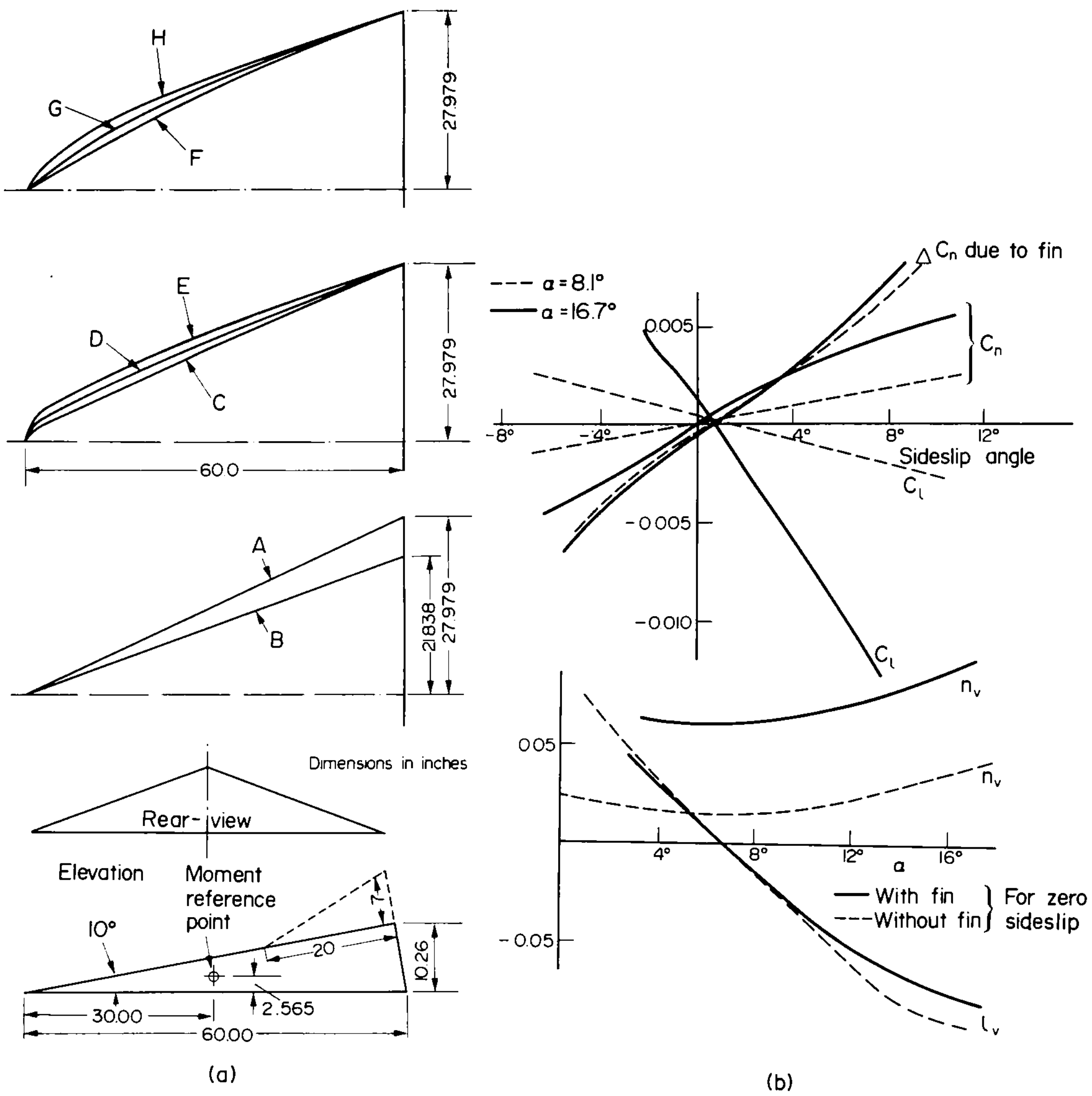


Fig. 63. Experiments on the lateral characteristics of delta and near-delta wings at subsonic speeds. (Data due to Keating and Mayne (114).)
(a) Wind tunnel models.
(b) Wind tunnel measurements for model E, (s/ℓ = 0.466, planform apex radius 4.224 in. (10.3 cm)).

The second is that the vehicle should be able to fly well at low speeds and be capable of conventional landings, preferably without a change in its geometry and thus without resort to high lift devices such as the switch-blade wings of the Hyper 3. It is suggested below that both objectives may be achievable by suitable design of the vehicle sides, that the influence of side flows should be included as part of the undersurface design exercise, and that the ability to do so may be eased by the use of a concave undersurface.

A concave undersurface with an attached shock wave encloses an undersurface flow, which is entirely distinct and isolated from the remainder of the flow. Thus, if the leading edges are aerodynamically sharp and the vehicle sides are inclined as with vehicle C in Fig. 49, they can support attached and uniform flows as discharged by the plane oblique shockwaves from swept wedges. Even if sideflow Reynolds numbers are low, much existing data will be applicable and the effects of side forces, for example on roll stability, will be calculable.

If the demand for attached shock waves is relaxed (and from both heating and stability considerations this may be desirable) it will, nonetheless, be possible to regard the flow as "known", because the thin shock layer solutions of Hillier can include the flows over side walls (see Fig. 47a) and will, in certain cases, predict the forces and pressure distributions upon them. Thus, Hillier can design the undersurface and sidewall as a single unit, provided only that the shock wave remains attached to the planform apex. Even if the vehicle sides are to be inclined, as for vehicle D in Fig. 49, they may still support attached flows and so permit simple design studies to be performed on vehicle stability and control. Experimental evidence of attached flows on such surfaces at reentry conditions is available, especially relevant data being found in the work of Allegre and Bisch (115) (nitrogen, $M_\infty = 18$, $R_{el} = 6.3 \times 10^5$), Stollery (116) (air, $M_\infty = 14.8$, $R_{el} = 1.1 \times 10^5$) and Fitzhugh (117) (air, $M_\infty = 15.1$). These data are selected because they were obtained in nitrogen or air at Reynolds numbers, which are roughly correct for reentry at realistic wing loadings, at Mach numbers where the heating problems are worst, for surfaces at negative angle of attack and with several leading edges and/or surface shapes.

Allegre and Bisch demonstrated the onset of separation above a nearly sharp flat plate as the angle of attack changed from 0 to -6°, but also showed that attached flow could be regained by very slightly blunting the leading edge. Even with highly conductive leading edge material (11,12) a lifting reentry vehicle will still require slight blunting of the edges, especially with attached flows on both sides. It seems that slight blunting could be of value to side-flow aerodynamics as well as on material grounds.

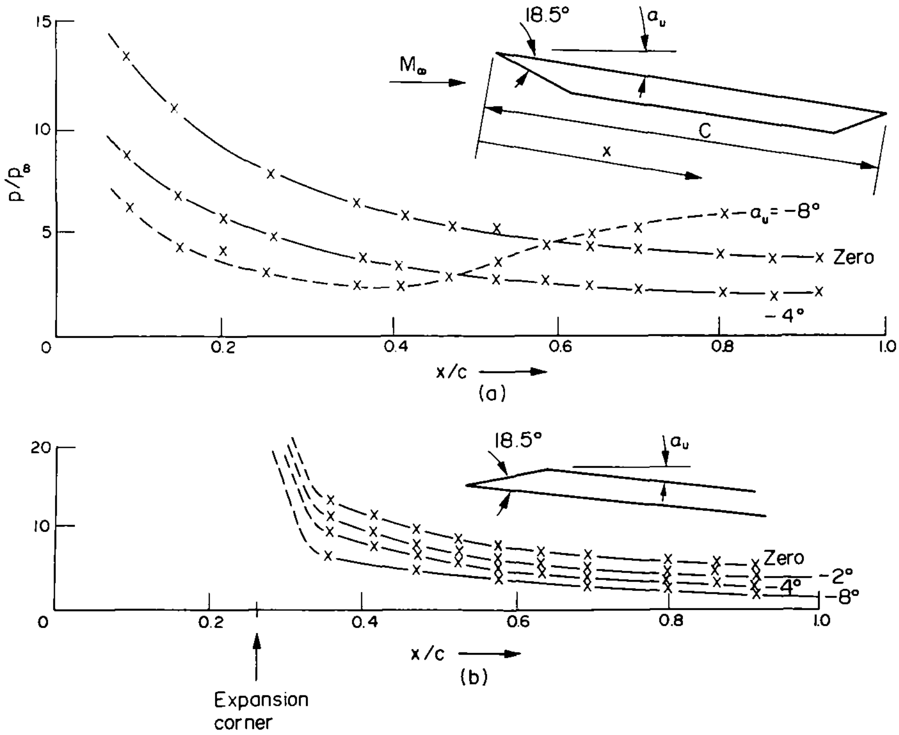


Fig. 64. Pressure measurements on the upper surface of (a) a flat plate at incidence and (b) a flat plate at incidence with the leading edge drooped by 18.5°. $M_\infty = 14.8$, $R_e = 1.1 \times 10^5$. (Data due to Stollery (116).)

Work at $M_\infty = 14.8$ by Stollery (116) has demonstrated another means for delaying separation on surfaces with negative angle of attack down to -8°. Separation from a flat plate started at -4° and by -8° had encroached on much of the surface. However, when the leading edge region was drooped by 18.5° (giving an intial angle of attack of $+ 10.5^\circ$) the entire flow remained attached (see Fig. 64). At main surface angles below -8° the flow started to separate near the trailing edge.

From the above, it is concluded that flow containment on lower surfaces may improve the chance of obtaining uniform heating patterns on sides and upper surfaces, and sufficiently simple flows to allow their design by available theory. Surfaces to provide control and stability can thus be designed from the start. In particular, Fitzhugh (117) has tested a surface at initial incidence of -6° and concave curvature through 28°. In spite of the negative initial incidence and subsequent compression, the flow remained attached and initial heating rates were correctly predicted by Cheng's theory, which has since been extended (118) to account for curvatures.

For those flows which assume attached shock waves structural credibility may in part depend upon the practicality of building "aerodynamically sharp" leading edges, which not only survive the fiercest regions of reentry, but resist prolonged heating of somewhat reduced intensity at post-reentry conditions and can do so without frequent refurbishing. Estimates of temperatures likely to be reached during reentry by solid graphite leading edges with internal conduction and surface radiation were made by Nonweiler (119) and presented in Ref. (8), together with those predicted by Capey (120) for post-reentry conditions (see Fig. 65). For reentry Nonweiler (119) showed that for a graphite leading edge about 7.5 cm thick and for $W/S = 25$ lb/ft^2 (123 kg/m^2), high C_L and high M_∞ the equilibrium leading edge temperatures would be

$$\left.\begin{array}{l}\text{about 1500 K at } \Lambda = 0^\circ \\ \text{about 1270 K at } \Lambda \simeq 70^\circ\end{array}\right\} \text{ at the worst case, i.e. } V/\sqrt{r_o g} = 0.8.$$

The variation of these temperatures in this region is shown in Fig. 65, but doubling the wing loading would have raised them by only about 100 K; thus, the use of graphite leading edges seems plausible even if wing loadings exceed 50 lb/ft^2 (226 kg/m^2). For the Shuttle orbiter reentry wing loading varies from about 55 lb/ft^2 to just over 70 lb/ft^2 (about 250 kg/m^2 to nearly 320 kg/m^2); by comparison, the Dynasoar X-20 A was to have had (8) a wing loading of just over 15 lb/ft^2 (about 73 kg/m^2), whereas in early Shuttle studies, wing loadings of nearly 90 lb/ft^2 (440 kg/m^2) were considered by Masaki and Yakura (121). Within these fairly wide limits on wing loading, aerodynamically sharp leading edges can probably be regarded as practicable provided that conductive or active cooling is accepted as a technique, that the weight penalties are also acceptable, that the temperature predictions are not invalidated by real gas effects on viscosity estimates (119,122), and that it is recognised that temperature limits may be as restrictive during the post-reentry glide as they are during reentry itself (see Fig. 65). Evaluation of the influence of these and other factors on cross-range performance will be greatly simplified by the recent solution of the reentry trajectory equations by Busemann *et al.* (123). The design of a lifting reentry vehicle having partially concave undersurfaces, stabilising sidewalls and aerodynamically sharp leading edges could be as shown in Fig. 66. Operation could be as follows. As reentry proceeds towards and through peak heating ($M_\infty \simeq 17$-21 say), the vehicle achieves partial flow containment and enhanced C_L, although the shock is detached from the leading edges (see Fig. 67a). The vehicle may also be pitch stable (if Hui and Hemdan's result for flat delta wings still holds for concave undersurfaces), but the main criterion is that heat transfer should be minimised. Roll and yaw stability due to the sideflows can be significant according to Hillier (104). The upper-

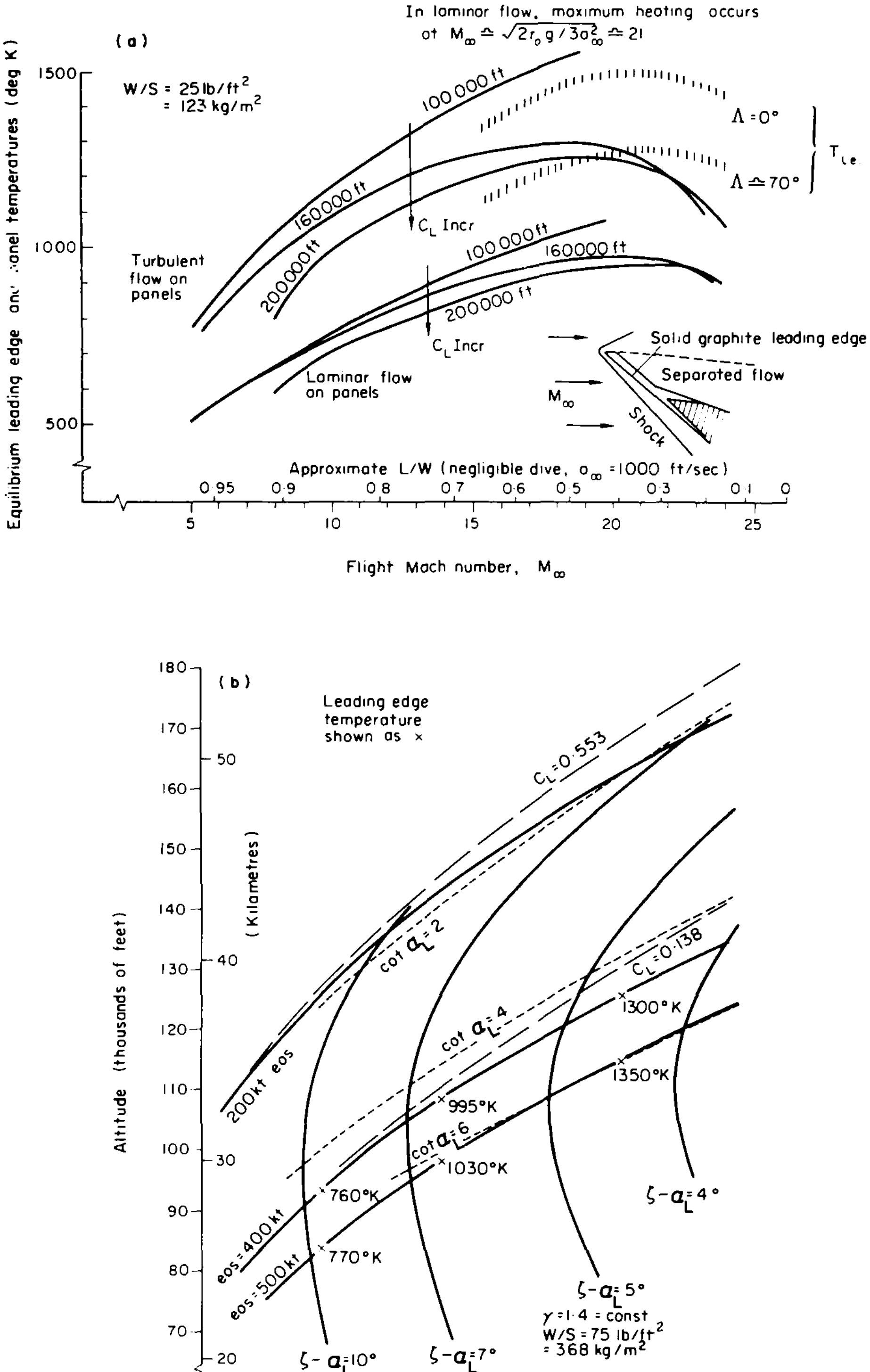


Fig. 65. Leading edge and panel temperatures ($5 < M_\infty < 25$). (Data due to Nonweiler (119) and Capey (120).)
(a) Leading edge and panel temperatures for emissivity = 0.8.
(b) Flight conditions for Nonweiler wings operating at design conditions and constant anhedral and root chord.

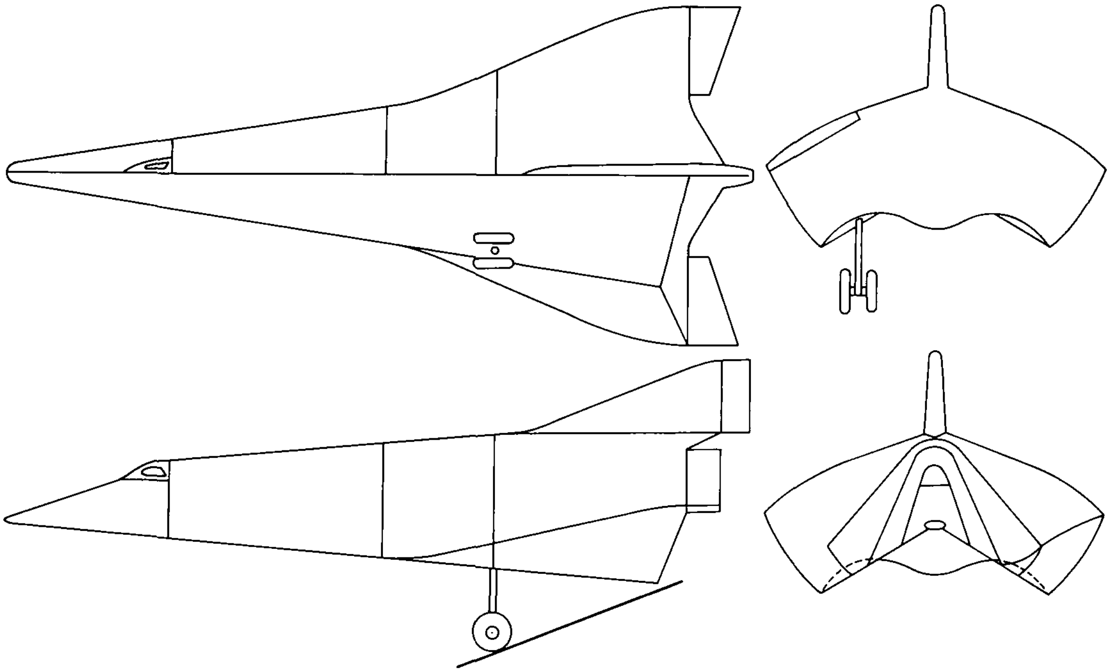


Fig. 66. High cross-range reentry glider based on waverider principles.

surface flows will presumably combine with the base flows and lead to relatively unimportant forces and heating rates. With the availability of thin shock-layer solutions due to Hillier, Hui and Squire, the undersurface and sidewalls can be designed and studied by computation, wind tunnel tests being confined to a small range of carefully selected models.

When the peak heating condition is past, the vehicle angle of attack is reduced in order to increase lift/drag ratio and the cross-range ability (see Fig. 67b). Even if the vehicle is operating at maximum reentry wing loading some bank angle may be applied at this stage. With shocks now attached to the leading edges the undersurfaces flow is fully contained, but quite large excursions in angle of attack will be permissible without shock detachment; thus, for at least some geometries, the sideflows will form as basically two dimensional flows behind swept, oblique shock waves and will contribute to roll and yaw stability. The fin on the upper surface will not yet be effective (to judge from data due to Arrington and Stone (124)), but the upper surface separated flows will probably be complex (55,125-129) and heat loads significant; aerodynamic forces will probably be small, due partly to the avoidance of geometric complications (such as a large fuselage above the wing which can, with at least some configurations (55), cause a considerable increase in the pressures in the region of a shock induced separation). The above situation could well persist throughout the remainder of hypersonic flight and the thin shock-layer theories of Hillier, Hui and Roe should once again permit sufficient computation for wind tunnel tests to be restricted in number or to be concentrated on understanding the upper surface flows.

At high supersonic Mach numbers, full flow containment should still be possible (see Fig. 67c); in consequence C_L enhancement and stability from simple sideflows should still be available, although at moderate angles of attack ($\alpha_L = 20^o$ say) the flow may become attached to the upper surface, subject to disturbance from the cockpit and the growth of vortices from leading edges.

At lower supersonic Mach numbers (see Fig. 67d), the undersurface shock wave will finally detach from the leading edges (although not necessarily eliminating C_L enhancement), the

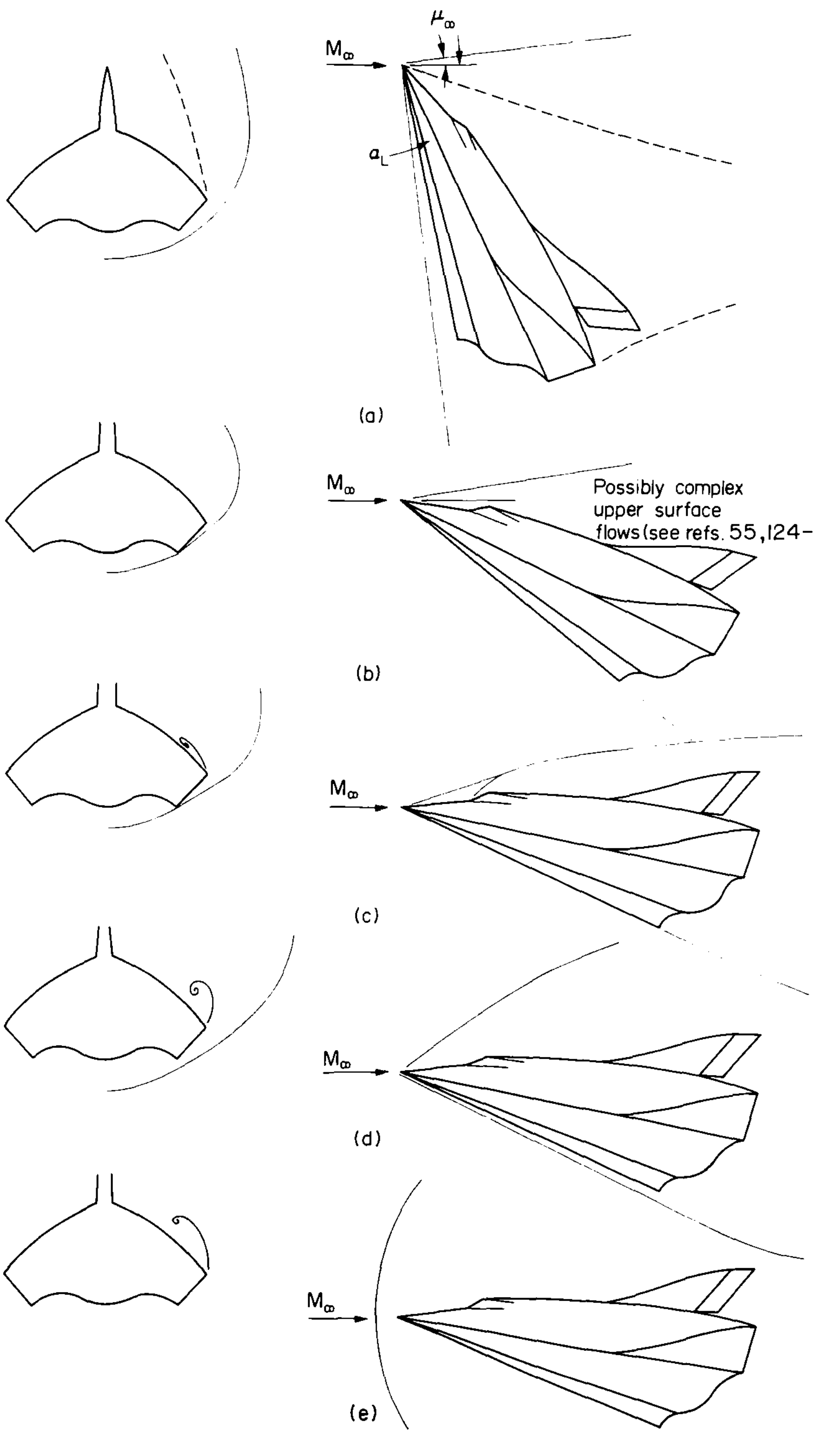


Fig. 67. Possible flow regimes for waverider reentry glider.
(a) Partial flow containment, shock attached to apex. $M_\infty \doteq 20 \pm 5$, $\alpha_L \doteq 50\text{-}60$, $L/W < 0.7$.
(b) Full flow containment — all shocks attached to leading edges. Possibly complex upper surface flows (see Refs 55, 124-129). $M_\infty \doteq 10 \pm 5$, $\alpha_L = 25\text{-}35^\circ$, $L/W > 0.7$.
(c) Full flow containment — all shocks attached to leading edges. $M_\infty \doteq 4 \pm 1$, $\alpha_L \doteq 15\text{-}25^\circ$, $L/W > 0.95$.
(d) Partial flow containment, shock attached to apex. $M_\infty \doteq 2.25 \pm 0.75$, $\alpha_L \doteq 15\text{-}25^\circ$.
(e) Nearly normal detached shock. $M_\infty = 1.25 \pm 0.25$, $\alpha_L \doteq 10\text{-}20^\circ$.

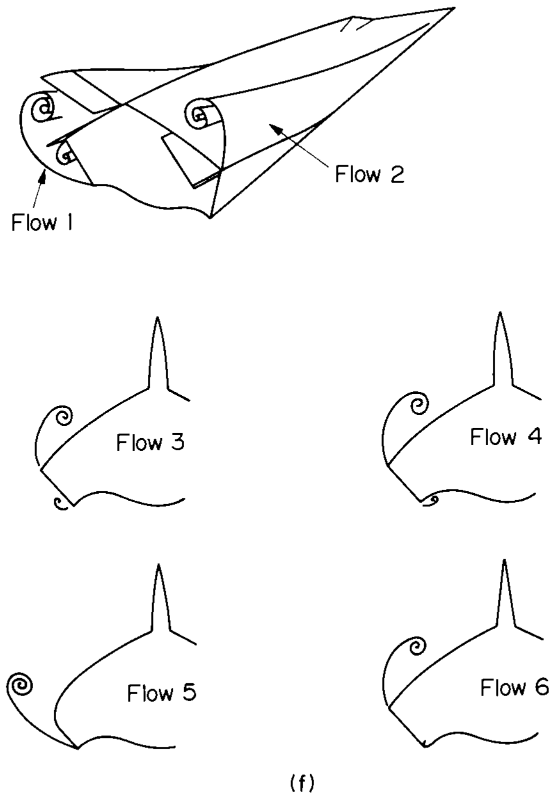


Fig. 67. (f) Possible variants for $M_\infty \doteq 0.25$, $\alpha_L \doteq 10\text{-}30^\circ$.

lower surface flow will merge with the sideflows (preferably without separation from the lower leading edges) and strong vortices may now feature above the upper surface; however, the associated flows may be far from conical, even for pure delta planforms (129).

These upper surface vortices should survive through the transonic region (at which the relatively simple vehicle configuration would probably perform in a placid manner (see Fig. 67e and Refs (8, 130 and 131)). Again, at subsonic speeds the vehicle may operate as a slender wing (see Fig. 67f Flows 1 and 2), though there could be complications due to possible secondary separations (see Flows 3 and 4). Should these secondary separations prove troublesome, it is possible that the vehicle cross-section could be modified so as to produce Flow 5 or 6; however, the first might complicate upper surface flows at higher speeds and the second could prevent full containment of the undersurface flow.

With vehicles such as that of Figs 66 and 67, there could also be stability problems due to the depth of the vehicle at and aft of the cockpit region, and unconventional control characteristics due to the elevons being mounted above a deep base area. However, by comparison with the Hypersonic Research Aircraft (see Fig. 68 and Ref. (132)) and the X-24C (see Fig. 69 and Ref. (133)), there seems no reason to suppose that low speed problems would be worse and, at higher speeds, the features noted in preceding paragraphs could offer at least comparable performance at the same flight speeds ($M_\infty < 7.5$ approximately).

By comparison with the Space Shuttle orbiter, the vehicle considered above should offer improved cross-range and/or higher lift coefficients at reentry and post-reentry conditions; the latter may increase the prospect of using a radiative heat shield. Also the basic shape incorporates surfaces intended to increase reentry stability, so that the vehicle could be

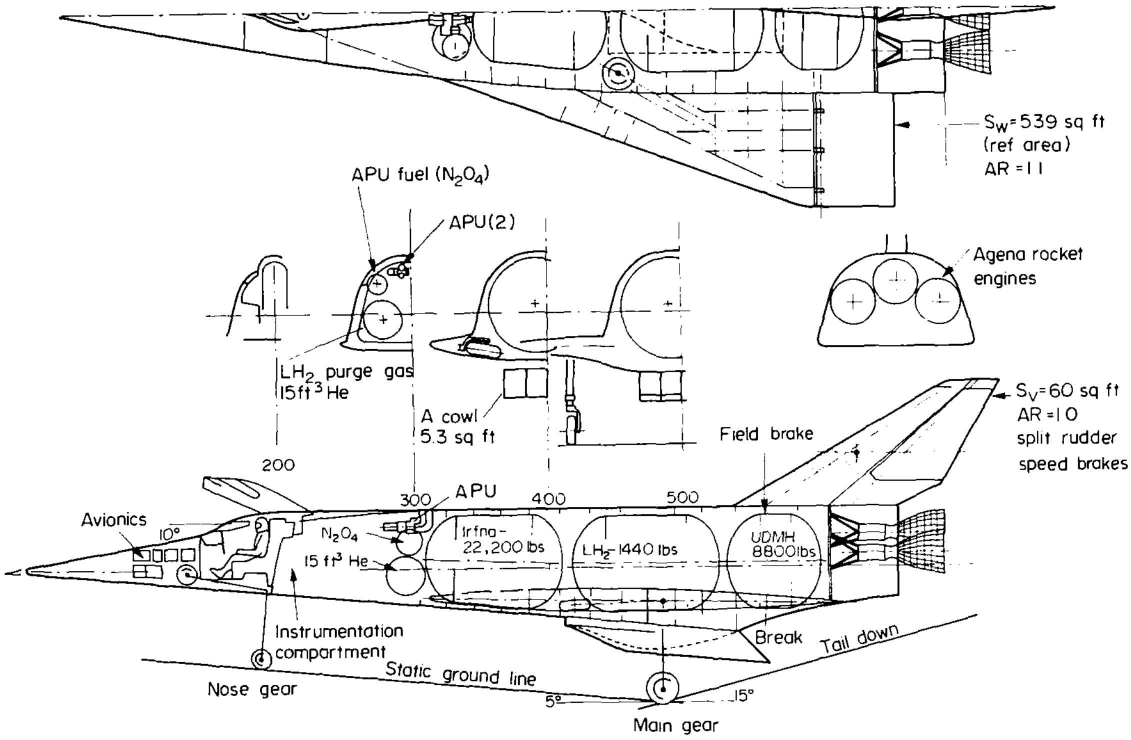


Fig. 68. Hypersonic research aeroplane — Rockwell International configuration (132).

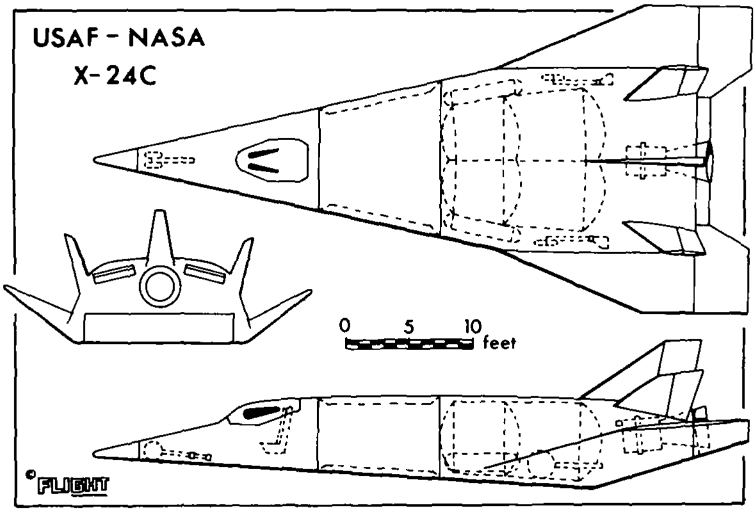


Fig. 69. The USAF-NASA X-24C (133).

safer than if it were reliant upon stability augmentation systems; finally, the positioning of these stabilising sidewalls allows the vehicle significant additional volume. Both structurally and aerodynamically this type of vehicle design seems to merit further study.

3.3. A reusable external tank for the Space Shuttle system

It may be recalled from Section 1. and Fig. 2 that fully reusable Space Shuttle systems were examined in the early stages of the Shuttle studies and more recently (1976) reexamined by Rockwell International (3) (see Fig. 2b). With the existing Shuttle system, the solid

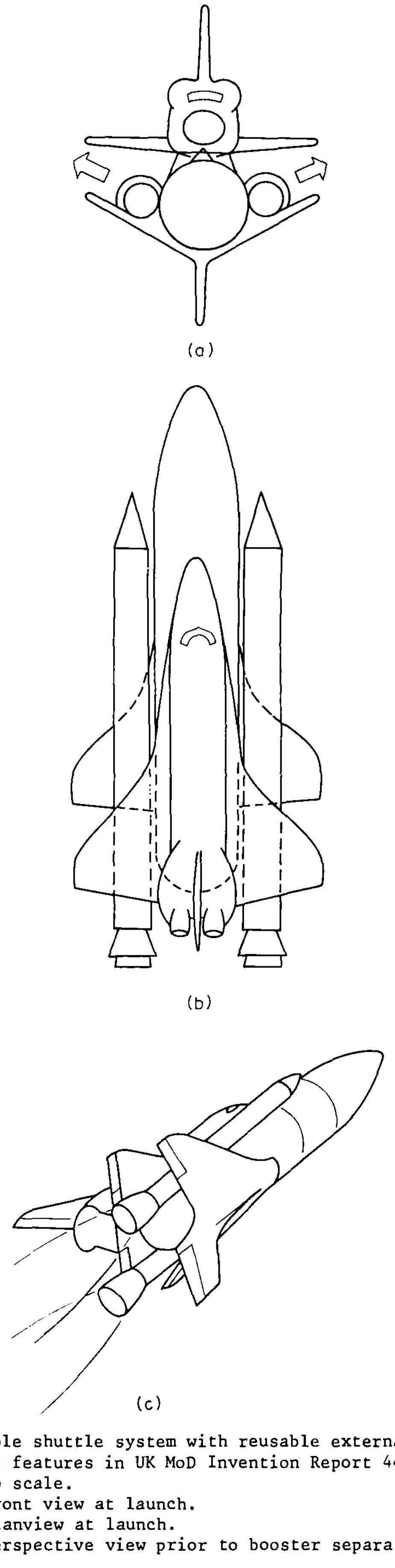


Fig. 70. Possible shuttle system with reusable external tank. (This scheme features in UK MoD Invention Report 4417.) Drawing not to scale.
(a) Front view at launch.
(b) Planview at launch.
(c) Perspective view prior to booster separation.

boosters are to be used for 20 missions, but the external tank is expended after only one launch, in spite of the fact that it accounts for nearly one quarter of the cost of each launching. In principle, a modified external tank could be made reusable within the constraints of the existing system and so might improve the operating economics without demanding a redesign in conceptual terms (134). A possible solution is shown in Fig. 70.

In Fig. 70, the external tank is shown fitted with two "orbiter wings" mounted on the tank so that pronounced anhedral is provided and the solid boosters are partly enclosed under the wing roots. The boosters would now be jettisoned in the directions indicated by the hollow arrows, after which the orbiter and winged tank would continue the launch to separate at higher altitude and Mach number, as already planned for the existing system.

The weight, drag and cost penalty of the added wings would be additional to those implied by the need to provide a reusable thermal protection system for the external tank and its wings; however, the aerodynamic lift and heat transfer on the "high wing tank" should be substantially assisted by flow containment (see the HWO tests of Section 3.1.1. and Figs 53-57) and subsonic flows could resemble those of Fig. 59.

With regard to subsonic performance the maximum loaded landing weight of the existing orbiter (at 200 kt landing speed) is about 200,000 lb (90700 kg). The existing empty external tank weighs about 100,000 lb (45350 kg) and so, with a somewhat higher landing speed to accommodate the weight penalties of the wings, fin, TPS and undercarriage, a winged reusable tank might need only half the wing area used for the orbiter; alternatively, if the tank were to use "orbiter wings", these could be rigged at about 8° of incidence and so, by permitting a low angle of attack (as measured at the tank axis), could permit the use of a short main undercarriage on the tank, with consequent reductions in the undercarriage weight penalty. This weight penalty would also be limited by mounting the undercarriage legs at existing strongpoints already provided for attaching the tank to the orbiter; this design approach is possible for the configuration of Fig. 70, since the tank and orbiter ride belly-to-belly rather than in piggy-back style.

4. CONCLUDING REMARKS

(a) Data on simple shapes have confirmed, that at lifting reentry conditions concave undersurfaces offer aerodynamic advantages such as superior values of C_{Lmax} and of C_L per unit L/D. Some data are also available on vehicle shapes which seem realistic and these data confirm that aerodynamic advantages are retained. It is suggested that there is now sufficient basic knowledge to justify detailed studies on configuration design and that "aimed research" should evaluate the practicality of various vehicle concepts over the entire speed range from reentry to approach and landing. Two types of vehicle have been suggested; one is a variant of the Space Shuttle orbiter and offers C_L increases of some 20%.

(b) Although there is now a large body of supersonic and hypersonic data on the relative-lifting performance of flat, concave and partly concave wings of delta and related planforms, the amount of data on heat transfer to such shapes remains extremely small, and most relates to low Reynolds number conditions. Since the case for concavity has been based on the claim that heating of fullscale vehicles would be moderated, it is important that this gap should be filled by appropriate theoretical and experimental research.

(c) Fundamental and aimed research should include more work on control and stability in pitch, roll and yaw at combinations of Mach and Reynolds number, which imply realistic wing loadings. The aims should include the achievement of vehicles, which are not only controllable, but stable at reentry conditions; thin shock layer theory suggests that certain partly concave shapes are suitable in this respect.

(d) The desirability, practicality and penalties of leading edges, which are aerodynamically sharp, should remain a major theme in aerodynamic and structural studies, and in research on high temperature materials.

(e) The practicality and cost of heatshields, which can survive many reentries without refurbishing, should be included in studies of heat transfer and of configuration design, based on partly concave undersurfaces.

(f) The fullscale influence of real gas effects should be assessed to the fullest extent that newly developing theoretical and experimental research permits. Existing data suggest that concave and partly concave undersurfaces retain their aerodynamic advantages over flat-bottomed wings.

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REFERENCES

1 Space Shuttle. Flight International, 108, 3485, 923-933, 25 Dec. 1975.

2 Baker, D. and Wilson, M. Space Shuttle debut. Flight International, 110, 3524, 975-981, 25 Sept. 1976.

3 Gregory, W. H. (Ed.) Special report: Space Shuttle. Aviation Week and Space Technology, 105, 19, pp. 9, 14-15, 36-102. 8 Nov. 1976.

4 Smith, T. W. An approach to economic space transportation. J. R. Aero. Soc., 70, 668. Aug. 1966.

5 Küchemann, D. The aerodynamic design of aircraft. To be published by Pergamon Press, 1978.

6 Nonweiler, T. R. F. Aerodynamic problems of manned space vehicles. J. R. Aero. Soc., 63, 585, 521-528. 1959.

7 Pennelegion, L. and Cash, R. F. Preliminary measurements in a shock tunnel of shock angle and undersurface pressure related to a Nonweiler wing. NPL Aero Rept. 1037. ARC CP 684. 1962.

8 Townend, L. H. Some design aspects of Space Shuttle orbiters. RAE Technical Report 70139 (1970). Also published in Progress in Aerospace Science, 13, 81-135, Pergamon Press, 1972.

9 Squire, L. C. A comparison of the lift of flat delta wings and waveriders at high angles of incidence and high Mach number. Ing. Archiv. 40 S.339-352, 1971.

10 Camarda, C. J. Analysis and radiant heating tests of a heat-pipe-cooled leading edge. NASA TN D-8468. August 1977.

11 Nonweiler, T. R. F. Conduction of heat within a structure subjected to kinetic heating. Aircraft Engineering, 28, 383-387, November 1956.

12 Nonweiler, T. R. F., Wong, H. Y. and Aggarwal, S. R. The role of heat conduction in leading edge heating. Ing. Archiv. 40 S. 107-117. 1971.

13 Davies, L., Cash, R. F., Regan, J. D., Townsend, J. E. G. and Catley, A. Experiments on flat delta wings and waveriders up to angles of incidence and Mach numbers suitable for lifting re-entry. Proceedings of 8th International Shock Tube Symposium, London, UK, 5-8 July 1971. Publ. Chapman and Hall, 1971.

14 Davies, L., Roe, P. L., Stollery, J. L. and Townend, L. H. Configuration design for high lift re-entry. AIAA paper 72-132. 1972.

15 Gregory, W. H. NASA analysis Shuttle economics. Av. Wk. and Sp. Tech., 99, 73-79, 24 Sept. 1973.

16 Heiss, K. Space Shuttle economics and US defence potentialities. Interavia 11/1976, 1071-1073. 1976.

17 Stollery, J. L. Private communications to author, 1970.

18 Davies, L. Private communications to author, 1970.

19 Rao, D. M. Hypersonic aerodynamic characteristics of flat delta and caret wing models at high incidence angles. J. Spacecraft and Rockets, 7, 12, 1475-1476. December 1970.

20 Roe, P. L. Private communications to author, 1970.

21 Squire, L. C. Private communications to author, 1970.

22 Houwink, R. and Richards, B. E. Experimental study of a high lift reentry vehicle configuration. AIAA J, 11, 5, 749-751. May 1973.

23 Townend, L. H. Private communication to RAE, 1972.

24 Townsend, J. E. G. and Davies, L. Private communication to author, 1972.

25 East, R. A. Private communications to author, 1976 and 1977.

26 Edwards, D. G., Burgin, K. and Davies, L. Private communication to author, 1976.

27 Townend, L. H. Ramjet propulsion for hypersonic aircraft. Presented to Euromech 3 Colloquium on "Supersonic flow with heat addition" Aachen. February 1966.

28 Seddon, J. and Spence, A. The use of known flow fields as an approach to the design of high speed aircraft. AGARD CP 30. 1968.

29 Bertram, M. H. Comments (on paper by Seddon and Spence (28)) in AGARD CP 30 (Supplement). 1968.

30 Plattner, C. M. NASA to begin unmanned tests of new types lifting shape for hypersonic manoeuvres. Av. Wk. and Sp. Tech., 91, 13, 52-58. 29 Sept. 1969.

31 Cole, J. D. and Brainerd, J. J. Slender wings at high angles of attack in hypersonic flows. In Hypersonic Flow Research (Ed. F. R. Riddell) 321-343. Academic Press, 1962.

32 Kennet, M. The inviscid hypersonic flow on the windward side of a delta wing. Inst. Aerospace Sci. Preprint 63-55. 1963.

33 Antonov, A. M. and Hayes, W. D. Calculation of the hypersonic flow about blunt bodies. J. Appl. Math. Mech., 30, 421-427. 1966.

34 Messiter, A. F. Lift of slender delta wings according to Newtonian theory. AIAA J., 1, 4, 794-802, April 1963.

35 Hayes, W. D. and Probstein, R. F. Hypersonic flow theory. Second edition, Vol. 1 - Inviscid flows. Academic Press, New York, 1966.

36 Hida, K. Thickness effects on the force of slender wings in hypersonic flow. AIAA J., 3, 3, 427-433, March 1965.

37 Squire, L. C. Calculated pressure distributions and shock shapes on thick conical wings at high supersonic speeds. Aero. Q., XVIII, 185-206, May 1967.

38 Squire, L. C. Calculation of pressure distributions on lifting conical wings with application to the off-design behaviour of wave riders. AGARD CP 30. 1968.

39 Hillier, R. The effects of yaw on conical wings at high supersonic speeds. Aero. Q., 21, 3, 199-210. August 1970.

40 Hillier, R. Some applications of thin shock layer theory to hypersonic wings. Ph. D. Thesis, University of Cambridge, 1970.

41 Squire, L. C. Calculated pressure distributions and shock shapes on conical wings with attached shock waves. Aero. Q., XIX, 31-50, February 1968.

42 Woods, B. A. Hypersonic flow with attached shock waves over delta wings. Aero. Q. XXI, 379-399, 1970.

43 Roe, P. L. A simple treatment of the attached shock layer on a delta wing. RAE TR 70246, 1970.

44 Roe, P. L. Aerodynamic problems of hypersonic aircraft. AGARD LS 42. 1972.

45 Babaev, D. A. Numerical solution of the problem of supersonic flow past the lower surface of a delta wing. (Translated from the Russian). AIAA J, 1, 9, 2224-2231, September 1963.

46 Babaev, D. A. Flow about a triangular wing for large values of M. (Translated from the Russian). USSR Computational Mathematics and Mathematical Physics, 3, 2, 528-532, 1963.

47 Voskresenskii, G. P. Numerical solution of the problem of a supersonic gas flow past an arbitrary surface of a delta wing in the compression region. Mekh. Zhid. i Gaza, 4, 134. 1968.

48 South, J. Methods for calculating non-linear conical flows. NASA SP 228, 1969.

49 Hillier, R. Three-dimensional wings in hypersonic flow. JFM, 54, 2, 305-337. 1972.

50 Shanbhag, V. V. Numerical studies on hypersonic delta wings with detached shock waves. ARC CP 1277, 1974.

51 Squire, L. C. Some extensions of thin shock layer theory. Aero. Q., 25, 1-13. February 1974

52 Squire, L. C. The effects of recessed lower surface shape on the lift and drag of conical wings at high incidence and high Mach number. Aero. Q., 26, 1, 1-10. February 1975.

53 Roe, P. L. Correlation between the heat transfer to a reentering space vehicle and the inviscid flow field. RAE TR to be published.

54 Bashkin, V. A. Experimental study of flow about flat delta wings at M=5 and angles of attack from 0^{o} to 70^{o}. Izv. an SSSR. Mekh. Zhid. i Gaza, 2, 3, 102-108. 1967.

55 Reding, J. P. and Ericsson, L. E. Effects of delta wing separation on shuttle dynamics. J. Spacecraft and Rockets, 10, 7, 421-428. July 1973.

56 Carr, C. J. Force measurements on caret and delta wings over the incidence range $27^{o} \leqslant \alpha \leqslant 55^{o}$ at M=12.2. Imperial College Aero. Rept. 71-22. October 1971.

57 Coleman, G. T. Force measurements on caret and delta wings at high incidence. Imperial College Aero. Rept. 72-16. July 1972.

58 Coleman, G. T. Aerodynamic force measurement on caret and delta wings at high incidence. J. Spacecraft and Rockets, 10, 11, 750-751. November 1973.

59 Jeffery, R. W. The aerodynamics of wings in low density hypersonic flow. Ph. D. Thesis. University of London, 1975.

60 East, R. A. and Lewis, K. J. Comparison of the measured lift coefficients of flat and caret delta wings from 20 degrees to 70 degrees incidence. ARC 35671, UK.

61 Richards, B. E. and Houwink, R. An experimental study of space vehicle configurations with high lift during reentry. VKI TN 117. 1976.

62 Davies, L., Regan, J. D. and Townsend, J. E. G. Private communication 1972.

63 Reggiori, A. Lift and drag of a wing-cone configuration in hypersonic flow. AIAA J., 9, 4, 744-745. April 1971.

64 Edge, E. R. and Powers, W. F. Shuttle ascent trajectory optimisation with function space quasi-Newton techniques. AIAA J., 14, 10, 1369-1376. October 1976.

65 Fey, W. A. Use of a lifting upper stage to achieve large offsets during ascent to orbit. AIAA Paper 66-60, 1966.

66 Elliot, J. R. and Rau, T. R. Optimal payload trajectory characteristics for winged booster vehicles. J. Spacecraft and Rockets, 5, 2, 218-220. February 1968.

67 Elliot, J. R. and Rau, T. R. Optimum use of the Earth atmosphere for Shuttle ascent. Deutsche Gesselschaft für Luft-und Raumfahrt Journal for Raumfahrtforschung. Band 16, Heft 2. March-April 1972.

68 Martin, J. A. Optimal payload ascent trajectories of winged vehicles. J. Spacecraft and Rockets, 11, 12, 860-862. December 1974.

69 Hidalgo, H. and Vaglio-Laurin, R. High altitude aerodynamics and its effects on lifting reentry performance. Proc. 18th. International. Astro. Cong., (Belgrade 1967), Vol. 3, Propulsion and reentry, 231-245. Pergamon Press 1968.

70 Hefer, G. An investigation of a waverider at low Reynolds numbers. Proc. 21st. Intern. Astron. Congress, North Holland, 1971.

71 Kipke, K. Experimental investigations of waveriders in the Mach number range from 8 to 15. AGARD CP 30. 1968.

72 Kipke, K. Experimentelle Untersuchungen an Wellenreiter-Flügeln in Hyperschallbereich. Abhandlungen der Brennschweigischen Wissenschaftlichen. Gesellschaft, 21, 407. 1970.

73 Stollery, J. L. Viscous interaction effects on reentry aerothermodynamics: theory and experimental results. AGARD LS 42, 1, 10-1 to 10-28. July 1972.

74 Metcalf, S. C. and Berry, C. J. Private communication to author, 1971.

75 Davies, L. and Berry, C. J. Private communication to author, 1973.

76 Galloway, E. M., Jeffery, R. W. and Harvey, J. K. Heat transfer to a delta wing and two waverider wings in a rarefied hypersonic flow. Imperial College Aero. Rept. 76-04. September 1976.

77 Vidal, R. J. and Bartz, J. A. Experimental studies of low-density effects in hypersonic wedge flows. In "Rarefied Gas Dynamics", Vol. 1, Academic Press, 1965.

78 Cheng, H. K., Hall, J. G., Golian, T. C. and Hertzberg, A. Boundary layer displacement and leading edge bluntness effects in high temperature hypersonic flow. J. Aerosp. Sci., 28, 5, 353-410, 1961.

79 Cheng, H. K. The blunt body problem in hypersonic flow at low Reynolds number. Cornell Aero. Lab., CAL Rept. AF-1285-A-10, 1963.

80 East, R. A. and Scott, D. J. G. An experimental determination of the heat transfer rates to a caret wing at hypersonic speed. Univ. of Southampton, AASU Rept. 273, November 1967.

81 Enkenhus, K. Intermittent facilities. Published by Von Karman Institute Rhôde-St. Génèse, 1969.

82 Bird, G. A. Some methods of evaluating imperfect gas effects in aerodynamic problems. Unpublished MOD(PE) material.

83 Bray, K. N. C. Real gas effects on lifting reentry aerothermodynamics. AGARD LS 42, Vol. 1, 9-1 to 9-19. July 1972.

84 Stalker, R. J. and Stollery, J. L. The use of a Stalker Tube for studying the high enthalpy non-equilibrium airflow over delta wings. Proc. Tenth Int. Shock Tube Symp., Japan, 1975.

85 Stalker, R. J. Development of a hypervelocity wind tunnel. Aeronautical Journal, 76, 738, 374-384, June 1972.

86 Pike, J. The flow past flat and anhedral delta wings with attached shockwaves. RAE TR 71081, 1971.

87 Hui, W. H. Exact theory for the stability of caret wings in hypersonic flow. University of Southampton AASU Rept. No. 284 (ARC 30663). 1968.

88 Hui, W. H. Stability of oscillating wedges and caret wings in hypersonic and supersonic flows. AIAA J., 7, 8, 1524-1530. 1969.

89 Liu, D. D. A unified unsteady flow theory of delta wings with attached shock waves. Ph. D. Thesis, Univ. of Southampton. July 1974.

90 Liu, D. D. and Hui, W. H. Oscillating delta wings with attached shock waves. AIAA J., 15, 6, 804-812. 1977.

91 Hui, W. H. Supersonic and hypersonic flow with attached shock waves over delta wings. Proc. Roy. Soc., London, Series A, 325, 251-267. 1971.

92 Hui, W. H. Methods for calculating nonlinear flow with attached shock waves over conical wings. AIAA J., 11, 10, 1443-1445. 1973.

93 Hui, W. H. and Hemdan, H. T. Unsteady hypersonic flow over delta wings with detached shock waves. AIAA J., 14, 4, 505-511. 1976.

94 East, R. A. Measurements of the dynamic pitching stability of delta wings at hypersonic speeds. ARC 35670. November 1974.

95 Orlik-Ruckemann, K. J. Dynamic viscous pressure interaction in hypersonic flow. NRC Aero. Rept. LR-535.

96 Enkenhus, K. R., Richards, B. E. and Culotta, S. Free-flight stability measurements in the Longshot Tunnel. Proc. Eight International Shock Tube Symposium, London, July 1971.

97 Freeman, D. C., Boyden, R. P. and Davenport, E. E. Supersonic dynamic stability of a modified 089 B Shuttle orbiter. NASA TMX-72630. October 1974.

98 Boyden, R. P. and Freeman, D. C. Subsonic and transonic dynamic stability of a modified 089 B Shuttle orbiter. NASA TMX-72631, January 1975.

99 Uselton, B. L. and Jenke, L. M. Pitch-, yaw-, and roll-damping characteristics of a Shuttle orbiter at $M_\infty = 8$. Arnold Eng. Dev. Centre, Arnold AFB, Tenn., AEDC-TR-74-129. May 1975.

100 Uselton, B. L., Freeman, D. C. and Boyden, R. P. Experimental dynamic stability characteristics of a Shuttle orbiter at $M_\infty = 8$. J. Sp. and Rockets, 13, 10, 635-640. October 1976. (N.B. In an earlier published form of this paper, sections of text were misplaced- the version quoted here is correctly printed).

101 Bagley, J. A. An estimate of the lateral forces and moments on yawed caret wings. Unpublished MOD(PE) material.

102 Crabtree, L. F. and Treadgold, D. A. Experiments on hypersonic lifting bodies. RAE TR 67004. 1967. Also presented at 5th International Congress of the Aeronautical Sciences, London, September 1966.

103 Kipke, K. Untersuchung an schiebenden Wellenreiter-Flugeln im Hyperschallbereich. ZFW, 21, 381,1973.

104 Hillier, R. Hypersonic flow over conical wing-body combinations. (To be published in Aero Q.1978).

105 Space Shuttle System: key issue-orbiter structural design. Grumman Aerospace Corp., December 1971.

106 Tamaki, F., Hinada, M. and Kurosaki, M. An application of the method of series truncation to the conical flow problem. Inst. of Space and Aero. Sci., Univ. of Tokyo. ISAS Rept. No. 469, 36, 11, 295-325. 1971.

107 Henderson, J. M. Low speed handling of a slender delta (HP 115). J. R. Aéro. Soc., 69, (653), 311-324, 1965.

108 Peckham, D. H. Low-speed wind-tunnel tests on a series of uncambered slender pointed wings with sharp edges. ARC R & M 3186. 1961.

109 Küchemann, D. A non-linear lifting surface theory for wings of small aspect ratio with edge separations. RAE Rept. Aero. 2540. April 1955.

110 Poisson-Quinton, Ph. Validitie de la soufflerie pour la prevision des performances et des qualités de vol. Facilities and techniques for aerodynamic testing at transonic speeds and high Reynolds number. AGARD CP 83. 1971.

111 Glass, K. J. and Whitnah, A. M. Space Shuttle: static aerodynamic characteristics of the MSC - 040A Space Shuttle orbiter with wedge centreline vertical and twin vertical tails at Mach numbers from 0.6 to 4.96. NASA CR 120050. March 1972.

112 Romero, P. O. and Chambliss, E. B. Space Shuttle: longitudinal aerodynamic characteristics of low aspect ratio wing configurations in ground effect for a moving and stationary ground surface. NASA CR 120082. November 1972.

113 Reding, J. P. and Emisson, L.E. Unsteady aerodynamic analysis of space shuttle vehicles. Part 4: Effect of control deflections on orbiter unsteady aerodynamics. NASA CR 120125. August 1973.

114 Keating, R. F. A. and Mayne, B. L. Low-speed characteristics of waverider wings. RAE TR 69051. 1969.

115 Allegre, J. and Bisch, C. Etude de l'écoulement à nombre de Mach de 18 autour d'une plaque plane en incidence et à borde d'attaque variable. AGARD CP 30 May 1968.

116 Stollery, J. L. Hypersonic viscous interaction - an experimental investigation of the flow over flat plates at incidence and around an expansion corner. Aerospace Research Lab (USAF) 70-0125. July 1970.

117 Fitzhugh, H. Hypersonic laminar boundary layer growth in an adverse pressure gradient. AGARD CP 30 May 1968.

118 Stollery, J. L. Hypersonic viscous interaction on curved surfaces. J. Fluid Mech, 43, 3, 497-511. September 1970.

119 Nonweiler, T. R. F. Private communication to the author, 1970.

120 Capey, E. C. Alleviation of leading edge heating by conduction and radiation. RAE TR 66311 1966.

121 Masaki, M. and Yakura, J. K. Transitional boundary layer considerations for the heating analysis of lifting re-entry vehicles. Jl. of Spacecraft and Rockets, 6, 9, 1048-1053. 1969.

122 Freel, R. Private communication to the author, 1974.

123 Busemann, A., Vinh, N. X. and Culp, R. D. Solution of the exact equations for three-dimensional atmospheric entry using directly matched asymptotic expansions. NASA CR-2643. March 1976.

124 Arrington, J. P. and Stone, D. R. Aerodynamic and flow visualisation studies of two delta-wing entry vehicles at a Mach number of 20.3. NASA TN D-7282. 1973.

125 Cross, E. J. Analytical investigation of the expansion flow field over a delta wing at hypersonic speeds. ARL 68-0027, Feb. 1968.

126 Seegmiller, H. L. Surface flow visualisation investigation of a high cross-range shuttle configuration at a Mach number of 7.4 and several Reynolds numbers. NASA TMX-62036. June 1970.

127 Rao, D. M. and Whitehead, A. H. Lee-side vortices on delta wings at hypersonic speeds. AIAA J., 10, 11, 1458-1465. November 1972.

128 Squire, L. C. Flow regimes over delta wings at supersonic and hypersonic speeds. Aeronautical Quarterly, 27, 1, 1-14, February 1976

129 Richards, I. C. Supersonic flow past a slender delta wing. An experimental investigation covering the incidence range $-5^{o} \leqslant \alpha \leqslant 50^{o}$. Aero.Q. 27, 2, 143-153, May 1976.

130 Küchemann, D. Die Aerodynamische Entwicklungs von schlanken Flügeln für den Uberschallflug. WGLR Jahrbuch, 66-77. 1962.

131 Ganzer, U. Experimentelle Ergebnisse zum Nonweiler- Wellenreiter im Unterschall-, Transschall- unt Ueberschallbereich, ZFW, 21, 153. 1973.

132 Camp, V. V. Van and Williams, E. T. Propulsion options for the hypersonic research airplane. J. Aircraft, 12, 7, 611-616. July 1975.

133 Towards hypersonics. Flight International, 108, 3477, 657-658. 30 Oct. 1975.

134 Haefeli, R. C., Littler, E. G., Hurley, J. B. and Winter, M. G. Technology requirements for advanced Earth-orbital transportation systems. Summary Report. NASA CR-2867. October 1977.